

SKIN AND SOFT TISSUES

SKIN

Dorsal skin versus palmar skin

The dorsal skin is thin, mobile and frequently hirsute over the proximal phalanges and the ulnar aspect of the dorsum of the hand. The skin of the palm and the palmar surface of the digits is thick and hairless, and has a well-defined stratum lucidum and a higher density of nerve endings and eccrine sweat glands, but no sebaceous glands.

Skin creases and fingerprints

Flexure lines crease the skin of the flexor surfaces of the wrist and hand (Fig. 50.1) where they are anchored to the deep fascia. Less regular, mainly transverse, creases cross the dorsal aspect of the radiocarpal, carpal, metacarpophalangeal and interphalangeal joints. The dorsal creases are more prominent in extension; the loose skin becomes stretched in flexion. (For a general review of 'skin lines', see p. 156.)

Three anterior transverse lines cross the wrist. The proximal line marks the proximal limit of the flexor synovial sheaths, the intermediate line overlies the radiocarpal joint, and a distal line marks the proximal border of the flexor retinaculum.



Fig. 50.1 The relation of the skin flexure lines and palmar arterial arches to the bones of the left hand. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

In the palm, a curved radial longitudinal line bounds the thenar eminence, ending close to the distal wrist crease. Proximal and distal transverse lines ascend medially across the palm. The proximal line begins at the distal end of the thenar crease and runs obliquely to the middle of the hypothenar eminence across the shafts of the metacarpals. The distal line begins at or near the cleft between the index and middle finger, and curves across the palm over the second to fourth metacarpal heads, near the proximal ends of the fibrous flexor sheaths.

The second to fifth digits show proximal, middle and distal sets of transverse lines. The proximal, often double, lies approximately 2 cm distal to the metacarpophalangeal joints. The middle creases are usually double, the proximal line lying directly over the proximal interphalangeal joint. The distal lines are usually single, and lie proximal to the distal interphalangeal joints. The metacarpophalangeal joint of the thumb is crossed by a crease that starts on the radial side and ends in the web space level with the base of the proximal phalanx. There is a second, shorter, crease, usually 1 cm distal to this line. Two creases cross the interphalangeal joint of the thumb (see Fig. 50.1).

Cutaneous vascular supply

The skin of the palmar aspect of the wrist is supplied by branches from the superficial palmar branch of the radial artery, the ulnar artery and, occasionally, the median artery (see Fig. 46.3). The skin over the thenar eminence is supplied by small perforating branches from the superficial palmar branch of the radial artery and the princeps pollicis. The skin over the hypothenar eminence is supplied by perforating branches from the ulnar artery, some of which pass through palmaris brevis. The remainder of the palm is supplied by branches from the common palmar digital arteries that pierce the palmar aponeurosis, and branches from the radialis indicis artery. The palmar digital skin is supplied by branches from each digital artery. In the distal phalanx, the two digital arteries typically form an H-shaped anastomosis, from which cutaneous perforators fan out within the pulp. Deep digital veins accompanying the digital arteries are usually very small and frequently absent. Superficial palmar veins tend to pass dorsally and drain into the larger superficial dorsal venous system.

The dorsal skin of the wrist is supplied by branches from an arterial plexus that lies over the extensor retinaculum, formed by branches of the radial artery, including its dorsal carpal branch, the dorsal carpal branch of the ulnar artery, and anterior and posterior interosseous arteries. The blood supply to the dorsum of the hand arises from longitudinal rows of four or five branches from each of the dorsal metacarpal arteries, which usually arise either directly from the radial artery or from the dorsal carpal arch. At the level of the neck of the metacarpals, where the second, third and fourth dorsal metacarpal arteries communicate with branches from the corresponding common palmar digital arteries, a large cutaneous perforating branch passes proximally to supply an area of skin as far as the dorsal aspect of the wrist.

The blood supply to the dorsum of the fingers arises proximally from the terminal branches of the dorsal metacarpal arteries, (supplying a region as far distally as the proximal interphalangeal joint) as well as from dorsal branches of the palmar digital arteries that are given off at each phalangeal level. At the level of the distal phalanx, the cutaneous supply comes from three dorsal arcades: a superficial arcade over the base of the distal phalanx, and two distal subungual arcades. The skin of the dorsum of the thumb is supplied by longitudinal axial branches of the princeps pollicis and dorsal branches from the palmar digital arteries.

Degloving injuries of the hand and wrist



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A degloving injury is one in which the skin is avulsed from its underlying structures, and usually occurs as the result of a low-velocity injury such as a ring avulsion or when the hand is caught in industrial machinery, typically between rollers or in a conveyor belt. The plane of cleavage is through the subcutaneous tissues, leaving the underlying structures essentially intact. The degloving may be complete or partial; complete avulsion is rare, whereas avulsion of the skin of all or part of the finger is relatively common.

Urbaniak et al (1981) grouped ring avulsion injuries into three classes: those in which the blood supply is intact (class I); those in which the blood supply is compromised (class II); and those in which digits are either completely degloved or amputated (class III). While these represent an increasing injury, the avulsion force required to achieve them does not occur in a linear manner (Kupfer et al 1999). The longitudinal force applied to a finger causes a traction injury to the blood vessels, which may extend over a considerable distance. The crushing element of a degloving injury may cause skin necrosis and thrombosis of the underlying vessels (Brooks et al 2007).

Treatment of these injuries depends on their extent (Krishnamoorthy and Karthikeyan 2011). In the acute phase, replantation is the best option in cases where the skin has been completely separated from the body. Where the skin remains attached but is devascularized, revascularization by microvascular anastomoses is preferred. If neither of these procedures is feasible, the avulsed skin may be defatted and reapplied to the affected part, where it is effectively acting as a split skin graft. If none of these techniques is practicable, primary amputation may be considered. Subsequently, reconstruction may be by split skin graft, skin/fasciocutaneous flap or free flap.

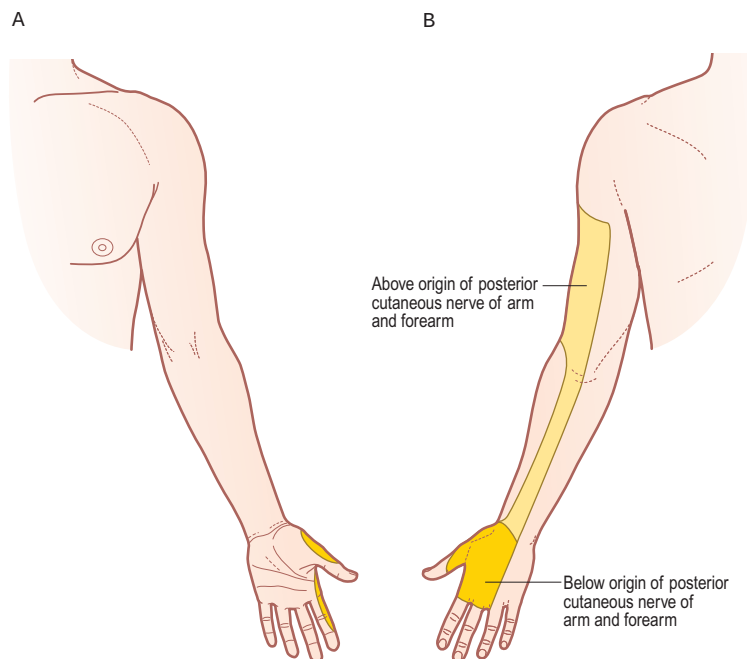


Fig. 50.2 The approximate area of sensory loss or disturbance after complete interruption of the radial nerve. **A**, Anterior. **B**, Posterior. The area of sensory disturbance after high lesions is very variable.

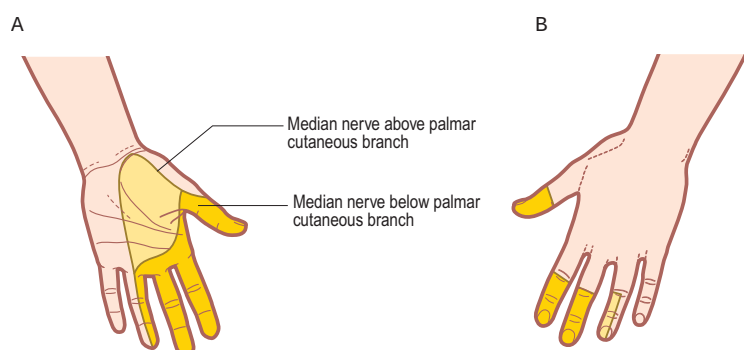


Fig. 50.3 The approximate area of sensory loss or disturbance after complete interruption of the median nerve. **A**, Anterior. **B**, Posterior.

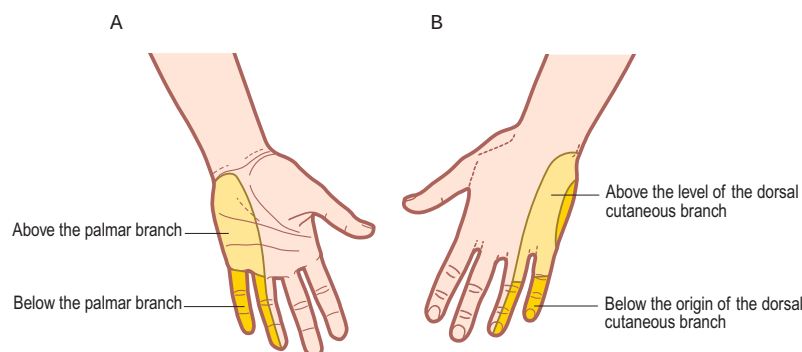


Fig. 50.4 The approximate area of sensory loss or disturbance after complete interruption of the ulnar nerve. **A**, Anterior. **B**, Posterior.

Basic skin flaps in reconstruction



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Cutaneous innervation

The skin of the anterior aspect of the wrist is innervated by the terminal branches of the lateral and medial cutaneous nerves of the forearm. The skin of the palm is innervated by the palmar branches of the ulnar and median nerves (see Fig. 50.40). The palmar skin of the thumb, index and middle fingers and the radial aspect of the ring finger is supplied by the median nerve, while that of the little finger and ulnar side of the ring finger is supplied by the ulnar nerve (Figs 50.3, 50.4).

The cutaneous innervation of the radial aspect of the dorsum of the wrist and hand, as well as the dorsal aspect of the radial three and a half digits as far distally as the nail beds, arises from the terminal branches of the radial nerve, namely: the dorsal digital nerves (Fig. 50.2). Between two and five dorsal digital nerves supply each digit. The cutaneous innervation of the ulnar aspect of the dorsum of the wrist

and hand, and the dorsal aspect of the ulnar one and a half digits as far distally as the nail beds, arises from the dorsal branch of the ulnar nerve, again ending as dorsal digital nerves. The skin of the dorsum of the middle and distal phalanges is also supplied by dorsal branches of the palmar digital nerves.

NAIL APPARATUS

The nail apparatus consists of the nail plate, proximal and lateral nail folds, nail matrix, nail bed and hyponychium; it is described on page 151; see also Figure 7.18.

SOFT TISSUES

Flexor retinaculum

The flexor retinaculum is a strong, fibrous band (see Figs 50.5, 50.25B). It crosses the front of the carpus and converts its anterior concavity into



The pedicled groin flap used to be regarded as the workhorse flap for reconstruction of the hand. This has gradually been replaced by the use of regional flaps based on the radial or posterior interosseous artery and by free flaps. Both radial and ulnar forearm flaps have been used to achieve soft tissue cover of the hand. As they rely on the sacrifice of one of the major arteries to the hand, it is essential, before considering one of these procedures, to demonstrate, using flow studies and angiography, that the remaining vessel, whether radial or ulnar, is adequate.

The radial forearm flap was originally described by surgeons from the Shenyang Military Hospital in 1978 and was consequently known as the 'Chinese flap'. It can be used as a free flap and also as a reversed pedicled fasciocutaneous island flap to cover soft tissue defects of the ipsilateral hand. Its blood supply is based on septocutaneous perforators that arise from the distal part of the radial artery and which supply the skin on the radial side of the lower forearm; these tend to become more sparse proximally. Venous drainage is by the venae comitantes and superficial subcutaneous veins. The ulnar forearm flap is a fasciocutaneous flap based on the ulnar artery distal to its common interosseous branch. A variant on this flap based on the ascending branch of the dorsal ulnar artery was described by [Becker and Gilbert \(1988\)](#). It has the disadvantage of a short pedicle but is useful for achieving cover of defects of the proximal and ulnar side of the palm.

The posterior interosseous island flap, distally based on the dorsal carpal arterial arch via the anastomoses between the posterior and anterior interosseous arteries just lateral to the head of the ulna ([Costa et al 2007](#)), was introduced by Zancolli in 1985. Its principal advantages are that it spares both of the major arteries of the forearm and is a good match for the skin colour and texture of the dorsum of the hand. Care must be taken to avoid injury to the adjacent posterior interosseous nerve when harvesting the flap. Doppler studies should be carried out preoperatively to ensure the presence and positioning of the anastomoses between the posterior and anterior interosseous arteries. It has been claimed that most defects of the hand can be treated using this flap; its particular advantage is in the treatment of hand injuries with severe accompanying vascular damage.

The anatomical distribution of the first dorsal metacarpal artery allows a flap of skin over the dorsum of the proximal phalanx of the index finger to be raised on the artery and its accompanying venae comitantes. This flap is particularly useful under certain circumstances for reconstruction of the thumb following injury.

The distribution of the second, third and fourth dorsal metacarpal arteries permits the surgical elevation of flaps of dorsal skin based either proximally, on the dorsal metacarpal arteries proper, or distally, on their direct cutaneous branches. These flaps may be used for reconstructing areas of missing tissue elsewhere in the hand.

The dorsal cutaneous branch of the ulnar artery is constant and can support a flap of skin on the medial aspect of the distal forearm just proximal to the wrist; it is useful in the reconstruction of areas of missing skin in this region.

the carpal tunnel that transmits the flexor tendons of the digits and the median nerve. The retinaculum is short and broad, measuring 2.5–3 cm both transversely and proximodistally. It is attached medially to the pisiform and the hook of the hamate. Laterally, it splits into superficial and deep laminae. The superficial lamina is attached to the tubercles of the scaphoid and trapezium. The deep lamina is attached to the medial lip of the groove on the trapezium. Together with this groove, the two laminae form a tunnel, lined by a synovial sheath, which contains the tendon of flexor carpi radialis. The retinaculum is crossed superficially by the ulnar vessels and nerve, immediately radial to the pisiform, and by the palmar cutaneous branches of the median and ulnar nerves (see Fig. 50.40A). A slender band of fascia, the superficial part of the flexor retinaculum, bridges the ulnar neurovascular bundle and attaches to the radial side of the pisiform, forming a tunnel (Guyon's canal) that is an occasional site of ulnar nerve entrapment. The tendons of palmaris longus and flexor carpi ulnaris are partly attached to the anterior surface of the retinaculum. Distally, some of the intrinsic muscles of the thumb and little finger are attached to the retinaculum.

Guyon's canal Guyon's canal is a fibro-osseous canal, approximately 4 cm long in the adult, on the anteromedial side of the wrist (Fig. 50.5). The roof is formed by the palmar carpal ligament and palmaris brevis. The floor is formed by the transverse carpal and pisohamate ligaments and, more distally, by the pisometacarpal ligaments and flexor digiti minimi. The ulnar side is bounded by flexor carpi ulnaris, the pisiform bone and abductor digiti minimi, and the medial side is bounded by the extrinsic flexor tendons, the transverse carpal ligament and the hook of the hamate. The fibrous arcade at the origin of the hypothenar muscles lies distally.

The canal transmits the ulnar nerve and artery, together with occasional venae comitantes. The ulnar nerve divides within it at the level of the hook of the hamate into a deep, radial motor branch and a superficial, ulnar sensory branch. If the nerve is compressed proximally, both modalities will be affected, whereas, distal to the bifurcation, only the sensory or motor branch will be compromised. Variations in the branching of the ulnar nerve may occur.

The most common variant involves the formation of abductor digiti minimi, particularly when an accessory head is present.

Extensor retinaculum

The extensor retinaculum is a strong, fibrous band that extends obliquely across the back of the wrist (Fig. 50.6). It is attached laterally to the anterior border of the radius, medially to the triquetral and pisiform bones, and, in passing across the wrist, to the ridges on the dorsal aspect of the distal end of the radius. It prevents bowstringing of the tendons across the wrist joint.

Palmar fascial complex

The palmar fascia is a three-dimensional ligamentous system composed of longitudinal, transverse and vertical fibres (Fig. 50.7).

Longitudinal fibre system

The longitudinal fibres run distally from the tendon of palmaris longus or the flexor retinaculum of the wrist to fan out across the whole width of the central third of the palm, producing four well-defined longitudinal bundles to the index, middle, ring and little fingers. A less well-defined bundle passes to the thumb. Distal to the transverse fibres of the palmar aponeurosis, the longitudinal fibres pass in three layers (McGrouther 1982). The most superficial longitudinal fibres (layer 1) are inserted superficially into the skin of the distal palm between the distal palmar crease and the proximal digital crease. Some superficial fibres pass distally into the palmar midline of the digit. Deeper longitudinal fibres (layer 2) pass deep to the natatory ligament and neurovascular bundles into the skin at the apex of the web space and into the fingers themselves, where they are continuous with Cleland's ligaments and the lateral digital sheet. Deeper still, the longitudinal fibres in layer 3 perforate the deep transverse metacarpal ligament to pass around the sides of the metacarpophalangeal joint and attach to the metacarpal bone and proximal phalanx, and extensor tendon.

Transverse fibre system

The transverse fibre system consists of the superficial transverse metacarpal (natatory) ligament, the transverse fibres of the palmar aponeurosis and the deep transverse metacarpal ligament.

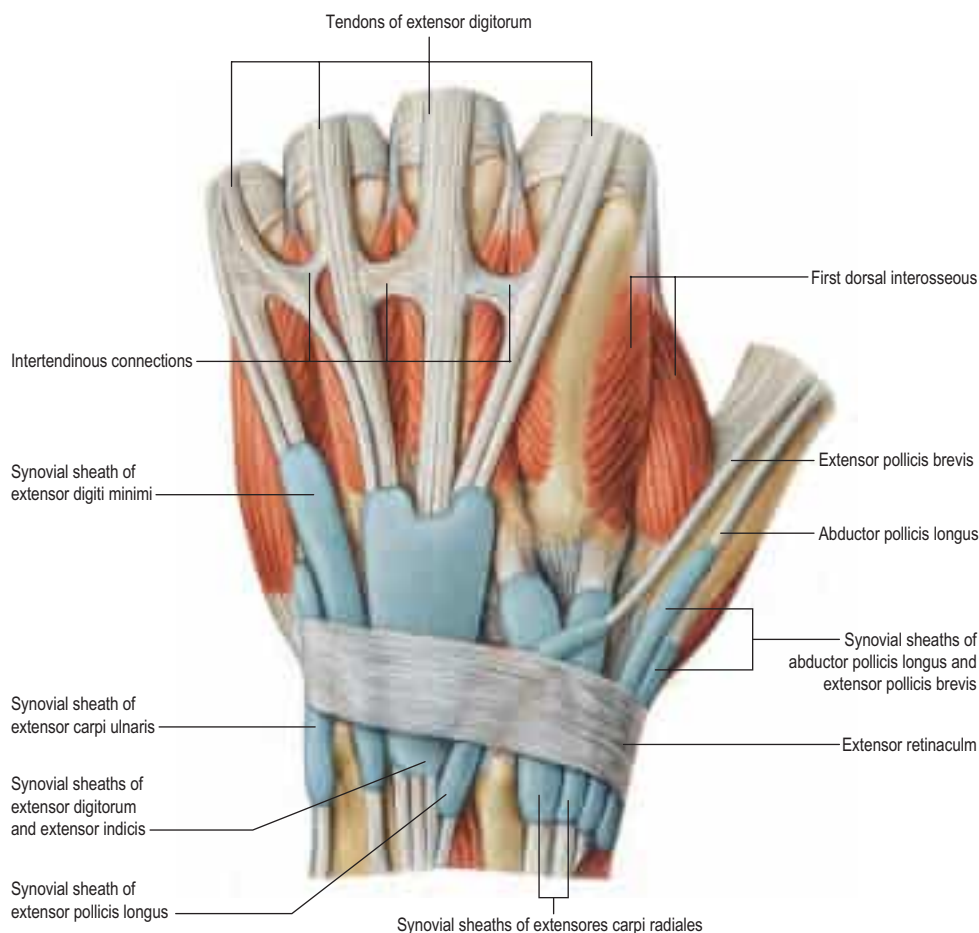


Fig. 50.6 The synovial sheaths of the tendons on the extensor aspect of the left wrist. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

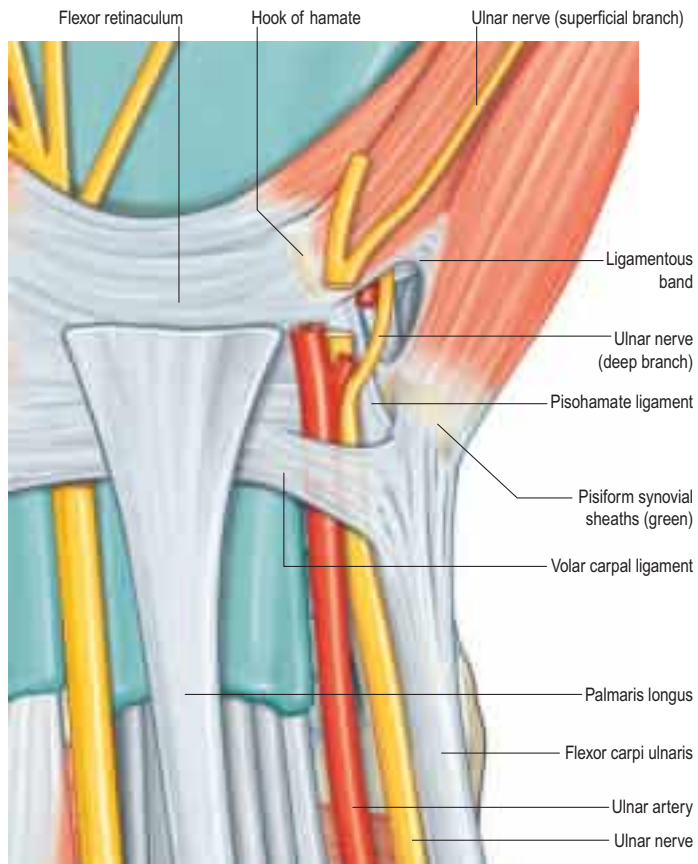


Fig. 50.5 Guyon's canal (space) at the wrist. Note the possibilities of entrapment at, and distal to, the volar carpal ligament. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*. 2nd edition, 2011. Springer-Verlag, London.)

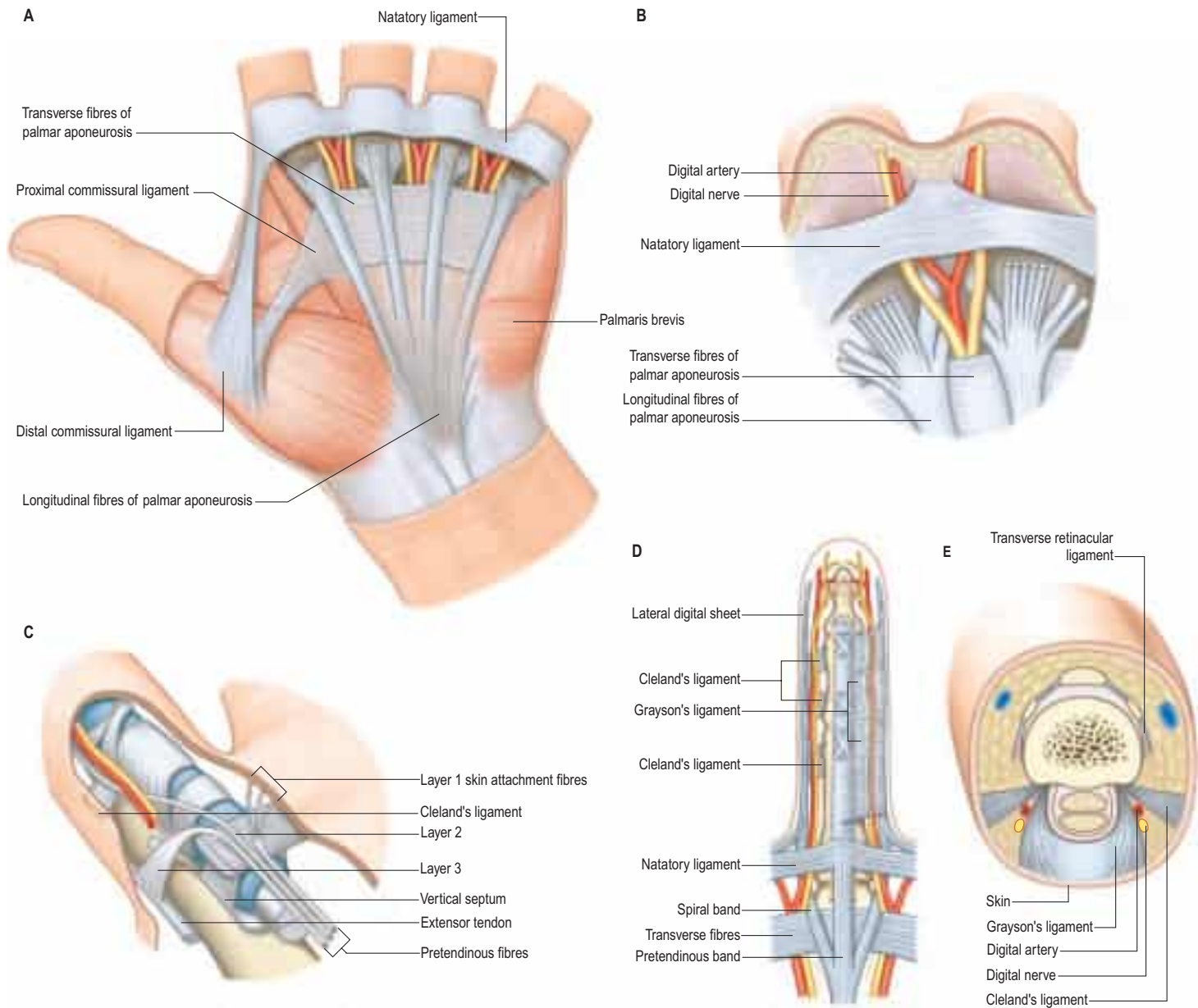


Fig. 50.7 The palmar aponeurosis and distal fascial complex. **A**, The palmar fascia. **B**, Structures at the web space. **C**, The fate of the distal longitudinal fibres. **D–E**, The normal digital fascia.

Superficial transverse metacarpal ligament (natatory ligament)

The fibres of the superficial transverse metacarpal ligament cross the apex of the web skin and extend into each digit to blend with the lateral digital sheet. They limit the separation of adjacent fingers and form the distal commissural ligament in the web space of the thumb.

Transverse fibres of the palmar aponeurosis

The transverse fibres of the palmar aponeurosis represent the deepest layer of the palmar fascia. They lie proximal to the distal palmar crease in a band approximately 2 cm wide, interconnecting the anterior fibres of the flexor tendon sheaths with the fasciae over the thenar and hypothenar muscles. The extension to the first ray forms the proximal commissural ligament.

Deep transverse metacarpal ligament

The strong fibres of the deep transverse metacarpal ligament lie deep to the palmar aponeurosis and flexor sheaths, and are attached to the volar plates of the metacarpal heads of the fingers.

Vertical fibre system

The vertical fibres are more delicate. They pass from the dermis, between the longitudinal and transverse fibres, to the fibrous flexor sheaths and the metacarpal bones. They are concentrated on either side of the palmar skin creases, as well as the thenar and hypothenar eminences. A series of vertical septa pass from the transverse fibres of the palmar

aponeurosis to the underlying deep transverse metacarpal ligament, forming compartments that contain the flexor tendons and the lumbricals and neurovascular bundles.

Digital fascial complex

The superficial fascia within a finger is fibrofatty in the palmar and dorsal aspects, but condenses laterally into the lateral digital sheet (see Fig. 50.7). The flexor sheath represents a well-defined thickening of the fascia and is discussed in detail on page 866. Cleland's ligaments extend from the sides of the phalanges, pass dorsal to the neurovascular bundles, and insert into the lateral digital sheet. Grayson's ligaments are more delicate, may even be discontinuous, and pass from the lateral sides of the phalanges anterior to the neurovascular bundles to insert into the lateral digital sheet. Landsmeer's ligaments are inconsistent anatomical structures made up of transverse and oblique retinacular ligaments.

Functions of the fascia of the hand

The fascial continuum of the hand performs a number of different, but interrelated, functions. It channels and provides a gliding surface for structures in transit between the forearm and the digits; transmits loads; anchors the skin; protects underlying vessels; and provides a framework for muscle attachments.

Channelling of structures in transit between forearm and digits

The vertical septa act as spacers between the tendons and neurovascular bundles of the individual digital rays. Where tendons change direction around a concave surface, the channels are thickened. The septa form sheaths with specialized pulleys to prevent the tendon springing away from the underlying skeleton.

Transmission of loads

At points where compressive loading is applied to the hand, such as the finger pulp and palm, loculi of fat act as shock absorbers. The loculi are contained within defined fibrous boundaries, which means that the shape, but not the volume, of each loculus can change. The compliance or deformability of the boundaries determines the amount of shock absorption. Local 'turgor' (deformability) and blood volume are measures of this anatomical property.

The palm also contains much larger fibrous compartments between skin and skeleton that transmit muscles, tendons and other structures. The honeycomb pattern of these compartments constitutes the palmar shock absorption system. The soft, padded parts of the hand are able to conform to the contours of objects that are grasped, and this permits better interpretation of sensation and better grip.

The hand must also resist tensile loading. Tendons and ligaments are particularly suitable for resisting such forces but many other parts of the fascial continuum, such as the anchorage system of the palm, also play a major role in resisting 'pulling' forces.

Anchorage

Skin is retained by fascial ligaments that allow the hand to flex while retaining the skin in position. The palmar creases have been described as skin 'joints'. Fascial anchors may be vertical (perpendicular to the palm), as in the mid-palm, where scattered vertical fibres run from the dermis down into the depths of the hand; horizontal (in the plane of the palm); or oblique to the skin surface.

The insertion of the longitudinal (pretendinous) fibres of the palmar aponeurosis is an example of a well-developed horizontal anchorage system. The most superficial longitudinal fibres insert into the dermis of the distal palm. This arrangement resists horizontal shearing force in gripping tasks, such as holding a hammer, where it prevents distal skin slippage or degloving of the palm during use. The characteristic blisters on the palms of those unaccustomed to such use map out the sites of the skin anchorage points. This anchorage system can be demonstrated by flexing the palm until the skin of the distal palm folds loosely. An attempt to pull the loose skin distally will reveal the anchoring longitudinal fibres of the palmar aponeurosis.

Oblique anchors occur in the fingers where Cleland's ligaments tether the skin of the proximal and middle segments of the digits to the region of the proximal interphalangeal joints.

Binding

Transversely orientated fascial structures help to maintain the transverse arch of the hand by 'binding' the underlying skeletal structures or the tendon sheaths.

Limiting or tethering

Joint motion is limited not only by joint ligamentous action, but also, in some cases, by skin tightness. Skin in the interdigital webs is generally reinforced by fascial ligamentous fibres that run just beneath the dermis in a direction that resists stretch; they are well developed in the thumb web.

Lubrication

There are many other gliding planes, e.g. between periosteum and the extensor apparatus, and between the latter and the skin, on the dorsum of the digits. The flexor tendon sheaths are lined by synovium.

Biomechanical studies of tendons gliding within their sheaths have emphasized the importance of avoiding extreme postures of a joint, which increase the friction between tendon and pulley, as well as the need to establish a smooth, well-lubricated tendon surface after repair. Increased friction within a tendon sheath may, in part, explain the cause of spontaneous rupture of extensor pollicis longus because the gliding resistance of the tendon has been shown to be significantly higher than that of other tendons (An 2007). Synergistic movements in the post-repair phase of recovery have been shown to improve tendon excursion and postoperative function.

Within the carpal tunnel and hand, the median nerve has been shown to glide by up to 3 mm, depending on the position of the elbow and the attitude of the forearm in pronation or supination (Echigo et al 2008). The greatest longitudinal excursion of the median

nerve (15.5 mm) is seen just proximal to the carpal tunnel; similar excursion of the ulnar nerve has been recorded as 14.8 mm in cadavers (Wilgis and Murphy 1986).

Vascular protection and pumping action

The blood vessels of the palm are surrounded by a cuff of tough fascia or by a fatty pad. When the hand is compressed, as in gripping, these relatively incompressible fascial structures function as a venous pumping mechanism to assist return of blood from the limb. By contrast, large capacitance veins on the dorsum of the hand lie in gliding skin, surrounded by loose areolar tissue, which allows the veins to dilate.

Framework for muscle attachments

Many of the small muscles of the hand, such as abductor pollicis brevis and palmaris longus, are attached to the fascial skeleton, at least in part. The fascial framework can be visualized as a harness by which muscles can act on the underlying skeleton. The metacarpophalangeal joint is moved by a ring of fascial and ligamentous structures that surrounds the joint and to which tendons are attached.

Dupuytren's disease



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Digital and palmar spaces

There are many potential spaces within the hand, often with ill-defined margins. The nail fold is a 'U-shaped' space made up of the eponychium and the lateral nail fold. The apical spaces at the tip of the finger are formed by the fibrous attachments of the distal phalanx to the tip of the digital pulp skin. The digital pulp spaces are confined compartments bounded by the digital creases that overlie the joints, and are attached to the underlying pulleys. The synovial flexor tendon sheaths are described on page 879. The web space is bounded distally by the skin and natatory ligament, by the deep transverse metacarpal ligament posteriorly, and by the deep attachments of the palmar fascia, together with their lateral attachments to the tendon sheaths, proximally. The deep palmar space is a complex, three-dimensional space, limited proximally by the carpal tunnel. It lies deep to the palmar aponeurosis, between the radial and ulnar condensations of vertical fibres that connect the palmar aponeurosis to the thenar and hypothenar eminences. Partitions that pass deeply from the longitudinal bands of the palmar aponeurosis form eight narrow compartments; four contain the digital flexor tendons and four contain the lumbricals and the neurovascular bundles.

Infections of the hand The spaces of the hand limit the spread of infection. Infections in the digit can occur in the nail fold (paronychia); the apical spaces at the very tip of the finger; the distal pulps (a felon); and the flexor sheaths. The flexor synovial sheath of the thumb and the little finger is continuous throughout the palm. Infection may spread to the palm and into other sheaths within the carpal tunnel but sepsis is as serious in those digits whose sheaths do not communicate with the carpal tunnel sheaths (index, middle, ring) as it is in those that do (thumb, little finger). Deep infections in the palm are usually not confined to any particular space.

Compartment syndrome of the hand



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BONES

The skeleton of the hand consists of the carpus, metacarpus and the phalanges.

CARPUS

The carpus contains eight bones: four each in proximal and distal rows (Figs 50.8–50.9). In radial to ulnar order, the scaphoid, lunate, triquetrum and pisiform make up the proximal row, and the trapezium, trapezoid, capitate and hamate make up the distal row. The pisiform articulates with the palmar surface of the triquetrum. The other three proximal bones form an arch that is convex proximally, and which articulates with the radius and articular disc of the distal radio-ulnar

Dupuytren's disease appears to be inherited as an autosomal dominant trait with incomplete penetrance (MacFarlane 1991, Ling 1963). It has been associated with heavy alcohol consumption, smoking, diabetes, manual work and vibration exposure, and possibly the ingestion of anticonvulsants (Descatha et al 2011). In the proliferative stage of the disease, there is considerable myofibroblast activity, which may be the result of upregulation of the musculo-aponeurotic fibrosarcoma oncogene homologue B (*MafB*), found in the diseased cord tissue (Lee et al 2006). Mechanical stimuli subsequently cause the alignment of myofibroblasts along the lines of traction, which are later replaced by collagen.

The pattern of contracture is determined by the anatomy of the basic constituents of the palmar and digital fascia. The pretendinous bands that constitute the superficial layer of the palmar aponeurosis (layer 1) are most frequently involved. The development of nodules usually precedes cord formation; the nodules tether the skin to the deep fascia. Contracture of the pretendinous cords causes flexion of the metacarpophalangeal joints. The cords run between the flexor sheaths and the neurovascular bundles. Shortening of the vertical bands (of Grapow) results in the appearance of skin pits and thickening of the skin. Involvement of the natatory ligament restricts abduction of the fingers. None of these changes usually displaces the neurovascular bundle because each component lies superficially in the palm.

The pretendinous cord in the palm extends into a finger as a central cord that lies superficially in the midline. The lateral digital sheet gives rise to a lateral cord that is attached to the skin or flexor tendon sheath near Grayson's ligament, causing a flexion contracture of the proximal interphalangeal joint and, sometimes, displacing the digital neurovascular bundle medially. Spiral cords arise from the pretendinous band in layer 1 of the palmar fascia and pass deeper through layer 2 to become attached to the middle phalanx by Grayson's ligament. They are so named because they spiral round the neurovascular bundle, displacing it proximally, palmarwards and towards the midline as contracture increases, and rendering it vulnerable to injury during fasciectomy (MacFarlane 1974). The distal interphalangeal joint is uncommonly involved because the palmar aponeurosis only extends as far as the sides of the middle phalanx.

An abscess in any of the digital or palmar spaces should always be drained. Incisions should, wherever possible, follow the line of the flexion creases and adhere to the principles of extensile exposure. The motor and palmar cutaneous branches of the median nerve are potentially at risk when draining a thenar space abscess through a palmar incision. A dorsal approach through the first web space, perpendicular to the web itself and between the first and second metacarpals, is to be preferred on anatomical grounds but may not always achieve adequate drainage on its own. The mid-palmar space is best approached through a combination of a transverse distal palmar incision with an oblique longitudinal extension (McDonald et al 2011).

Hypothenar space infections are very rare. They tend to be confined by the hypothenar fascia and result in swelling of the ulnar side of the hand with some flexion and adduction of the little finger. They may be drained by an incision along the ulnar border of the space without significant risk of damage to underlying structures.

There are 10 identifiable compartments in the hand: four contain the dorsal interossei, three contain the palmar interossei, and three contain the thenar and hypothenar muscles and adductor pollicis. Any or all of these compartments may be injured by a rise in intracompartmental pressure. The absolute pressure at which damage is caused has not been determined but is thought to be less than that required in the lower limb. Injury may be caused by blast, burn or crush, and by injection, whether of therapeutic substances at low pressure or of industrial agents at high pressure. Symptoms may be as non-specific as aching of the hand at rest or on exercise, or the hand may become painful with increasing swelling and loss of movement in the digits. On examination, an intrinsic minus-type hand (claw hand) may be seen, with extension of the metacarpophalangeal joints and flexion of the proximal interphalangeal joints. Intracompartmental pressure measurements should be taken; the means by which this is done is not critical (Ortiz and Berger 1998). Any suspicion of a compartment syndrome should be followed by prompt decompression. It is usually possible to decompress thoroughly through two dorsal longitudinal incisions over the second and fourth metacarpals, as well as release of the flexor retinaculum. The dorsal compartments are addressed directly and the volar compartments opened deep by longitudinal incision. (See also Eardley and Stewart (2010).)

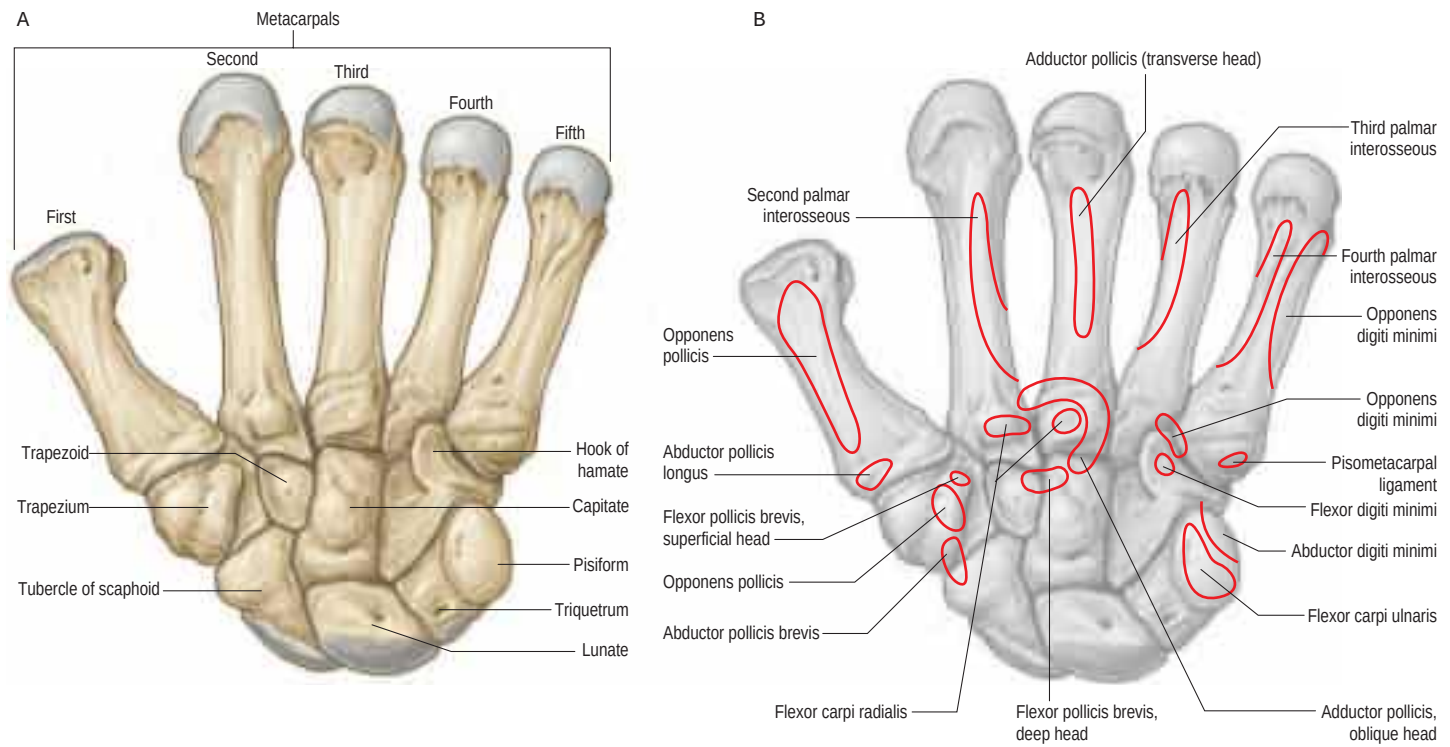


Fig. 50.8 **A**, The carpal and metacarpal bones of the left hand: palmar aspect. **B**, Sites of muscle attachments (except the dorsal interossei).

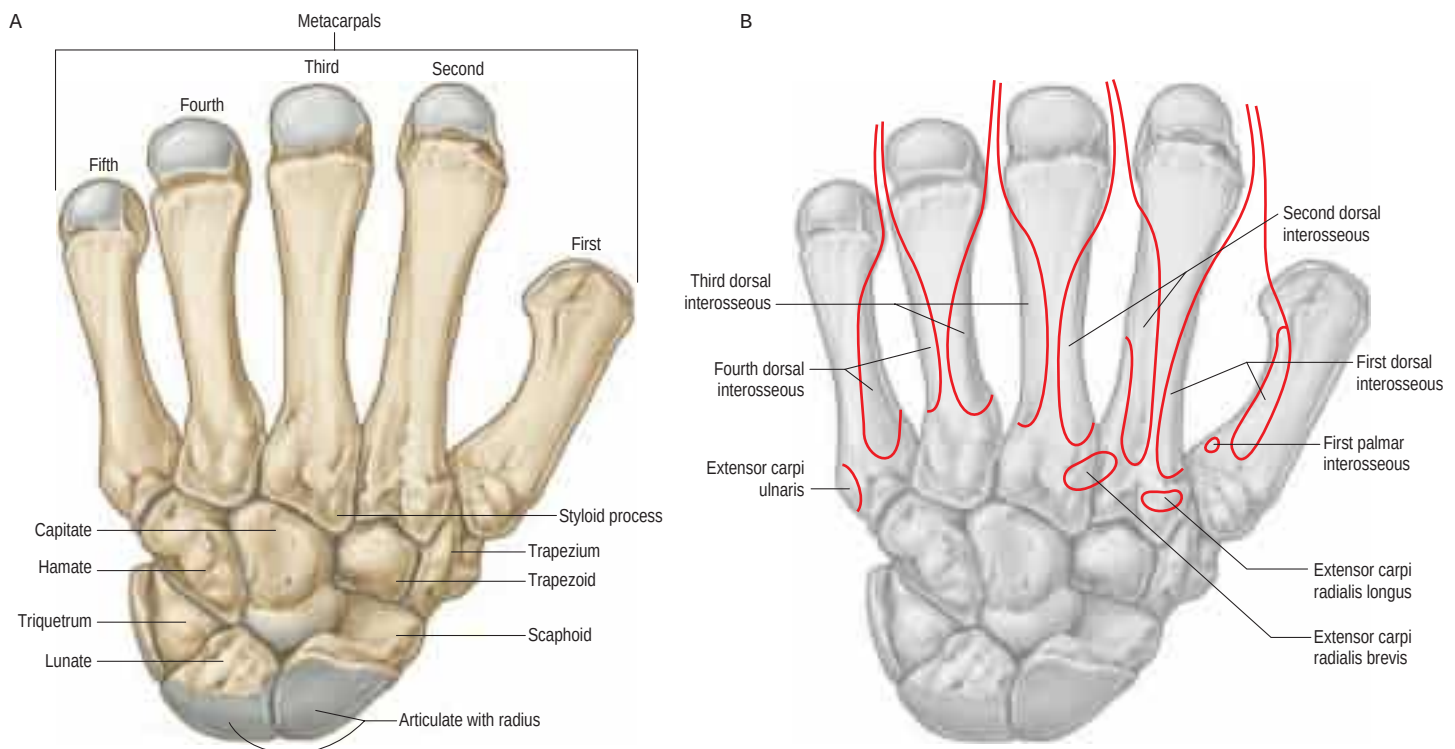


Fig. 50.9 **A**, The carpal and metacarpal bones of the left hand: dorsal aspect. **B**, Sites of muscle attachments.

joint. The distal concavity of the arch articulates with the capitate and hamate.

The dorsal carpal surface is convex. The palmar surface forms a deeply concave carpal groove, accentuated by the palmar projection of the radial and ulnar borders. The ulnar projection is formed by the pisiform and the hook (hamulus) of the hamate. The pisiform is at the proximal border of the hypothenar eminence, on the ulnar side of the palm, and it is easily felt in front of the triquetrum. The hamulus is concave in a radial direction; its tip is palpable 2.5 cm distal to the pisiform, in line with the radial border of the ring finger. The superficial division of the ulnar nerve can be rolled over it. The radial border of the carpal groove is formed by the tubercles of the scaphoid and trapezium. The former is distal on the anterior scaphoid surface and palpable (sometimes, also visible) as a small medial knob at the proximal

border of the palmar thenar eminence, radial to the tendon of flexor carpi radialis. The tubercle of the trapezium is a vertically rounded ridge on the anterior surface of the bone, slightly hollow medially and just distal and radial to the scaphoid tubercle; it is difficult to palpate. Both the scaphoid and trapezium may be grasped individually, and moved passively, by firm pressure between an opposed index finger and thumb applied to the palmar surface and 'anatomical snuff-box' simultaneously. The carpal groove is made into an osseofibrous carpal tunnel by the flexor retinaculum attached to its margins. The tunnel carries flexor tendons and the median nerve into the hand. The retinaculum strengthens the carpus and augments flexor efficiency. Radiocarpal, intercarpal and carpometacarpal ligaments are attached to the palmar and dorsal surfaces of all of the carpal bones, except the triquetrum and pisiform.

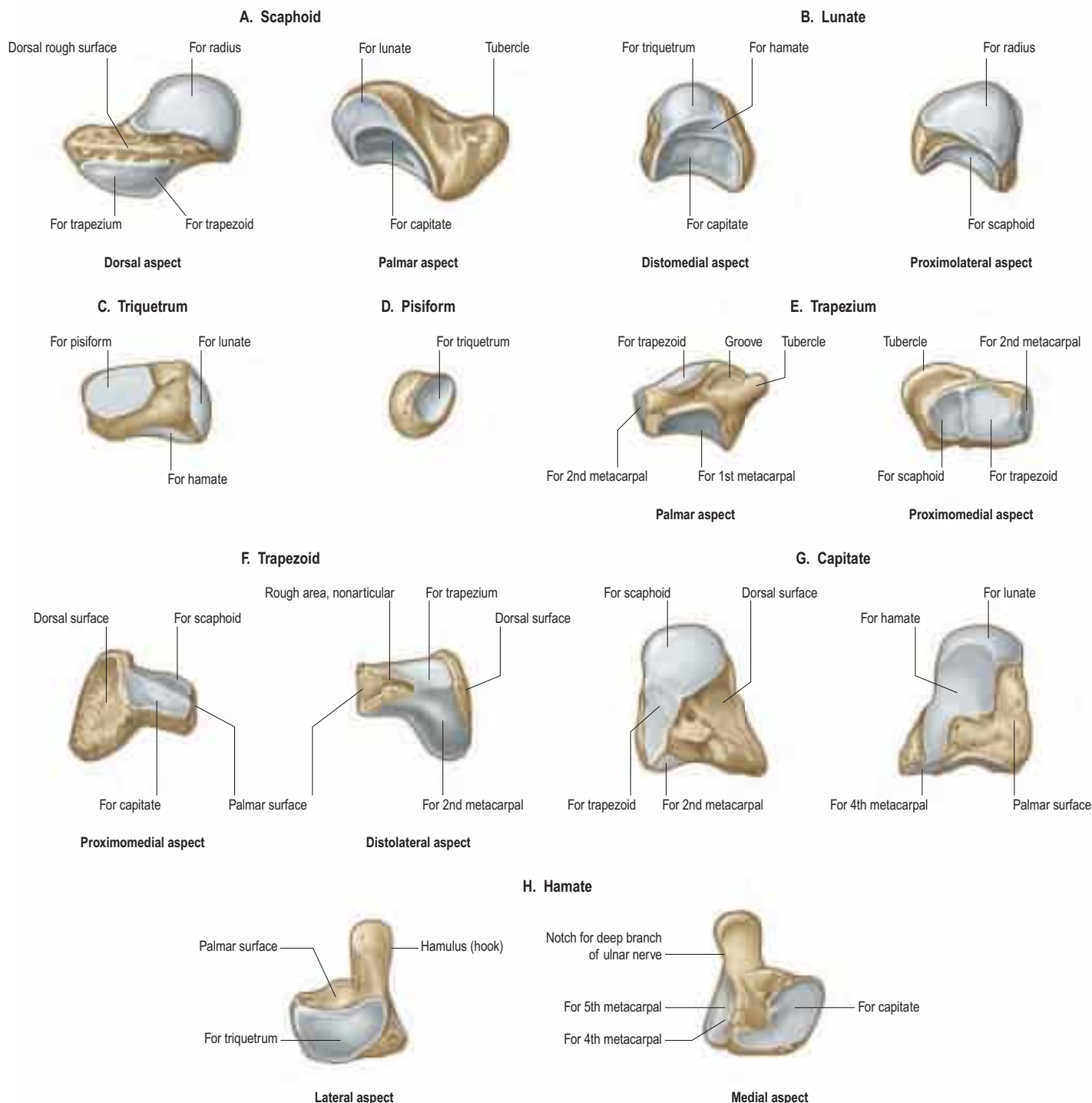


Fig. 50.10 Disarticulated carpal bones.

Individual carpal bones

Scaphoid

The scaphoid is the largest element in the proximal carpal row (Fig. 50.10A, see Figs 50.8, 50.9). It has a long axis that is distal, radial and slightly palmar in direction. A round tubercle on the distolateral part of its palmar surface is directed anterolaterally (see Fig. 50.8A), and provides an attachment for the flexor retinaculum and abductor pollicis brevis; it is crossed by the tendon of flexor carpi radialis. The rough dorsal surface is slightly grooved, narrower than the palmar, and pierced by small nutrient foramina, which are often restricted to the distal half. The radial collateral ligament is attached to the lateral surface, which is also narrow and rough. The remaining 42% of surfaces are all articular. The radial (proximal) surface is convex, proximal and directed proximolaterally; the lunate surface is flat and semilunar, and faces medially; the capitate surface is large, concave, distal, and directed distomedially. The surface for the trapezium and trapezoid is continuous, convex and distal.

The proximal, palmar surface of the non-articulating portion of the scaphoid is supplied by branches from the palmar carpal artery or, in

its absence, by branches of the superficial palmar artery (Oehmke et al 2009). The middle and distal thirds of the palmar surface are supplied by the superficial palmar artery and small radial rami arteries. The distal third may also receive branches from the first dorsal metacarpal artery. The dorsal surface is principally supplied by the dorsal scaphoid artery and by branches of the styloid artery. Variation in this pattern is not uncommon. Intraosseous blood vessels have been found close to the articulating surface.

Scaphoid bone fractures



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Lunate

The lunate is approximately semilunar and articulates between the scaphoid and triquetrum in the proximal carpal row (see Figs 50.8–50.10B). Its rough palmar surface, almost triangular, is larger and wider than the rough dorsal surface. Its smooth convex proximal surface articulates with the radius and the articular disc of the distal radio-ulnar joint. Its narrow lateral surface bears a flat semilunar facet for the

[Gelberman and Menon \(1980\)](#) describe the proximal 70–80% of the scaphoid as being supplied by branches of the radial artery that enter along its dorsal ridge, either just distal to the waist (14%), directly over the waist (59%) or just proximal to the waist (27%). They estimated that 14% of their specimens would have had significant interruption and 59% a partial interruption to the vascularity of the proximal pole of the scaphoid from a fracture through the waist, and that failure to unite would have resulted in avascular necrosis of the proximal fragment. It has been argued that an inadequate blood supply is unlikely to be the cause of scaphoid non-union after fracture ([Oehmke et al 2009](#)), but see also [Sendher and Ladd \(2013\)](#).

The scaphoid is the most frequently fractured carpal bone, typically as a result of a fall on to the outstretched hand. [Compson \(1998\)](#), in a study of the true, rather than the radiological, anatomy of scaphoid fractures, identified three distinct fracture patterns: transverse through the 'surgical waist'; oblique (45° to the surgical waist) in the plane of the dorsal sulcus; and through the proximal pole.

scaphoid. The medial surface, almost square, articulates with the triquetrum and is separated from the distal surface by a curved ridge, usually somewhat concave for articulation with the edge of the hamate in adduction (see Fig. 50.10B, left). The distal surface is deeply concave to fit the medial part of the head of the capitate.

Kienböck's disease



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Triquetrum

The triquetrum is somewhat pyramidal and bears an oval isolated facet for articulation with the pisiform on its distal palmar surface (see Figs 50.8–50.10C). Its medial and dorsal surfaces are confluent, and marked distally by the attachment of the ulnar collateral ligament, but smooth proximally to receive the articular disc of the distal radio-ulnar joint in full adduction. The hamate surface, lateral and distal, is concavo-convex, broad proximally and narrow distally. The lunate surface, almost square, is proximal and lateral.

Pisiform

The pisiform is a sesamoid bone and is shaped like a pea, with a distolateral long axis (see Figs 50.8, 50.10D). It bears a dorsal flat articular facet for the triquetrum. The tendon of flexor carpi ulnaris and the distal continuations of the tendon, the pisometacarpal and pisohamate ligaments, are all attached to the palmar non-articular area, which surrounds and projects distal to the articular surface.

Trapezium

The trapezium has a tubercle and groove on its rough palmar surface (see Figs 50.8A, 50.10E). The groove, which is medial, contains the tendon of flexor carpi radialis, and two layers of the flexor retinaculum are attached to its margins. The tubercle is obscured by the thenar muscles that are attached to it (opponens pollicis, flexor pollicis brevis and abductor pollicis brevis) (see Fig. 50.8B). The elongated, rough dorsal surface is related to the radial artery. The large lateral surface is rough for attachment of the radial collateral ligament and capsular ligament of the thumb carpometacarpal joint. A large sellar surface faces distolaterally and articulates with the base of the first metacarpal. Most distally, it projects between the bases of the first and second metacarpal bones and carries a small, quadrilateral, distomedially directed facet that articulates with the base of the second metacarpal. The large medial surface is gently concave for articulation with the trapezoid. The proximal surface is a small, slightly concave facet for articulation with the scaphoid. Its ridge, or 'summit', fits the concavity of the first metacarpal base, and extends in a palmar and lateral direction, at an angle of approximately 60° with the plane of the second and third metacarpals. Abduction and adduction occur in the plane of the ridge, which is shorter than the corresponding metacarpal groove. Their contours vary reciprocally: they are more curved near the second metacarpal base, whereas the radius of curvature is longer further away from this site. The two surfaces are not completely congruent, and the area of close contact probably moves towards the palm in adduction and dorsally in abduction. While the axis of flexion/extension passes through the trapezium, that for adduction/abduction is in the metacarpal base. Flexion is accompanied by medial rotation, and extension by lateral rotation.

Trapezoid

The trapezoid is small and irregular. It has a rough palmar surface that is narrower and smaller than its rough dorsal surface (see Figs 50.8–50.10F). The distal surface, which articulates with the grooved base of the second metacarpal, is triangular, convex transversely and concave at right angles to this. The medial surface articulates by a concave facet with the distal part of the capitate, the lateral surface articulates with the trapezium, and the proximal surface articulates with the scaphoid.

Capitate

The capitate is the central and largest carpal bone. It articulates with the base of the third metacarpal via its triangular distal concavo-convex surface (see Figs 50.8–50.10G). Its lateral border is a concave strip for articulation with the medial side of the base of the second metacarpal. Its dorsomedial angle usually bears a facet for articulation with the base of the fourth metacarpal. The head projects into the concavity formed by the lunate and scaphoid; the proximal surface articulates with the lunate, and the lateral surface with the scaphoid. The facets for the scaphoid and trapezoid, though usually continuous on the distolateral surface, may be separated by a rough interval. The medial surface bears

a large facet for articulation with the hamate, which is deeper proximally where it is partly non-articular. Palmar and dorsal surfaces are roughened for carpal ligaments, the dorsal being the larger.

Hamate

The hamate is cuneiform and bears an unciform hamulus (hook) that projects from the distal part of its rough palmar surface (see Fig. 50.10H). The hamulus is curved with a lateral concavity and its tip inclines laterally, contributing to the medial wall of the carpal tunnel (see Figs 50.8–50.9). The flexor retinaculum is attached to the apex of the hamulus. Distally, on the hamular base, a slight transverse groove may be in contact with the terminal deep branch of the ulnar nerve. The remaining palmar surface, like the dorsal, is roughened for attachment of ligaments. A faint ridge divides the distal surface into a smaller lateral facet that articulates with the base of the fourth metacarpal, and a medial facet for articulation with the base of the fifth. The proximal surface, the thin margin of the wedge, usually bears a narrow facet that contacts the lunate in adduction. The medial surface is a broad strip, convex proximally, concave distally, which articulates with the triquetrum; distally, a narrow medial strip is non-articular. The lateral surface articulates with the capitate by a facet covering all but its distal palmar angle.

Ossification

Carpal bones are cartilaginous at birth, although ossification may have started in the capitate and hamate. Each carpal bone is ossified from one centre – capitate first and pisiform last; the order in the others varies (Figs 50.11–50.15). The capitate begins to ossify in the second month; the hamate at the end of the third month; the triquetrum in the third year; and the lunate, scaphoid, trapezium and trapezoid in the fourth year in females and fifth year in males. The pisiform begins to ossify in the ninth or tenth year in females, and the twelfth in males. The order varies according to sex, nutrition and, possibly, race. Occasionally, an os centrale occurs between the scaphoid, trapezoid and capitate bones; during the second prenatal month, it is a cartilaginous nodule that usually fuses with the scaphoid. Occasionally, lunate and triquetral elements may fuse. Other fusions and accessory ossicles have also been described.

METACARPUS

The metacarpus consists of five metacarpal bones, conventionally numbered in radio-ulnar order. These are small long bones, with a distal head, shaft and expanded base. The rounded heads articulate with the proximal phalanges. Their articular surfaces are convex, although less so transversely, and extend further on the palmar surfaces, especially at their margins. The knuckles are produced by the metacarpal heads. The metacarpal bases articulate with the distal carpal row and with each other, except the first and second. The shafts have longitudinally concave palmar surfaces, which form hollows for the palmar muscles. Their dorsal surfaces bear a distal triangular area, which is continued proximally as a round ridge. These flat areas are palpable proximal to the knuckles.

The medial four metacarpals gently diverge proximodistally. The first metacarpal is more anterior and rotated medially on its axis through 90°, so that its morphologically dorsal surface is lateral, its radial border palmar, its palmar surface medial, and its ulnar border dorsal. Hence, the thumb flexes medially across the palm and can be rotated into opposition with each finger. Opposition depends on medial rotation and is important in manual dexterity; when an object is grasped, fingers and thumb encircle it from opposite sides, greatly increasing the power and skill of the grip.

Individual metacarpal bones

First metacarpal

The first metacarpal is short and thick (see Figs 50.8–50.9; Fig. 50.16A). Its dorsal (lateral) surface can be felt to face laterally; its long axis diverges distolaterally from its neighbour. The shaft is flattened, dorsally broad and transversely convex. The palmar (medial) surface is longitudinally concave and divided by a ridge into a larger lateral (anterior) and smaller medial (posterior) part. Opponens pollicis is attached to the radial border and adjoining palmar surface; the first dorsal interosseous muscle (radial head) is attached to its ulnar border and adjacent palmar surface. The base is concavo-convex and articulates with the trapezium. Abductor pollicis longus is attached on its lateral

Kienböck's disease, first described in 1910, is avascular necrosis of the lunate. Aetiology remains unclear but is thought to be associated with the arterial blood supply to the lunate. Up to eight small vessels usually enter both the dorsal and the palmar aspects of the bone, three on the dorsal surface and five on the palmar; they divide into smaller branches within the bone. However, in some cadaveric studies, only one vessel has been identified, and it has been inferred that this pattern would render an individual at increased risk of osteonecrosis, particularly if the branching pattern within the bone is limited. Other possible aetiological factors include disruption or obstruction of the venous outflow from the lunate, causing increased intraosseous pressure; negative ulnar variance; trauma; variations in the shape of the lunate; and mechanical factors. The natural history of the disease process is unknown.

While patients generally present with pain and stiffness of the wrist, these symptoms correlate poorly with the radiological findings and, indeed, patients may be asymptomatic. (For further reading, see [Cross and Matullo \(2014\)](#).)



Fig. 50.11 A radiograph of a hand at 2½ years (male), dorsopalmar projection. Note the early stages of ossification in the epiphyses at the proximal ends of the phalanges and first metacarpal; the distal ends of the remaining metacarpals and radius; and for the capitate, hamate and lunate. Typically, the centre for the lunate is preceded by the centre for the triquetrum. Compare with [Figures 50.12](#) and [50.13](#).



Fig. 50.13 A radiograph of a hand at 11 years (female), dorsopalmar projection. Compare the maturing ossification with that seen in [Figures 50.11](#) and [50.12](#). Note the addition of the pisiform.



Fig. 50.12 A radiograph of a hand at 6½ years (male), dorsopalmar projection. Note the more advanced state of the centres of ossification that were already visible in [Figure 50.11](#), and the appearance of additional centres in the distal ulnar epiphysis and in the triquetrum, scaphoid, trapezium and trapezoid.



Fig. 50.14 A radiograph of an adult hand, dorsopalmar projection. Key: 1, head of middle phalanx of middle finger; 2, head of proximal phalanx of ring finger; 3, shaft of proximal phalanx of ring finger; 4, head of fifth metacarpal; 5, shaft of fifth metacarpal; 6, base of fifth metacarpal; 7, hook of hamate; 8, triquetrum; 9, hamate; 10, pisiform; 11, styloid process of ulna; 12, lunate; 13, head of ulna; 14, distal phalanx of index finger; 15, distal phalanx of thumb; 16, proximal phalanx of thumb; 17, sesamoid bone; 18, trapezoid; 19, trapezium; 20, capitate; 21, scaphoid; 22, styloid process of radius; 23, distal radio-ulnar joint; 24, radius.

(palmar) side, the first palmar interosseous muscle to its ulnar side. The head is less convex than in other metacarpals and is transversely broad. Sesamoid bones glide on radial and ulnar articular eminences on its palmar aspect.

Second metacarpal

The second metacarpal has the longest shaft and largest base (see [Figs 50.8–50.9](#); [Fig. 50.16B](#)). The latter is grooved in a dorsopalmar direction for articulation with the trapezoid. Medial to the groove, a deep ridge articulates with the capitate; laterally, nearer the dorsal surface of the base, there is a quadrilateral facet for articulation with the trapezium, and just dorsal to this facet, a rough impression marks the attachment of extensor carpi radialis longus. On the palmar surface, a small tubercle or ridge receives flexor carpi radialis. The medial side of the base articulates with that of the third metacarpal by a long facet, centrally narrowed. The shaft is prismatic in section and longitudinally curved, convex dorsally, concave towards the palm. Its dorsal surface is distally broad but proximally narrows to a ridge that is covered by extensor tendons of the index finger. Its converging borders begin at the tubercles, one on each side of its head for the attachment of collateral ligaments. Proximally, the lateral surface inclines dorsally for the ulnar head of the first dorsal interosseous. The medial surface inclines similarly, and is divided by a faint ridge into a palmar strip for attachment

of the second palmar interosseous, and a dorsal strip for attachment of the radial head of the second dorsal interosseous.

Third metacarpal

The third metacarpal has a short styloid process, projecting proximally from the radial side of the dorsal surface (see [Figs 50.8–50.9](#); [Fig. 50.16C](#)). Its base articulates with the capitate by a facet anteriorly convex but dorsally concave where it invades the styloid process on the lateral aspect of its base. A strip-like facet, constricted centrally, articulates with the bases of the second metacarpal (laterally) and the fourth metacarpal (medially), the latter by two oval facets. The palmar facet may be absent; less frequently, the facets are connected proximally by a narrow bridge. The palmar surface of the base receives a slip from the tendon of flexor carpi radialis; extensor carpi radialis brevis is attached

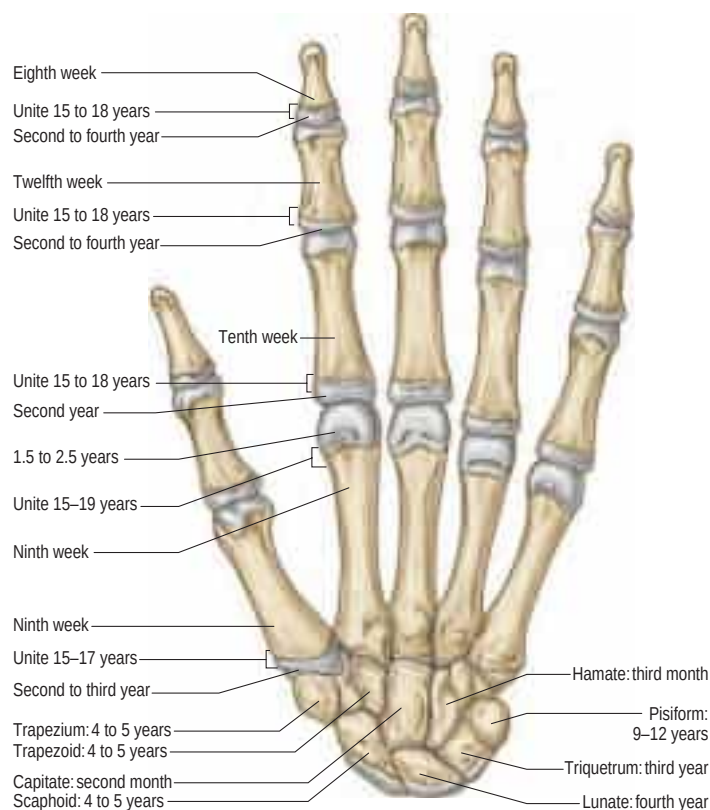


Fig. 50.15 The approximate dates of ossification of the individual bones of the hand and their epiphyses.

to its dorsal surface, beyond the styloid process. The shaft resembles that of the second metacarpal. The ulnar head of the second dorsal interosseous is attached to its lateral surface; the radial head of the third dorsal interosseous is attached to its medial surface, and the transverse head of adductor pollicis is attached to the intervening palmar ridge in its distal two-thirds. Its dorsal surface is covered by the extensor tendon.

Fourth metacarpal

The fourth metacarpal is shorter and thinner than the second and third (see Figs 50.8–50.9; Fig. 50.16D). On its base, it displays two lateral oval facets for articulation with the base of the third metacarpal; the dorsal is usually larger and proximally in contact with the capitate. A single medial elongated facet is for articulation with the base of the fifth metacarpal. The quadrangular proximal surface articulates with the hamate; it is anteriorly convex and dorsally concave. The shaft is like the second, but a faint ridge on its lateral surface separates the attachments of the third palmar interosseous and the ulnar head of the third dorsal interosseous. The radial head of the fourth dorsal interosseous is attached to the medial surface.

Fifth metacarpal

The fifth metacarpal (see Figs 50.8–50.9; Fig. 50.16E) differs in its medial basal surface, which is non-articular and bears a tubercle for extensor carpi ulnaris. The lateral basal surface is a facet, transversely concave, convex from palm to dorsum, for articulation with the hamate. A lateral strip articulates with the base of the fourth metacarpal. The shaft bears a triangular dorsal area that almost reaches the base; the lateral surface inclines dorsally only at its proximal end. Opponens digiti minimi is attached to the medial surface. The lateral surface is divided by a ridge, which is sometimes sharp, into a palmar strip for the attachment of the fourth palmar interosseous and a dorsal strip for the ulnar part of the fourth dorsal interosseous.

Ossification

Each metacarpal ossifies from a primary centre for the shaft and a secondary centre that is in the base of the first metacarpal and in the heads of the other four (see Figs 50.11–50.15). Ossification begins in the midshaft about the ninth week. Centres for the second to fifth metacarpal heads appear in that order in the second year in females, and between 1½ to 2½ years in males. They unite with the shafts about the fifteenth or sixteenth year in females, and the eighteenth or nineteenth in males. The first metacarpal base begins to ossify late in the second

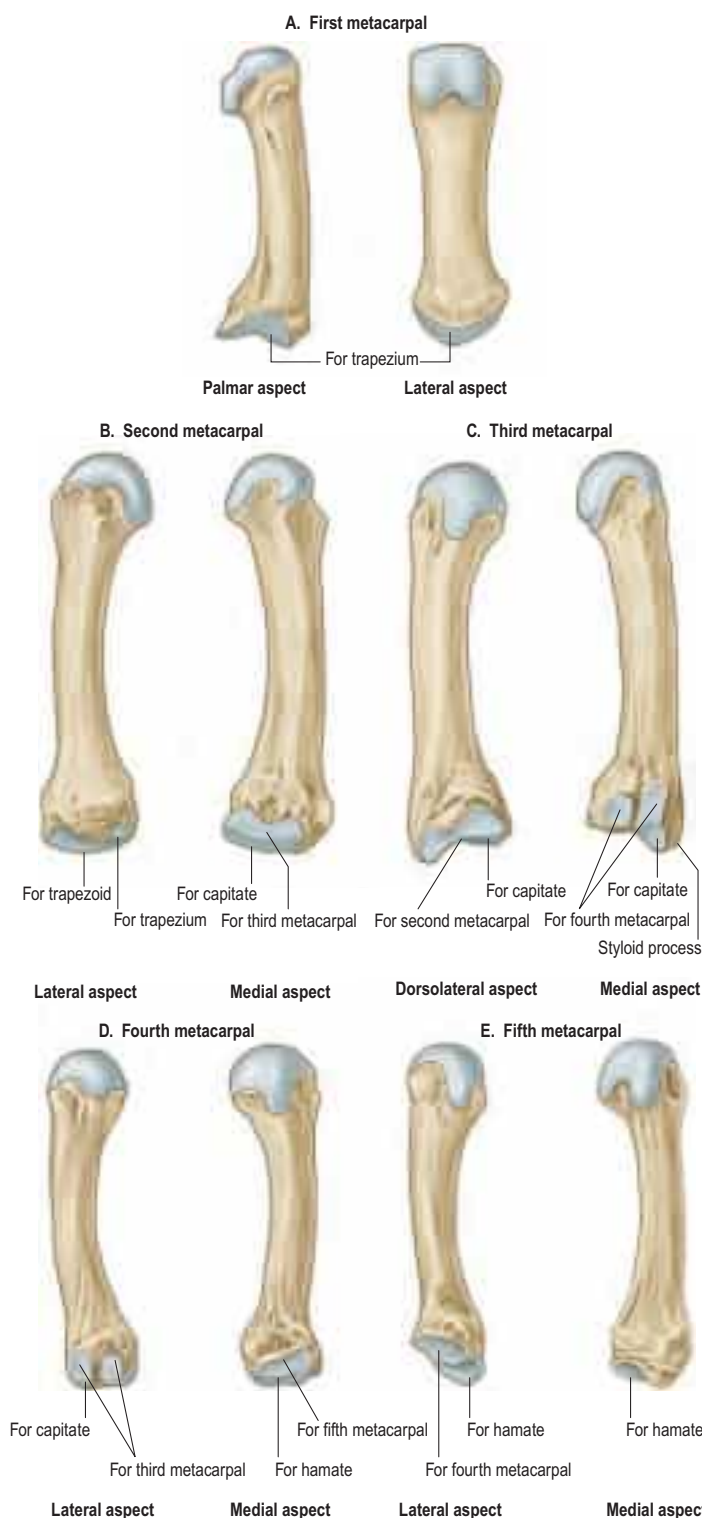


Fig. 50.16 Disarticulated metacarpal bones.

year in females, early in the third year in males, uniting before the fifteenth year in females and seventeenth in males. Sometimes, the styloid process of the third metacarpal is a separate ossicle. The thumb metacarpal ossifies like a phalanx, and some authorities therefore consider that the thumb skeleton consists of three phalanges. Others believe that the distal phalanx represents fused middle and distal phalanges, a condition occasionally observed in the fifth toe. When the thumb has three phalanges, the metacarpal has a distal and a proximal epiphysis. It occasionally bifurcates distally. When it does, the medial branch has no distal epiphysis and bears two phalanges, while the lateral branch shows a distal epiphysis, and three phalanges. The existence of only a distal metacarpal epiphysis may be associated with a greater range of movement at the metacarpophalangeal joint. In the thumb, the carpometacarpal joint has the wider range, and the first metacarpal has a basal epiphysis. A distal epiphysis may appear in the first, and a proximal epiphysis in the second, metacarpal.

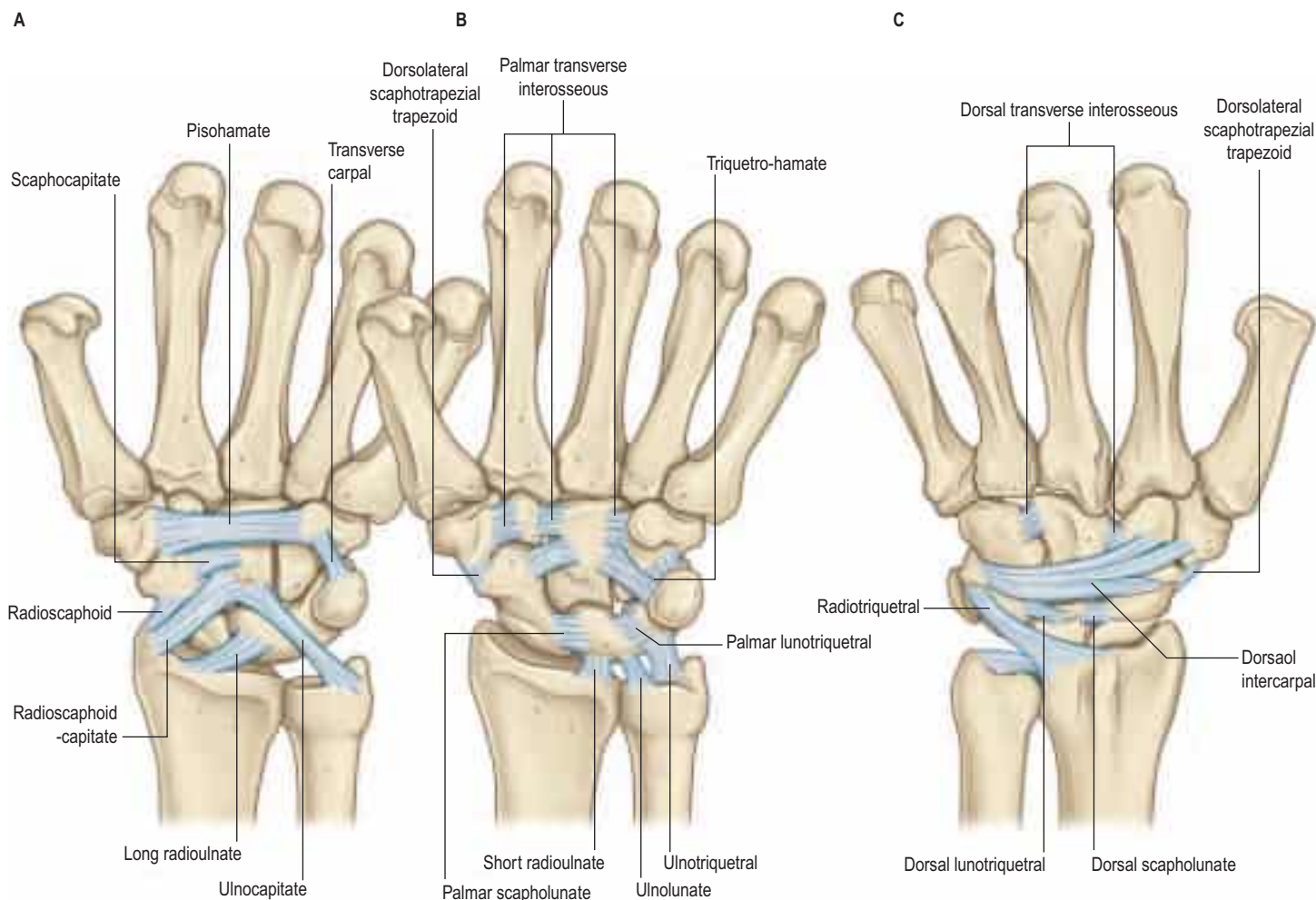


Fig. 50.17 Joints and ligaments of the left hand. (A) Palmar aspect: extrinsic ligaments. (B) Palmar aspect: intrinsic ligaments. (C) Dorsal aspect.

PHALANGES

There are 14 phalanges, three in each finger and two in the thumb (see Fig. 50.15). Each has a head, shaft and proximal base. The shaft tapers distally, its dorsal surface transversely convex. The palmar surface is transversely flat but gently concave anteriorly in its long axis. The bases of the proximal phalanges carry concave, oval facets adapted to the metacarpal heads. Their own heads are smoothly grooved like pulleys and encroach more on to the palmar surfaces. The bases of the middle phalanges carry two concave facets separated by a smooth ridge, conforming to the heads of the proximal phalanges. The bases of the distal phalanges are adapted to the pulley-like heads of the middle phalanges. The heads of the distal phalanges are non-articular and carry a rough, crescentic palmar tuberosity, to which the pulps of the fingertips are attached.

Articular ligaments and numerous muscles are attached to the phalanges. The tendon of flexor digitorum profundus is attached to the palmar aspect of the base of the phalanx and the tendon of extensor digitorum to its dorsal surface. The tendon of flexor digitorum superficialis and its fibrous sheath are attached to the sides of the middle phalanx, and a part of extensor digitorum is attached to the base dorsally. The fibrous flexor sheath is attached to the sides of a proximal phalanx, part of the corresponding dorsal interosseous is attached to its base laterally, and another dorsal interosseous is attached medially.

The phalanges of the little finger and the thumb differ. Abductor and flexor digiti minimi are attached to the medial side of the base of the proximal phalanx of the little finger. The tendon of extensor pollicis brevis and the oblique head of adductor pollicis (dorsally), and the oblique and transverse heads of adductor pollicis, sometimes conjoined with the first palmar interosseous (medially), are attached to the base of the proximal phalanx of the thumb.

Ossification

Phalanges are ossified from a primary centre for the shaft and a proximal epiphysal centre (see Figs 50.11–50.15). Ossification begins prenatally in shafts as follows: distal phalanges in the eighth or ninth week,

proximal phalanges in the tenth, and middle phalanges in the eleventh week or later. Epiphyseal centres appear in proximal phalanges early in the second year (females), or later in the same year (males), and in middle and distal phalanges in the second year (females), or third or fourth year (males). All epiphyses unite about the fifteenth to sixteenth year in females, and seventeenth to eighteenth year in males.

JOINTS

RADIOCARPAL (WRIST) JOINT

Articulating surfaces The radiocarpal joint is a synovial, biaxial and ellipsoid joint formed by articulation of the distal end of the radius and the articular disc of the triangular fibrocartilage with the scaphoid, lunate and triquetrum (Figs 50.17–50.18). In the neutral position of the wrist, only the scaphoid and lunate are in contact with the radius and articular disc; the triquetrum comes into apposition with the disc only in full adduction of the wrist joint. The radial articular surface and distal discal surface form an almost elliptical, concave surface with a transverse long axis. The radial surface is bisected by a low ridge into two concavities. A similar ridge usually appears between the medial radial concavity and the concave distal discal surface. The proximal articular surfaces of the scaphoid, lunate and triquetrum and their interosseous ligaments form a smooth convex surface that is received into the proximal concavity.

Fibrous capsule The fibrous capsule is lined by synovial membrane that is usually separate from that of the distal radio-ulnar and intercarpal joints. A protruding prestyloid recess (recessus sacciformis), anterior to the articular disc, is present and ascends close to the styloid process. The recess is bounded distally by the fibrocartilaginous meniscus, which projects from the ulnar collateral ligament between the tip of the ulnar styloid process and the triquetrum; both are clothed with hyaline articular cartilage. The meniscus may ossify. The capsule is strengthened by palmar radiocarpal and ulnocarpal, dorsal radiocarpal and radial and ulnar collateral ligaments.

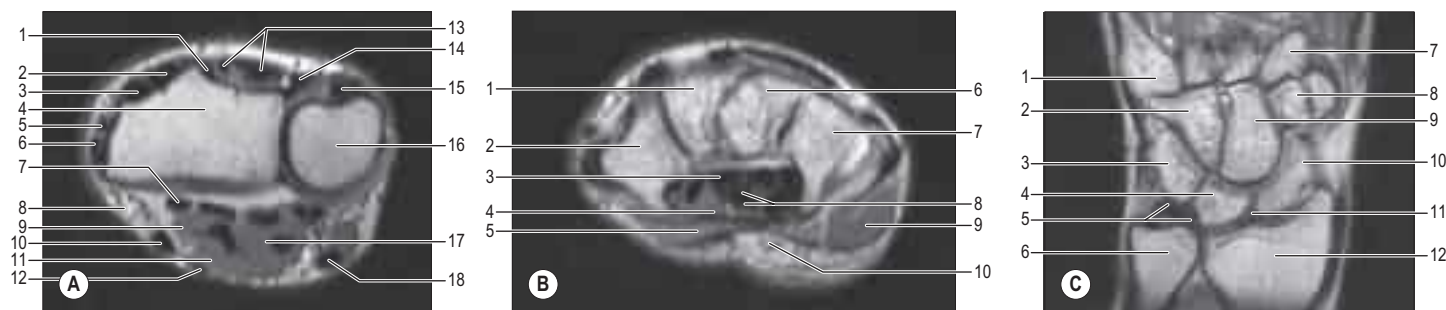


Fig. 50.18 Magnetic resonance image (MRI) of the pronated left wrist. **A**, An axial view at the level of the distal radio-ulnar joint (compare with Fig. 49.8). Key: 1, extensor pollicis longus; 2, extensor carpi radialis brevis; 3, extensor carpi radialis longus; 4, radius; 5, extensor pollicis brevis; 6, abductor pollicis longus; 7, flexor pollicis longus; 8, radial artery; 9, median nerve; 10, flexor carpi radialis; 11, flexor digitorum superficialis; 12, palmaris longus; 13, extensor digitorum and extensor indicis; 14, extensor digiti minimi; 15, extensor carpi ulnaris; 16, ulna; 17, flexor digitorum profundus; 18, flexor carpi ulnaris. **B**, An axial view at the level of the carpal tunnel. Key: 1, trapezoid; 2, trapezium; 3, flexor digitorum profundus; 4, median nerve; 5, thenar muscles; 6, capitate; 7, hamate; 8, flexor digitorum superficialis; 9, hypothenar muscles; 10, ulnar artery and nerve. **C**, Coronal view demonstrating normal bony anatomy and the triangular fibrocartilage: dorsal aspect. Key: 1, base of fifth metacarpal; 2, hamate; 3, triquetrum; 4, lunate; 5, triangular fibrocartilage; 6, distal ulna; 7, base of second metacarpal; 8, trapezoid; 9, capitate; 10, scaphoid; 11, scapholunate ligament; 12, distal radius. (Courtesy of Drs Allen S Prober and Robert J Ward, Tufts University, Boston, MA, USA.)

Synovial membrane The synovial membrane lines the fibrous capsule.

Muscles producing movement Movements accompany those of the intercarpal and midcarpal joints, and are described on page 874.

CARPAL JOINTS

The intercarpal joints connect the carpal bones. Their articular surfaces are saddle-shaped, ellipsoid or spheroidal. The carpal bones are connected by an extensive array of ligaments, not all of which are specifically named. The flexor retinaculum is an accessory intercarpal ligament.

Joints of the proximal carpal row

Joints of the proximal carpal row are between the scaphoid, lunate and triquetrum. In addition, the pisiform articulates with the palmar surface of the triquetrum at a small, oval, almost flat, synovial pisotriquetral joint. A thin capsule surrounds the joint. The synovial cavity is usually separate but may communicate with that of the radiocarpal joint.

Joints of the distal carpal row

Joints of the distal carpal row are between the trapezium, trapezoid, capitate and hamate. There is little movement at these joints.

Midcarpal joint

The midcarpal joint, between the scaphoid, lunate and triquetrum (proximally), and trapezium, trapezoid, capitate and hamate (distally), is a compound articulation that may be divided descriptively into medial and lateral parts. Throughout most of the medial compartment, the convexity formed by the head of the capitate and lunate articulates with a reciprocal concavity formed by the scaphoid, lunate and much of the triquetrum. However, most medially, the curvatures are reversed, forming a compound sellar joint. In the lateral compartment, the trapezium and trapezoid articulate with the scaphoid, forming a second planosellar compound articulation.

Carpal synovial membrane The extensive carpal synovial membrane lines an irregular articular cavity. Its proximal part is between the distal surfaces of the scaphoid, lunate and triquetrum, and the proximal surfaces of the second carpal row. It has proximal prolongations between the scaphoid and lunate, and lunate and triquetrum, and three distal prolongations between the four bones of the second row. The prolongation between the trapezium and trapezoid and/or between the trapezoid and capitate is often continuous with corresponding carpometacarpal joints, either from the second to the fifth, or from the second and third only. In the latter case, the joint between the hamate and fourth and fifth metacarpal bones has a separate synovial membrane and the carpometacarpal interosseous ligament is interposed.

Synovial cavities of carpometacarpal joints are prolonged slightly between the metacarpal bases. The synovial joint between the pisiform and triquetrum is usually isolated.

Wrist ligaments

The distal radio-ulnar joint and triangular fibrocartilage complex are described on pages 845–848.

Intracapsular ligaments lie between the fibrous and synovial layers of the wrist joint; their margins may not be distinct. Extracapsular ligaments are superficial to the fibrous layer. Most wrist ligaments lie within the joint capsule; the only exceptions are the flexor and extensor retinacula and the pisotriquetral ligament. The wrist ligaments are further classified into extrinsic and intrinsic named ligaments. Some of the extrinsic ligaments have superficial and deep parts; the latter are identifiable at wrist arthroscopy but the former are not. Wrist ligaments are conventionally named from proximal to distal and from radial to ulnar, e.g. the radioscaphocapitate ligament passes between the radius, scaphoid and capitate.

Extrinsic ligaments

The extrinsic ligaments connect the carpus with the forearm bones. They tend to be longer than the intrinsic ligaments and are approximately one-third as strong.

Extrinsic palmar carpal ligaments

When the synovial lining of the carpal tunnel is dissected away, two V-shaped ligamentous bands are visible with their apices lying distally (see Fig. 50.17A). The limbs of the 'V' take origin from the radius and ulna, respectively; the apex of one 'V' attaches to the distal row and that of the second 'V' to the proximal row.

Radioscaphocapitate ligament The radioscaphocapitate ligament originates from the radial styloid and the palmar lip of the radius, and has three parts. The radial part inserts on to the lateral aspect of the waist of the scaphoid (radial collateral ligament). The middle part continues as part of the distal 'V' and inserts on to the distal pole of the scaphoid. The ulnar part passes over the proximal pole of the scaphoid towards the mid-carpus and blends with the fibres that originate from the ulnar side of the triangular fibrocartilage complex to form the arcuate ligament over the palmar aspect of the capitate. A few of the fibres of the radioscaphocapitate ligament attach to the body of the capitate. There is a discrete interval between the inferior margin of this ligament and the palmar horn of the lunate (the space of Poirier).

Long radiolunate ligament The long radiolunate ligament arises adjacent to the radioscaphocapitate ligament on the palmar lip of the radius, passes over and supports the proximal pole of the scaphoid, and inserts into the palmar horn of the lunate. It is discrete from the radioscaphocapitate ligament, from which it is separated by the interligamentous sulcus, continuous with the space of Poirier.

Radioscapholunate (ligament of Testut) The radioscapholunate ligament is covered by a thick synovial lining and is a visible landmark during wrist arthroscopy.

Short radiolunate ligament The short radiolunate ligament is part of the proximal 'V'. It arises from the palmar lip of the lunate fossa of the radius and passes directly to the palmar horn of the lunate. On the ulnar side, its fibres blend with those of the palmar triangular fibrocartilage complex as they pass to their insertion on the lunate.

Ulnolunate ligament The ulnolunate ligament originates from the palmar aspect of the ulna adjacent to the short radiolunate ligament and inserts on to the palmar horn of the lunate. Part of this fibre complex arches radially and blends with part of the radioscapohcapitate complex to form the arcuate ligament.

Ulnotriquetral (ulnar collateral) ligament The ulnotriquetral ligament arises from the palmar aspect of the ulna, inserts into the medial aspect of the triquetrum, and continues distally to a further attachment to the medial aspect of the hamate. The ulnolunate and ulnotriquetral ligaments are also attached to the marginal ligament of the triangular fibrocartilage complex.

Extrinsic dorsal carpal ligaments

The dorsal wrist ligaments are comparatively thin and are reinforced by the floor and septa of the fibrous tunnels for the six dorsal compartments. The extrinsic dorsal carpal ligaments and the intrinsic dorsal intercarpal ligaments have a 'Z'-shaped configuration (see Fig. 50.17C).

Dorsal radiolunotriquetral ligament The dorsal radiolunotriquetral ligament has superficial and deep components. The superficial part connects the radius and triquetrum, and the deep part connects the radius, lunate and triquetrum. The wide superficial component arises from the dorsal margin of the distal radius and courses ulnarwards to insert on the dorsal edge of the triquetrum. The deep component takes a narrower origin from the ulnar aspect of the distal dorsal radius and passes in an ulnar direction to attach to part of the lunotriquetral articulation and the intrinsic lunotriquetral ligament.

Intrinsic ligaments

The intrinsic ligaments of the wrist are attached to carpal bones (see Fig. 50.17B). They are stronger and shorter than extrinsic ligaments and are connected with the extrinsic ligament complexes by interdigitating fibres. The intrinsic ligaments are subdivided into ligaments that connect the carpal bones of the proximal and distal rows, respectively, and ligaments that connect the rows by crossing over the midcarpal joint.

Proximal row interosseous ligaments The scapholunate and lunotriquetral ligaments are clinically and biomechanically important structures. In the sagittal plane, they have an approximately horseshoe-shape configuration with palmar, midcarpal and dorsal components. The scapholunate ligament contains short transverse fibres connecting the dorsal aspect of the respective bones and more obliquely orientated fibres connecting the palmar aspect. The tighter dorsal component of the ligament acts as a hinge facilitating flexion and extension of the scaphoid. The ligament continues in its midcarpal section as an interosseous membrane. The lunotriquetral ligament has similar dorsal, midcarpal interosseous and palmar components, but the fibres of the dorsal and palmar components are similarly orientated, which precludes the same pattern of preferential movement as occurs between the lunate and scaphoid. The interosseous membranes of the scapholunate and lunotriquetral ligaments separate the midcarpal from the radiocarpal joint spaces. Dye injected into one of these joint spaces that leaks to the other denotes a tear of one of these ligaments.

Distal row interosseous ligaments The strong distal row interosseous ligaments connect and stabilize the capitate, hamate, trapezium and trapezoid. They have superficial and deep components, and, unlike the ligaments of the proximal row, are seldom torn.

Palmar midcarpal ligaments The fan-shaped palmar scaphocapitate–trapezoid ligament arises from the scaphoid. It has two parts, the scaphotrapeziotrapezoidal ligament and the more ulnar scaphocapitate ligament. The triquetrohamate and triquetrocipitate ligaments lie towards the ulnar side of the carpus.

Dorsal midcarpal ligaments The dorsal intercarpal ligament assists in stabilization of the proximal carpal row. It arises from the trapezoid and distal pole of scaphoid, and passes across the dorsal horn of the lunate to be attached to the triquetrum. The ligament forms the floor of the fourth and fifth extensor compartments. The lateral scaphotrapeziotrapezoidal ligament is on the radial side.

Arthroscopy of the wrist and hand

Arthroscopy of the wrist joint may be carried out through up to 11 described portals, although only a few of these are favoured; most are dorsally situated. Some are, as de Smet (2002) has described, 'frighteningly' close to key anatomical structures. Consequently, only the skin should be incised when establishing a portal, and the remaining dissection should be carried out with blunt artery forceps and using blunt probes presaged by the introduction of a 22 gauge needle.

MOVEMENTS AND LOADING OF THE WRIST JOINT

Movements of the wrist (carpal kinematics)

The movements at the radiocarpal and intercarpal joints are considered together. The approximate ranges of active movement are flexion (85°), extension (85°), adduction (ulnar deviation) (45°), abduction (radial deviation) (15°) and circumduction (all measurements are approximate).

The range of flexion is greater at the radiocarpal joint, while in extension there is more movement at the midcarpal joint (Fig. 50.19). Hence, the proximal surfaces extend further posteriorly on the lunate and scaphoid bones. These movements are limited chiefly by antagonistic muscles; therefore, the range of flexion is perceptibly diminished when the fingers are flexed, due to increased tension in the extensors. Only when the joints are forced to the limits of flexion or extension are the dorsal or palmar ligaments fully stretched.

During extension of the wrist, the distal carpal bones rotate into extension with slight radial deviation, while during the 'dart-throwing' movement (from radial deviation and extension into ulnar deviation and flexion), movement occurs almost entirely at the distal row of the carpus.

Most adduction occurs at the radiocarpal joint. The lunate articulates with both the radius and articular disc when the hand is in the midposition but, in adduction, it articulates solely with the radius, and the triquetrum now comes into contact with the articular disc (Fig. 50.20A). Much of the proximal articular surface of the scaphoid becomes subcapsular beneath the radial collateral ligament and forms a smooth, convex, palpable prominence in the floor of the 'anatomical snuff-box'.

Abduction from the neutral position occurs at the midcarpal joint. Radiographs of abducted hands show that the capitate rotates round an anteroposterior axis so that its head passes medially and the hamate conforms to this; the distance between the lunate and the apex of the hamate is increased (Fig. 50.20B). The scaphoid rotates around a transverse axis, and its proximal articular surface moves away from the capsule to articulate solely with the radius. Movements are limited by antagonistic muscles and, at extremes, by the carpal collateral ligaments.

The proximal row (scaphoid, lunate and triquetrum) is an intercalated segment; no tendons insert on to the bones of the row. It is inherently unstable and controlled by specific retaining and gliding ligaments. Its relative position is determined by the spatial configurations of the radius, triangular fibrocartilage complex and ulna on one side, and the rigid distal carpal row on the other. The proximal carpal row is subject to two opposing moments: the scaphoid straddles the proximal and distal rows and tends to rotate the proximal row into flexion under axial load and radial deviation. At the same time, there is a force tending to extend the proximal row, which is initiated by the distal row and transmitted via the midcarpal ligaments to the triquetrum (Fig. 50.21). Stability of the midcarpal joint is thus ensured during both movement and loading.

The distal carpal row (trapezium, trapezoid, capitate and hamate) can be regarded as one rigid structure. The scaphoid bridges the proximal and distal carpal rows and provides a functional couple between the two.

Muscles producing movements

Flexion Flexors carpi radialis and ulnaris and palmaris longus produce flexion, assisted by flexors digitorum superficialis and profundus, and flexor pollicis longus.

Extension Extensors carpi radialis longus, brevis and ulnaris, assisted by extensors digitorum, digiti minimi, indicis and pollicis longus, produce extension.

Adduction (ulnar deviation) Flexor and extensor carpi ulnaris produce adduction.

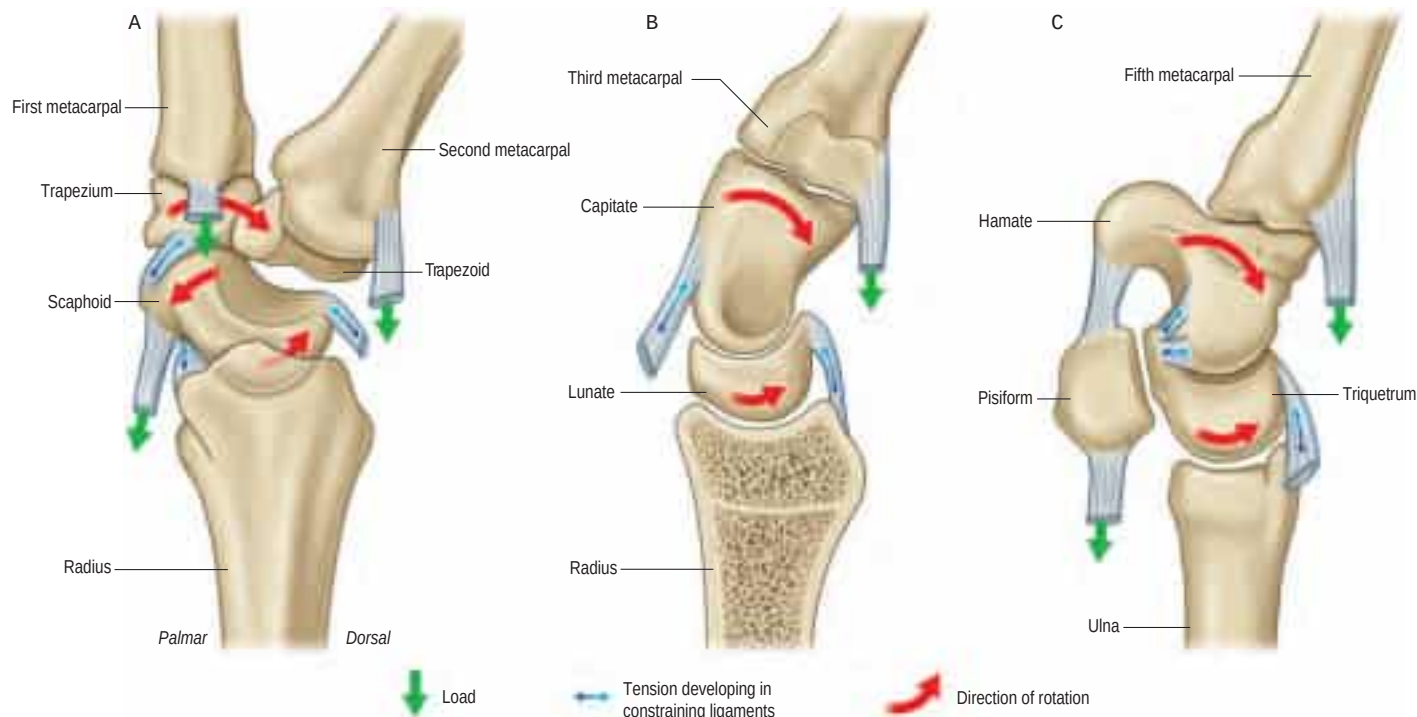


Fig. 50.21 The effect of axial loading on the rotatory movements of the bones of the wrist. **A**, The scaphoid flexes but is constrained on its palmar aspect by the scaphotrapeziotrapezoidal ligaments and dorsally by the scapholunate ligament. **B**, The lunate follows the scaphoid into flexion and is restrained by the dorsal radiolunate ligament. **C**, The triquetrum flexes with the scaphoid and lunate; it is constrained on its palmar aspect by the triquetrohamate and triquetrocapitate ligaments, and dorsally by the radiotriquetral ligament.

There are five possible dorsal portals (1–2, 3–4, 4–5, 6–R, 6–U) for viewing the radiocarpal joint and two (MCR, MCU) for the midcarpal joint. These are named by their relation to the six extensor compartments. The 1–2 portal is usually avoided because the radial artery lies a mean 3 mm (range 1–5 mm) radially, as do branches of the superficial radial nerve (range 1–6 mm) (Abrams et al 1994). The 3–4 portal is normally used as the main viewing portal and the 4–5 portal for instrumentation. The dorsal cutaneous branch of the ulnar nerve lies within 2.5 mm of the 6–R portal. Inflow is via the arthroscope (in the 3–4 portal) and outflow through a separate 6–U portal. The articular surfaces of the scaphoid and lunate are seen distally, linked by the scapholunate ligament. Proximally, the triangular fibrocartilage complex may be viewed. Tears of its central portion are commonly seen in older patients; congenital perforations are occasionally found. Bucket handle tears of the triangular fibrocartilage complex may be excised, or repaired if they occur in the peripheral vascularized portion. The radial midcarpal portal is used to view the scaphocapitate joint. The ulnar midcarpal portal is used for instrumentation and for viewing the ulnar portion of the midcarpal joint.

Volar portals (radial, VR, midcarpal, VMC and ulnar, VU) are less favoured but retain certain advantages, including the ability to view the dorsal capsule, the dorsal radiocarpal ligament and the palmar sub-regions of the scapholunate and lunotriquetral interosseous ligaments. Use of the volar radial portal carries with it the potential for damage to flexor carpi radialis, the palmar cutaneous branch of the median nerve and the lateral cutaneous nerve of the forearm, all of which can run close to, but not within 3 mm of, the portal. The volar ulnar portal lies more than 5 mm away from the ulnar neurovascular bundle, provided the entry point in the capsule is deep to the ulnar edge of the profundus tendons (Slutsky 2002).

The carpus was originally thought to move simply as proximal and distal rows (row or rigid body theory) (Johnston 1907). According to this view, during the composite movement of wrist flexion and extension, approximately two-thirds of movement occurs at the radiocarpal joint and one-third at the midcarpal joint. The carpus was later judged to move in lateral central and medial columns more than it did in rows, and the radius–lunate–capitate was described as a three-bar linkage system (column theory) (Navarro 1921). This theory was modified (Taleisnik 1976) to incorporate the specific stabilizing role of the scaphoid as it bridges the proximal and distal rows. A further theory proposed that the bones were linked by their ligaments in a ring configuration, so that any breakage of the key links leads to instability (ring theory). Most recently, the ‘four-unit’ theory suggests that the distal carpal row moves as a single unit, and the scaphoid, lunate and triquetrum move in complex but characteristic relationships that are dependent on the given movement. Clinical observation provides some support for each of these theories.



Fig. 50.19 **A**, A radiograph of the hand and wrist in full flexion: lateral aspect. Compare with **B**, and note the relative positions of the capitate and lunate, and the lunate and radius. **B**, A radiograph of the hand and wrist: lateral aspect. The long axes of the third metacarpal, capitate and lunate are, approximately, in line with the long axis of the radius. Note the relative positions of the capitate and lunate, and the lunate and radius. **C**, A radiograph of the hand and wrist in full extension: lateral aspect. Compare with **B** and note the alterations in the relative positions of the capitate and lunate, and the lunate and the radius. Key: 1, capitate; 2, lunate; 3, tubercle of the trapezium; 4, tubercle of the scaphoid.

Abduction (radial deviation) Flexor carpi radialis, extensors carpi radialis longus and brevis, with abductor pollicis longus and extensor pollicis brevis, produce abduction.

Flexor carpi ulnaris is stronger than extensor carpi ulnaris, which, in turn, is stronger than extensors carpi radialis longus and brevis, and flexor carpi radialis. The other muscles are all weaker. It is for this reason that these muscles tend to be selected for the major tendon transfers around the wrist (Brand 1985). During flexion and extension, flexor carpi radialis, flexor carpi ulnaris and extensor carpi radialis brevis have a greater excursion than extensor carpi radialis longus and extensor



Fig. 50.20 **A**, A radiograph of the hand in full adduction (ulnar deviation): dorsopalmar projection. The arrows point to the scaphoid on the radial side and to the pisiform on the ulnar side. Compare with **B** and observe that the movements occur at both the radiocarpal and intercarpal joints. **B**, A radiograph of the same hand in full abduction (radial deviation). The arrows point to the hamate and pisiform. Compare with **A** and note that: the scaphoid and lunate have passed medially (ulnarly) so that the latter articulates to a large extent with the articular disc of the distal radio-ulnar joint; the pisiform is now widely separated from the styloid process of the ulna; the scaphoid, having rotated round a transverse axis, is much foreshortened; the apex of the hamate has been thrust away from the lunate by the rotation of the capitate around an anteroposterior axis; a gap has opened up between the distal portions of the hamate and triquetrum; and the long axes of the capitate and lunate are now almost in the same straight line.

carpi ulnaris. During radial and ulnar deviation, extensor carpi radialis longus and extensor carpi ulnaris have a greater excursion, whereas that of flexor carpi radialis is relatively small. The axes of rotation for flexion/extension and radial/ulnar deviation intersect at a point in the proximal part of the capitate bone. Flexor carpi radialis and flexor carpi ulnaris are most efficient in flexion, extensor carpi radialis brevis in extension, extensor carpi radialis longus in radial deviation and extensor carpi ulnaris in ulnar deviation (Hori et al 1993).

Loading of the wrist (carpal kinetics)

Axial loading, i.e. load or force applied along a line parallel to the long bones of the arm, occurs in clenching the fist. In most activities, loading is multiplanar with a combination of vectors of force. Grasping an object whilst lifting it against gravity with the elbow in flexion causes transverse and axial loading.

Force transmission Radiocarpal joint

The radiocarpal and ulnocarpal articulations can be described in terms of specific fossae, namely: the scaphoid and lunate fossae of the radius,

and the facet formed by the distal aspect of the triangular fibrocartilage complex. The scaphoid fossa contributes 43%, the lunate fossa 46%, and the triangular fibrocartilage complex 11% of the total area of this articulating surface. Under physiological conditions, the contact area between this surface and the proximal carpal row is about 20%. Greater contact is seen in forearm supination, radial deviation and dorsiflexion of the wrist. Lesser contact occurs in forearm pronation, ulnar deviation and palmar flexion of the wrist. Axial loading increases the contact area to a maximum of about 40%. Force across the joint also varies with the position of the forearm and the degree of wrist flexion. For a given load in wrist-neutral position, 50% of force passes across the scaphoid fossa, 35% through the lunate fossa and 15% across the triangular fibrocartilage complex. With a power grip, there is slight ulnar deviation and the proportion of force passing across the scaphoid and lunate fossae reverses.

Midcarpal joint

With the wrist in neutral position, 50–60% of a given load is transmitted from the distal row through the capitate to the scaphoid and lunate. Up to 30% of the load is transmitted via the scaphotrapeziotrapezoid joint and up to 20% via the hamate–triquetral joint.

Distal radio-ulnar joint (DRUJ)

See page 845.

Specific actions

 See Video 50.1.

Power grip

In power grip, the wrist is positioned in mid-dorsiflexion and the object is tightly gripped. Force is transmitted from the hand and wrist via the radiocarpal joint and triangular fibrocartilage complex to both bones of the forearm, through the elbow joint to the humerus, and from there, via the glenohumeral joint, to the pectoral girdle. Dynamic muscle action accounts for transmission of a large part of the force; a smaller proportion is transmitted through the osseoligamentous structures. The distribution of force between the radius and ulna varies dynamically and depends on the absolute value of load, degree of forearm rotation and position of the carpus relative to the forearm.

Hammer action

The action exemplified by hammering, where the wrist alternates between the dorsoradial and the palmar–ulnar position, is an important movement for a wide variety of activities. It is believed that load distributes dynamically as the position of the carpus moves relative to the forearm and there are consequent changes in the relative position of the carpal bones.

Carpal instability

The upper limb has evolved to facilitate the placing of the hand in space for prehension. However, mobility of the hand on the forearm has evolved at the expense of stability. A wrist is said to be kinetically unstable when it cannot bear physiological loads without giving way or causing injury. A wrist is kinematically unstable when it exhibits sudden changes in carpal alignment, i.e. during a specific movement, there is a 'clunk' as one or more of the carpal bones moves abnormally with respect to the others. Stability of the various joints, so that the bones maintain normal anatomical relations with respect to each other throughout the normal range of motion, is essential for physiological load-bearing.

CARPOMETACARPAL JOINTS

The first carpometacarpal joint (CMCJ), or basal joint of the thumb, is a saddle joint that permits opposition of the thumb. The second to fifth carpometacarpal joints exhibit an increasing range of movement, progressing from the radial to the ulnar side. Thus, there is little mobility at the base of the index ray but considerable mobility at the base of the small finger ray, which facilitates 'cupping' of the palm of the hand. Compressive forces between the metacarpals and the distal carpal row are estimated to be upwards of ten times the forces at the tips of the fingers during 'pinch' or 'chuck' grip.

Carpometacarpal joint of the thumb

Articulating surfaces The carpometacarpal joint of the thumb is a sellar (saddle) joint between the first metacarpal base and trapezium. Its extensive articular surfaces and their topology ensure that the joint enjoys a wide mobility.

Ligaments The first metacarpal and trapezium are connected by lateral, anterior and posterior ligaments and a fibrous capsule. The broad lateral ligament runs from the lateral surface of the trapezium to the radial side of the metacarpal base. The palmar and dorsal ligaments are oblique bands that converge to the ulnar side of the metacarpal base from the palmar and dorsal surfaces of the trapezium, respectively.

Synovial membrane The synovial membrane lines the joint capsule and is separate from it.

Joint movements Except at initiation, flexion is accompanied by medial rotation, and medial rotation involves flexion. Linkage of movements is due largely to the shape of the articular surfaces and to the obliquity of the dorsal ligament (which, when taut, anchors the ulnar side of the metacarpal base while its radial side continues to move). Contraction of flexor pollicis brevis, assisted by opponens pollicis, produces medial rotation with flexion; combined with abduction, this opposes the thumb pulp to the pulps of the slightly flexed fingers. Full extension of the thumb metacarpal entails slight lateral rotation, attributable to the saddle shape of the joint and to the action of the palmar ligament (which is similar to that of the dorsal ligament in flexion). (For further reading, see [Edmunds \(2011\)](#), [Ladd et al \(2013\)](#).)

Muscles producing movements The muscles producing movements at the carpometacarpal joint of the thumb are as follows.

Flexion Flexion is produced by flexor pollicis brevis and opponens pollicis, aided by flexor pollicis longus when the other joints of the thumb are flexed.

Extension Extension is produced by abductor pollicis longus and extensors pollicis brevis and longus.

Abduction Abduction is produced by abductors pollicis brevis and longus.

Adduction Adduction is produced by adductor pollicis.

Opposition Opponens pollicis and flexor brevis pollicis simultaneously flex and medially rotate the abducted thumb. Interpulpal pressure, or that generated by digital grasping, is increased by adductor pollicis and flexor pollicis longus.

Circumduction Circumduction is produced by extensors, abductors, flexors and adductors acting consecutively in this, or reverse, order.

Second to fifth carpometacarpal joints

The second to fifth carpometacarpal joints are synovial ellipsoid joints between the carpus and second to fifth metacarpals. Although widely classed as plane, they have curved articular surfaces that are often of complex saddle shape. The bones are united by articular capsules and by dorsal, palmar and interosseous ligaments.

Ligaments The dorsal ligaments are the strongest, and connect the dorsal surfaces of the carpal and metacarpal bones. The second metacarpal has two ligaments, from the trapezium and trapezoid; the third has two, from the trapezoid and capitate; the fourth has two, from the capitate and hamate; and the fifth has a single band from the hamate, which is continuous with a similar palmar ligament, forming an incomplete capsule.

The palmar ligaments are similar, except that the third metacarpal has three: a lateral from the trapezium, superficial to the tendon sheath of flexor carpi radialis, an intermediate from the capitate, and a medial from the hamate.

The interosseous ligaments consist of two short, thick, fibrous bands. They are limited to one part of the carpometacarpal articulation and connect contiguous distal margins of the capitate and hamate with adjacent surfaces of the third and fourth metacarpal bones; they may be united proximally.

Some degree of carpal instability may occur as part of inherent hypermobility in affected individuals. Usually, it occurs only after rupture or attenuation of intrinsic and extrinsic carpal ligaments and may be detected either clinically or radiologically as malalignment of the carpal bones (**Fig. 50.22**). In theory, any pattern of traumatic carpal instability is possible. In practice, certain patterns predominate.

When axially loaded, the distal row of the carpus compresses the proximal row, causing the scaphoid to rotate into flexion and pronation. Provided the scapholunate and lunotriquetral ligaments remain intact, the load is transmitted to the lunate and triquetrum. If the mid-carpal stabilizers are disrupted, the proximal row of the carpus will be loaded into flexion, producing the characteristic pattern of volar intercalated segment instability (VISI). In the proximal row, the scaphoid exhibits the greatest range of movement, the lunate less and the triquetrum the least. Consequently, the scapholunate and lunotriquetral ligaments tighten as the scaphoid is progressively flexed, thereby increasing the intrinsic stability of the proximal row. If, however, the scapholunate ligaments have been disrupted (scapholunate dissociation), the scaphoid can flex excessively and loading of the lunate and triquetrum by the distal row of the carpus will force them into abnormal extension (dorsal intercalated segment instability, or DISI).



Fig. 50.22 Scapholunate dissociation. **A**, Anteroposterior view. **B**, Lateral view. Dissociation between the scaphoid and lunate, as shown here, leads to collapse of the scaphoid into flexion and the lunate into extension (dorsal intercalated segment instability, or DISI), thus removing the inbuilt tension across the proximal row. This causes the gap between the scaphoid and lunate in **A**.

Synovial membranes The synovial membranes are often continuous with those of the intercarpal joints. Occasionally, the joint between the hamate and fourth and fifth metacarpal bones has a separate synovial cavity, bounded laterally by the medial interosseous ligament and its extensions to the palmar and dorsal parts of the capsule.

Muscles producing movements Slight gliding movements are effected by the long flexor and extensor muscles of the digits.

INTERMETACARPAL JOINTS

Articulating surfaces The second to fifth metacarpal bases articulate reciprocally by small cartilage-covered facets connected by dorsal, palmar and interosseous ligaments.

Fibrous capsule The intermetacarpal joints have fibrous capsules.

Ligaments The dorsal and palmar ligaments pass transversely from bone to bone. The interosseous ligaments connect contiguous surfaces just distal to their articular facets.

Synovial membranes The synovial membranes are continuous with those of the carpometacarpal articulations.

Movements at the carpometacarpal and intermetacarpal joints Movements at the carpometacarpal and intermetacarpal articulations are limited to slight gliding, sufficient to permit some flexion–extension and adjunct rotation; ranges vary in different joints. They are partly accessory movements occurring when the palm is ‘cupped’, as in grasping an object. The fifth metacarpal is most movable and the second and third are the least mobile. These variations are easily demonstrated by opposing each digit to the thumb over the palmar centre. About two-thirds of the movements are those of the thumb, as described above, but during opposition, the carpometacarpal and metacarpal joints of the little finger become flexed, abducted and laterally rotated, accounting for the remaining third of the movement. The close-packed position probably coincides with carpal extension. A further accessory movement is spiral twisting of the whole metacarpus on the carpus.

Muscles producing movements The flexors and extensors of the second to fifth digits move the carpometacarpal and intermetacarpal joints.

METACARPOPHALANGEAL JOINTS

The metacarpal heads are adapted to shallow concavities on the phalangeal bases; they are not regularly convex but are partially divided on their palmar aspects and, thus, almost bicondylar.

Fibrous capsule The metacarpophalangeal joints all have fibrous capsules.

Ligaments Each metacarpophalangeal joint has a palmar and two collateral ligaments.

Palmar ligaments The palmar ligaments (volar plates) are unusual. They are thick, dense and fibrocartilaginous, and are sited between, and connected to, the collateral ligaments. They are attached loosely to the metacarpals but firmly to the phalangeal bases. Their palmar aspects are blended with the deep transverse palmar ligaments and are grooved for the flexor tendons, whose fibrous sheaths connect with the sides of the grooves. Their deep surfaces increase articular areas for the metacarpal heads.

Deep transverse metacarpal ligaments The deep transverse metacarpal ligaments are three short, wide, flat bands that connect the palmar ligaments of the second to fifth metacarpophalangeal joints. They are related anteriorly to the lumbricals and digital vessels and nerves, and posteriorly to the interossei. Bands from the digital slips of the central palmar aponeurosis join their palmar surfaces. On both sides of the third and fourth metacarpophalangeal joints, but only the ulnar side of the second and radial side of the fifth joints, transverse bands of the dorsal digital expansions join the deep transverse metacarpal ligaments. The lumbricals and the phalangeal attachments of the dorsal interossei lie anterior to this band; the remaining attachments of the dorsal and palmar interossei are posterior to it (see Fig. 50.29).

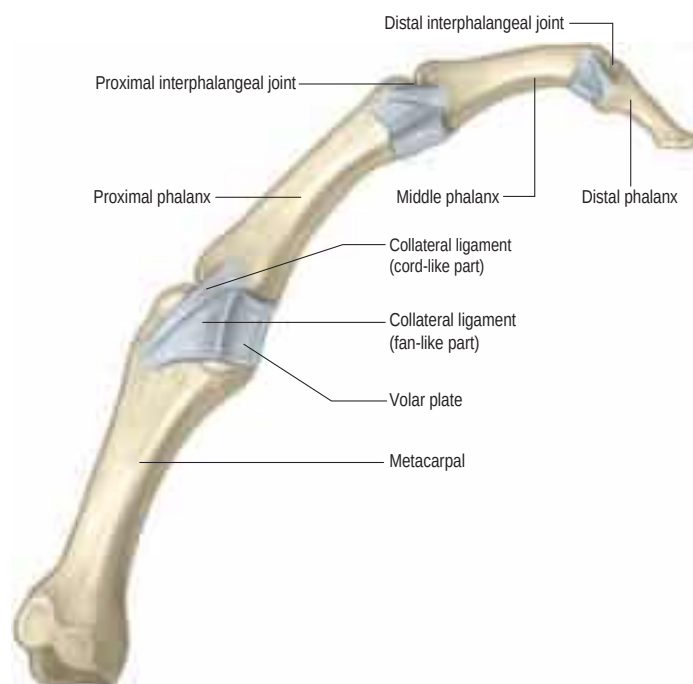


Fig. 50.23 The metacarpophalangeal and digital joints of the left third finger: medial aspect.

Collateral ligaments The collateral ligaments are strong, round cords that flank the joints. Each is attached to the posterior tubercle and adjacent pit on the side of its metacarpal head, and each passes disto-anteriorly to the side of the anterior aspect of its phalangeal base (Fig. 50.23).

Synovial membrane The metacarpophalangeal joints are lined by a synovial membrane.

Joint movements These include flexion, extension, adduction, abduction, circumduction and limited rotation. Rotation accompanies flexion–extension, e.g. when the finger flexes and rotates to place its tip near the centre of the palm. The range of rotation is frequently increased as a result of the resistance of a grasped object. Active flexion is 90° or more, whereas extension ranges from 10° (index) to 30° (little); both movements are limited mostly by antagonistic muscles. The metacarpophalangeal joint of the thumb has a flexion–extension range of approximately 60°, which is almost entirely flexion. Slight lateral rotation accompanies digital flexion of digits 3–5. Flexion of the index finger may be accompanied by minimal lateral rotation or no rotation; a small degree of medial rotation is frequently observed. Other movements are adduction–abduction (maximal range 25°), which invariably accompanies the corresponding carpometacarpal movements and increases their combined range; and slight rotation, which accompanies flexion–extension. Of the second to fifth metacarpophalangeal joints, the second is most mobile in adduction–abduction (approximately 30°), followed by the fifth, fourth and third.

Accessory movements Accessory movements are further rotation (most marked in the thumb), anteroposterior and lateral translation of a phalanx or metacarpal, and distraction.

Muscles producing movements The muscles producing movements at the metacarpophalangeal joints are as follows.

Flexion Flexion is produced by flexors digitorum superficialis and profundus, assisted by the lumbricals, interossei and flexor digiti minimi brevis (in the little finger). In the thumb, flexors pollicis longus and brevis, and the first palmar interosseous, are involved.

Extension Extension is produced by extensor digitorum, assisted in the second and fifth digits by extensor indicis and extensor digiti minimi, respectively. In the thumb, extensors pollicis longus and brevis are involved.

Adduction In extended fingers, adduction is produced by the palmar interossei; the long flexors are predominant during flexion. In the

thumb, limited metacarpophalangeal adduction is possible and may be attributable to adductor pollicis and the first palmar interosseous.

Abduction In extended fingers, abduction is produced by the dorsal interossei, assisted by the long extensors (except in the middle finger), and abductor digiti minimi in the minimus. In the thumb, abductor pollicis brevis (which also contributes to opposition) is involved. When the fingers are flexed at the interphalangeal joints, active abduction is restricted (it is virtually zero when metacarpophalangeal and interphalangeal joints are flexed); if the long digital flexors are inactive, passive abduction is free. Inability to abduct actively in this position may be due to shortening of the dorsal interossei and abductor digiti minimi by flexion, but the altered line of pull of the interossei relative to the axis of movement is probably the determining factor.

Gamekeeper's thumb



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INTERPHALANGEAL JOINTS

The interphalangeal joints are uniaxial hinge joints (see Fig. 50.23).

Fibrous capsule Each interphalangeal joint has a fibrous capsule.

Ligaments Each interphalangeal joint has a palmar ligament (the volar plate) and two collateral ligaments (Fig. 50.24). The long extensor tendons take the place of the dorsal capsular ligaments. Extensions from the extensor expansion, each collateral ligament and the palmar ligament all pass into the joint cavity and provide a significant increase to the articular surface area of the phalangeal base; their deformable nature improves joint congruence.

Palmar ligament (volar plate) The volar plate constitutes the floor of the interphalangeal joint. In the proximal interphalangeal joint, the distal end of the volar plate is thickened laterally where it is firmly attached to the base of the middle phalanx at the position of the true collateral ligament attachment. Centrally, it is more delicate and blends with the volar periosteum of the middle phalanx. Proximally, the volar plate is also very delicate in its central portion, but it is thickened laterally to form the 'check rein ligaments', which attach to the periosteum of the proximal phalanx just within the walls of the A2 pulley component of the fibrous flexor sheath. Nutrient branches of the digital

arteries pass underneath these check ligaments to reach the vinculae (see Fig. 50.24).

Collateral ligaments The collateral ligaments pass from the lateral aspect of the head of one phalanx to the volar aspect of the base of the adjacent phalanx. An accessory component to the collateral ligament arises in continuity with the main ligament and passes in a volar direction to attach to the volar plate.

Synovial membrane Each interphalangeal joint has a synovial lining.

Factors maintaining stability Stability is conferred by the articular contours of the joint surfaces and the collateral ligaments. The flexor and extensor tendons and retinacular ligaments provide secondary stabilization. Stability against hyperextension of the proximal interphalangeal joint is enhanced by the three-dimensional box arrangement that is produced by the collateral ligament–volar plate complex.

Joint movements Active movements at the interphalangeal joints are flexion and extension, and are greater in range at the proximal joints. Flexion is considerable, whereas extension is limited by tension of the digital flexors and terminated by tension in the palmar ligaments and non-articular compression. Full extension is the close-packed position. Flexion and extension are accompanied by slight rotation; during flexion, this turns the digital pulps slightly laterally, i.e. to face the opposed thumb, and an opposite rotation occurs during extension.

Accessory movements Accessory movements are limited rotation, abduction, adduction and anteroposterior translation. They permit the fingers to adapt to the shapes of gripped objects and help to protect against stresses and strains.

Muscles producing movements The muscles producing movements at the interphalangeal joints are as follows.

Flexion Flexion is produced at the proximal interphalangeal joints by the flexors digitorum superficialis and profundus; at distal interphalangeal joints, by flexor digitorum profundus; and at the thumb interphalangeal joint, by flexor pollicis longus.

Extension Extension is produced by extensors digitorum, digiti minimi and pollicis longus, in association with abductor pollicis longus and extensor pollicis brevis. Extension occurs simultaneously in both joints in digits 2–5.

Simultaneous flexion at the metacarpophalangeal joints and extension at the interphalangeal joints of a digit are essential for the fine movements of writing, drawing, threading a needle, etc. The lumbricals and interossei are primary agents in flexing the metacarpophalangeal joints and also in extending the interphalangeal joints via their attachments to the dorsal digital expansions.

Mallet finger



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Swan neck deformity



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Boutonnière deformity



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Fig. 50.24 Anatomy of the volar plate at a proximal interphalangeal joint. The blood vessel supplying the flexor tendons via the vincula brevum is shown; the mesentery forming the vinculum has been omitted.

VASCULAR SUPPLY OF THE JOINTS OF THE WRIST AND HAND

The radiocarpal joint is supplied by branches of the anterior interosseous artery, anterior and posterior carpal branches of the radial and ulnar arteries, palmar and dorsal metacarpal arteries, and recurrent rami of the deep palmar arch. The carpal joints are supplied by the posterior carpal branches of the radial and ulnar arteries, and by the anterior interosseous artery. The carpometacarpal joint of the thumb receives its blood supply from the radial artery and its first dorsal metacarpal branch. The second to fifth carpometacarpal joints are supplied by the posterior carpal branches of the radial and ulnar arteries, by twigs from

The ulnar collateral ligament of the metacarpophalangeal joint of the thumb may be avulsed from its distal attachment to the base of the proximal phalanx by a combination of forced extension and radial deviation of the joint. The injury was reported to occur in gamekeepers while breaking the necks of game, but can occur following simple falls on to the outstretched hand, especially in skiers who have the straps of ski poles wrapped around their thumbs.

The ulnar collateral ligament is normally covered by the aponeurotic insertion of the tendon of adductor pollicis. Following avulsion, the proximal end can become superficially displaced around the proximal free edge of the aponeurotic insertion of the adductor. This soft tissue interposition prevents the ends of the avulsed ligament from healing. If left untreated, a firm, palpable, inflammatory swelling occurs. Surgical repair requires division of the aponeurotic insertion of adductor pollicis, followed by bony reattachment of the avulsed ligament and repair of the aponeurosis.

The comparable injury to the radial collateral ligament is less common and constitutes only 10–42% of collateral ligament injuries (Edelstein et al 2008). It is caused by sudden, forced adduction of the metacarpophalangeal joint as the result of a fall or during ball games. The ligament may be ruptured in its mid-substance or avulsed from either bony attachment. Early identification is essential because the proximal phalanx of the thumb will subluxate ulnarwards and may rotate, leading to degeneration of the articular surface. Interposition of the torn or avulsed portion is uncommon because the adductor aponeurosis lies over the radial collateral ligament.

Mallet finger is a traumatic rupture, avulsion fracture or laceration of the terminal slip of the extensor tendon to the terminal phalanx of the finger that can result from forced flexion of the distal phalanx of the fully extended finger; it is a well-known injury in cricketers. The patient is unable to extend the terminal phalanx actively, although the distal interphalangeal joint can be extended passively.

Swan neck deformity involves hyperextension of the proximal interphalangeal joint, together with distal interphalangeal joint flexion, caused by relative overactivity of the extensors acting at the proximal interphalangeal joint compared with the flexors. It arises as a result of intrinsic muscle spasm, long-standing mallet finger or dysfunction of flexor digitorum superficialis; it can only occur if there is also some laxity of the volar plate.

Boutonnière deformity is a flexion deformity of the proximal interphalangeal joint following either division or laxity in the central slip of the extensor tendon (which normally inserts into the base of the middle phalanx). An essential feature is hyperextension at the distal interphalangeal joint. The deformity usually occurs as a result of either trauma or rheumatoid arthritis, causing the lateral bands of the extensor tendon to migrate in a volar direction and the head of the proximal phalanx to migrate dorsally. Initially, the deformity is passively correctable, but, with time, the soft tissues around the joint contract and a fixed deformity results.

the anterior interosseous artery, and from the palmar digital arteries. The metacarpophalangeal joints receive their blood supply from the dorsal and palmar metacarpal arteries, the *arteria princeps pollicis*, and the *arteria radialis indicis*. The interphalangeal joints are supplied by branches from the palmar digital arteries.

INNERVATION OF THE JOINTS OF THE WRIST AND HAND

The radiocarpal joint is innervated by the anterior and posterior interosseous nerves, with contributions from the median, ulnar and radial nerves. The anterior interosseous nerve and posterior interosseous nerve innervate the central two-thirds of the anterior and posterior wrist joint capsule, including the radiocarpal joint. These appear to be pure afferent nerve fibres with mechanoreceptor endings, of which there are an average of 10 in each dorsal radiocarpal ligament (Tomita et al 2007). In the palmar radiocarpal ligaments, however, Golgi organs, Pacinian corpuscles, Ruffini endings and free nerve endings are found (Petrie et al 1997). An uneven distribution of these throughout the ligaments would suggest specific functional adaptation but the precise nature of this is unclear. The carpometacarpal joint of the thumb is innervated by articular twigs from the posterior interosseous nerve and the superficial branch of the radial nerve. The second to fifth carpometacarpal joints are innervated by branches of the ulnar nerve, the anterior interosseous, and the superficial radial and dorsal ulnar nerves. The metacarpophalangeal joints are innervated by twigs from the palmar digital branches of the median and ulnar nerve, the deep terminal branch of the ulnar nerve, and the posterior interosseous nerve. The interphalangeal joints are innervated by the palmar digital branches of the median nerve (to the thumb, index, middle and ring fingers), and the ulnar nerve (to the ring and little fingers).

MUSCLES

EXTRINSIC LONG FLEXORS AND EXTENSORS

The extrinsic long flexors and extensors are described on pages 848–855. The flexor and extensor retinacula are described on pages 863, 864.

Long flexor tendon apparatus Flexor tendon sheaths

The fibrous sheaths of the flexor tendons are specialized parts of the palmar fascia. Each finger has an osseo-aponeurotic tunnel that extends from midpalm to the distal phalanx. The thumb has a tunnel for flexor pollicis longus that extends from the metacarpal to the distal phalanx. The proximal border is, to some extent, a matter of definition because the transverse fibres of the palmar aponeurosis may be considered to be a part of the pulley system. The sheath consists of arcuate fibres that arch anteriorly over bone, tendons (where the sheath is required to be stiff) and the centres of joints (where a bucket-handle of arcuate fibres is a mechanically favourable arrangement). In contrast, where the sheath is required to fold to permit joint flexion, it consists of cruciate fibres. These fibrous sheaths are lined by a thin synovial membrane that provides a sealed lubrication system containing synovial fluid. The synovial membrane extends from the distal phalanx to midpalm in the case of the index, middle and ring fingers, and further proximally in the case of the little finger (Fig. 50.25). The sheaths around the thumb and little finger are continuous with the flexor sheaths in front of the wrist. The parietal synovial membrane is reflected on to the surface of the flexor tendon, forming a visceral synovium.

A standard nomenclature for the anular (A) and cruciform (C) pulleys is used (Doyle and Blythe 1975). The usual pattern is as follows (see Fig. 50.25). The A1 pulley is situated anterior to the palmar cartilaginous plate of the metacarpophalangeal joint and may extend over the proximal part of the proximal phalanx. The A2 overlies the middle third of the proximal phalanx. It is the strongest pulley and arises from well-defined longitudinal ridges on the palmar aspect of the phalanx. Its distal edge is well developed. A pouch or recess of synovium extends superficial to the free edge of the pulley fibres so that the free edge forms a lip protruding into the synovial space. A3 is a narrow pulley lying palmar to the proximal interphalangeal joint. A4 overlies the middle third of the middle phalanx, and A5 overlies the distal interphalangeal joint. The cruciate fibres are numbered in a slightly different manner. C0 is palmar to the metacarpophalangeal joint. There are two cruciate zones, C1 and C2, at the proximal interphalangeal joint, and they lie

just proximal and distal, respectively, to A3. At the distal interphalangeal joint, there is one pronounced cruciate system, C3, which lies between A4 and A5. Variations occur frequently. During flexion, the cruciate fibres become orientated more transversely in the digits, and the edges of adjacent anular pulleys approximate so that they form, in full flexion, a continuous tunnel of transversely orientated fibres. Surgically, the most important pulleys that prevent bowstringing of the flexor tendons are the A2 and A4 pulleys.

In the thumb, there are three constant pulleys: two anular and one oblique (Fig. 50.26). The A1 pulley is located at the metacarpophalangeal joint. The oblique pulley is located over the mid-portion of the proximal phalanx and its fibres pass from the ulnar aspect proximally to the radial aspect dorsally. The A2 pulley is thinner than the A1 pulley and is situated just proximal to the interphalangeal joint. The oblique pulley is the most important pulley in the thumb for maintaining the action of flexor pollicis longus.

Synovial sheaths of the carpal flexor tendons

Two synovial sheaths envelop the flexor tendons as they traverse the carpal tunnel: one for flexors digitorum superficialis and profundus, the other for flexor pollicis longus. These sheaths typically extend into the forearm for 2.5 cm proximal to the flexor retinaculum and, occasionally, communicate with each other deep to it. The sheath of the flexors digitorum tendons reaches about halfway along the metacarpal bones, where it ends in blind diverticula around the tendons to the index, middle and ring fingers (see Fig. 50.25). The sheath is prolonged around the tendons to the little finger and is usually continuous with their digital synovial sheath. A transverse section through the carpus shows that the tendons are invaginated into the sheath from the lateral side (Fig. 50.27). The parietal layer lines the flexor retinaculum and the floor of the carpal tunnel, and is reflected laterally as the visceral layer over the tendons of flexor digitorum superficialis ventrally and flexor digitorum profundus dorsally. Medially, a recess formed by the visceral layer of the sheath insinuates between the two groups of tendons and passes laterally for a variable distance. The sheath of flexor pollicis longus, which is usually separate, is continued along the thumb as far as the insertion of the tendon.

Vincula

The phenomenon of tendon gliding within a fibrous sheath requires a very specialized arrangement of the vascular supply. Folds of synovial membrane containing a loose plexus of fascial fibres carry blood vessels to the tendons at certain defined points. These folds, *vincula tendinum* (Fig. 50.28), are of two kinds. *Vincula brevia*, of which there are two in each finger, are attached to the deep surfaces of the tendons near to their insertions. There is, thus, one *vinculum brevium* attaching flexor digitorum profundus to the region of the distal interphalangeal joint, and a more proximal *vinculum* deep to flexor digitorum superficialis at the proximal interphalangeal joint. *Vincula longa* are filiform; usually two are attached to each superficial tendon, one to each deep tendon.

Trigger finger



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Synovial sheaths of the carpal extensor tendons

Six tunnels deep to the extensor retinaculum transmit the extensor tendons, each containing a synovial sheath. The tendons of abductor pollicis longus and extensor pollicis brevis lie in a tunnel on the lateral side of the styloid process of the radius; there may be a separate synovial sheath for each, or the tendon of the abductor may be double. The tendons of extensors carpi radiales longus and brevis lie behind the styloid process; the tendon of extensor pollicis longus lies on the medial side of the dorsal tubercle of the radius; the tendons of extensors digitorum and indicis lie in a tunnel on the medial side of the tubercle; the tendon of extensor digiti minimi lies opposite the interval between the radius and ulna; and the tendon of extensor carpi ulnaris lies between the head and the styloid process of the ulna. The tendon sheaths of abductor pollicis longus, extensors pollicis brevis and longus, extensors carpi radiales and extensor carpi ulnaris stop immediately proximal to the bases of the metacarpal bones, while those of extensors digitorum, indicis and digiti minimi are sometimes prolonged a little more distally along the metacarpus.

De Quervain's tenosynovitis



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Trigger finger is a stenosing tenovaginitis that affects the fibrous flexor sheaths of the fingers or thumb within the palm. The affected sheath thickens and entraps the contained tendons, which become constricted at the site of entrapment and bulge distal to it. This produces a distinct nodule in the palm of the hand and the finger now snaps as the tendon nodule passes through the constriction on flexing the finger. The corresponding extensor muscle is insufficiently powerful to extend the affected finger, so the patient does this passively, accompanied by a painful snap. Treatment frequently requires surgical division of the A1 pulley of the flexor sheath to relieve the stricture.

De Quervain's tenovaginitis is a stenosing tenovaginitis of unknown aetiology that occurs in the first dorsal compartment at the level of the radial styloid. It involves the common extensor sheath containing the tendons of abductor pollicis longus and extensor pollicis brevis. There is palpable thickening of the tendon sheath with painful limitation of extension of the thumb. Treatment frequently requires division of the thickened sheath, care being taken to avoid the adjacent superficial radial nerve; division of the sheath produces no functional impairment.

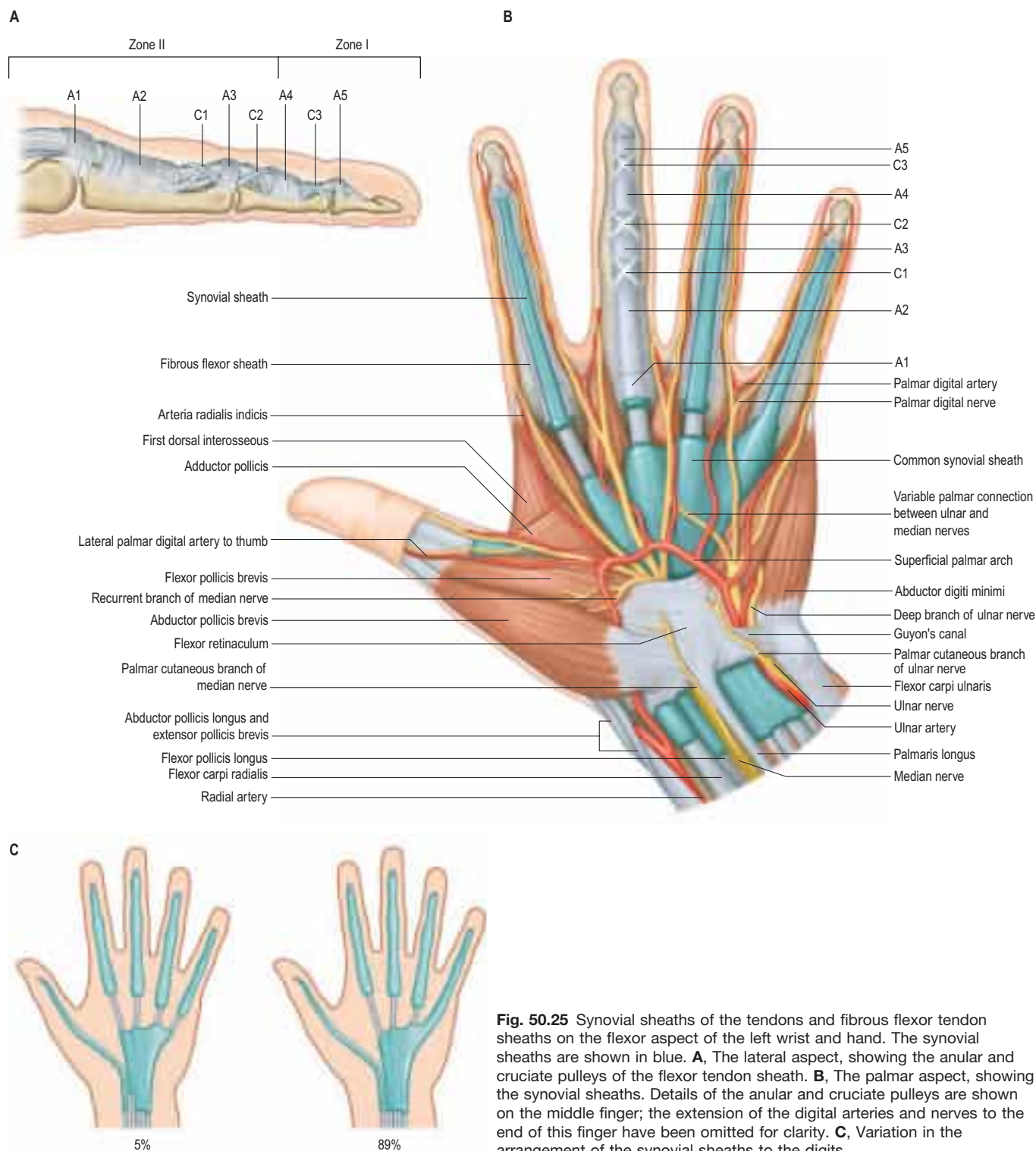


Fig. 50.25 Synovial sheaths of the tendons and fibrous flexor tendon sheaths on the flexor aspect of the left wrist and hand. The synovial sheaths are shown in blue. **A**, The lateral aspect, showing the anular and cruciate pulleys of the flexor tendon sheath. **B**, The palmar aspect, showing the synovial sheaths. Details of the anular and cruciate pulleys are shown on the middle finger; the extension of the digital arteries and nerves to the end of this finger have been omitted for clarity. **C**, Variation in the arrangement of the synovial sheaths to the digits.

Extensor tendon apparatus

The extensor digitorum tendons emerge through the fourth dorsal compartment on to the dorsum of the hand, where they are joined together distally by a varying pattern of oblique interconnections, the juncturae tendinae. These typically pass in a distal direction from middle finger to index finger, and from ring finger to middle and little fingers. Proximal lacerations to the middle finger extensor tendon may result in only partial loss of extension because of these tendinous interconnections.

At the level of the metacarpophalangeal joint, each extensor tendon is held in a central position over the dorsum of the joint by a flat, fibrous extensor expansion (Fig. 50.29). The expansion extends on to the dorsum of the proximal phalanx of each digit, forming a movable hood that moves distally when the metacarpophalangeal joint is flexed, and proximally when it is extended. Each extensor tendon blends with the extensor expansion along its central axis and is separated from the metacarpophalangeal joint by a small bursa. The expansion is triangular

in shape, with its base proximal. It receives the conjoined tendons of the interosseous and lumbrical muscles and is almost translucent between its margins and the tendon of extensor digitorum. Transverse fibres (the sagittal bands) pass to the volar plate and transverse metacarpal ligaments, separating the phalangeal attachment of the dorsal interosseous from the rest of the muscle, and the palmar interosseous from the lumbrical muscle. Injuries to the sagittal bands can lead to subluxation of the extensor tendon.

The margins of the extensor expansions are thickened on the radial side by the tendons of the lumbrical and interosseous muscles, and on the ulnar side either by the tendon of an interosseous alone or, in the case of the fifth digit, by abductor digiti minimi.

The tendons of the interossei join the extensor expansion at the level of the proximal portion of the proximal phalanx, while the tendons of the lumbricals join the extensor mechanism further distally at the mid-portion of the proximal phalanx. Their line of pull is anterior to the axis of rotation at the metacarpophalangeal joint, but dorsal to the axis

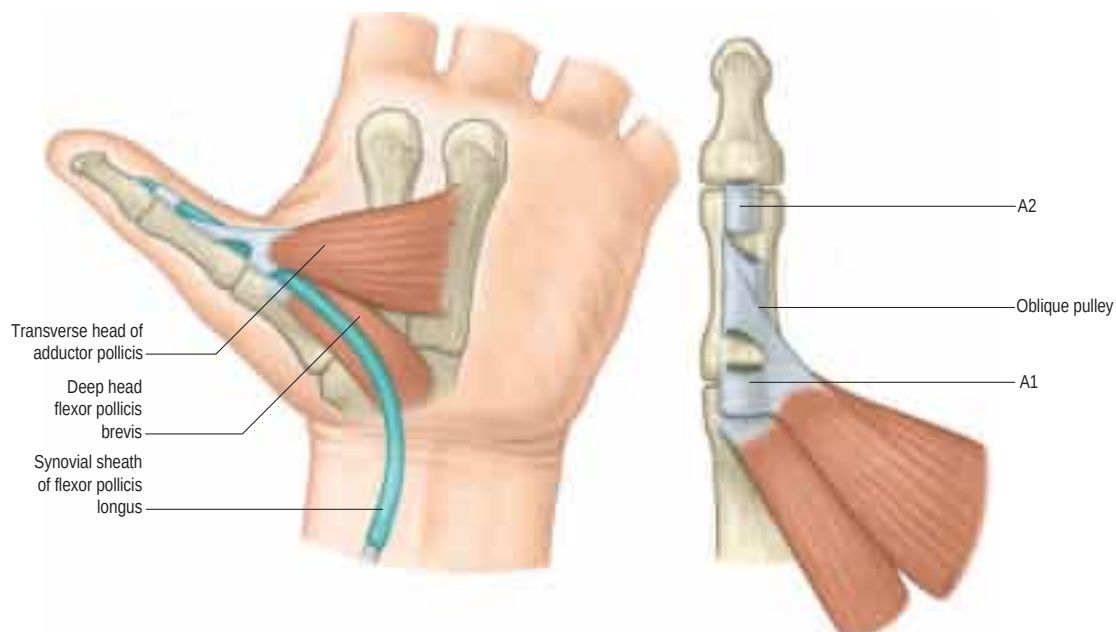


Fig. 50.26 The flexor sheath of the left thumb, showing the anular and oblique pulleys. (From Doyle JR, Blythe WF 1977 Anatomy of the flexor tendon sheath and pulleys of the thumb. J Hand Surg 2:149-51. With permission from the American Society for Surgery of the Hand.)

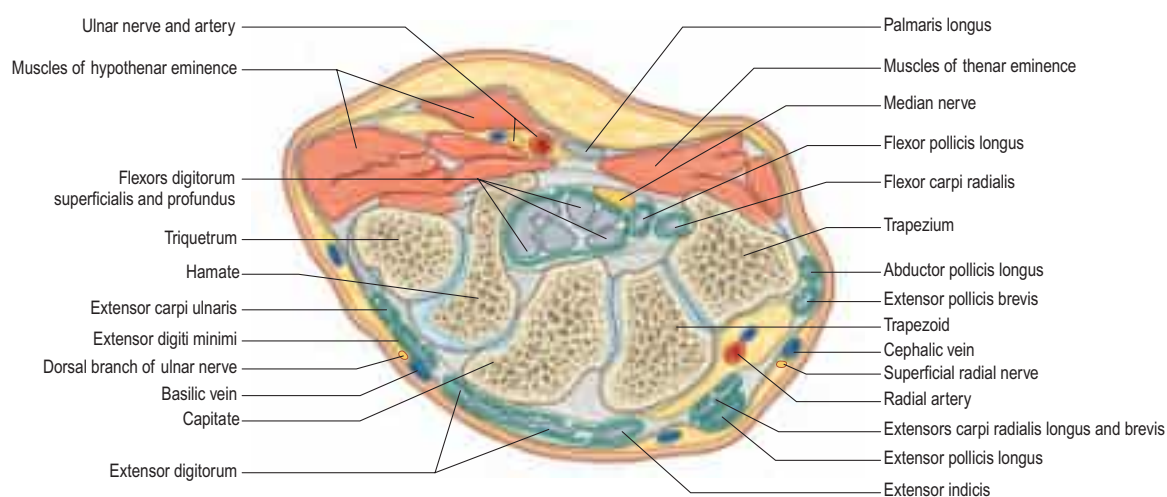


Fig. 50.27 A transverse section through the left wrist at the level of the distal row of the carpus, showing the muscle tendons (in grey) and their synovial sheaths (in green).

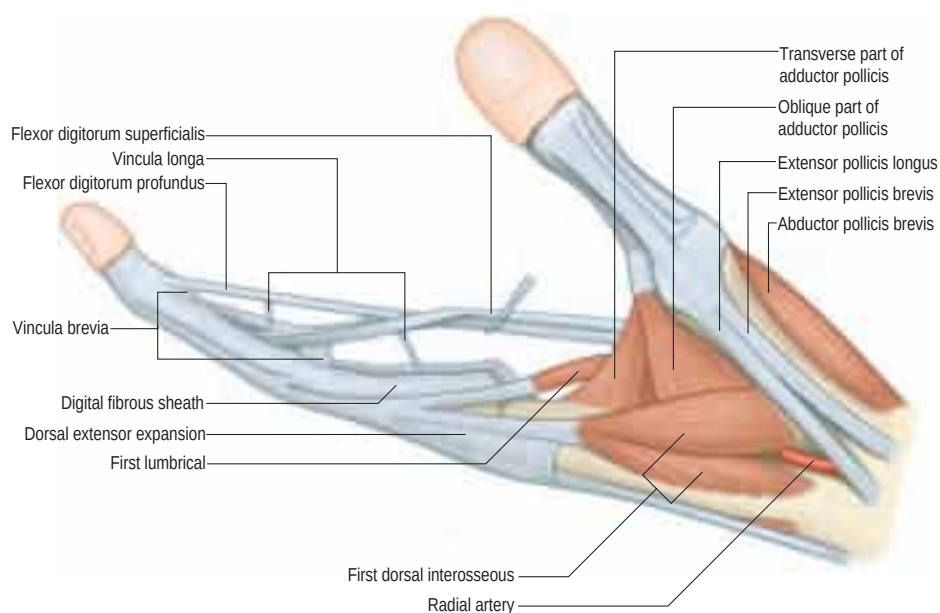


Fig. 50.28 The tendons and vincula tendinum of the index finger and the muscles in the first intermetacarpal space of the left hand: lateral aspect.

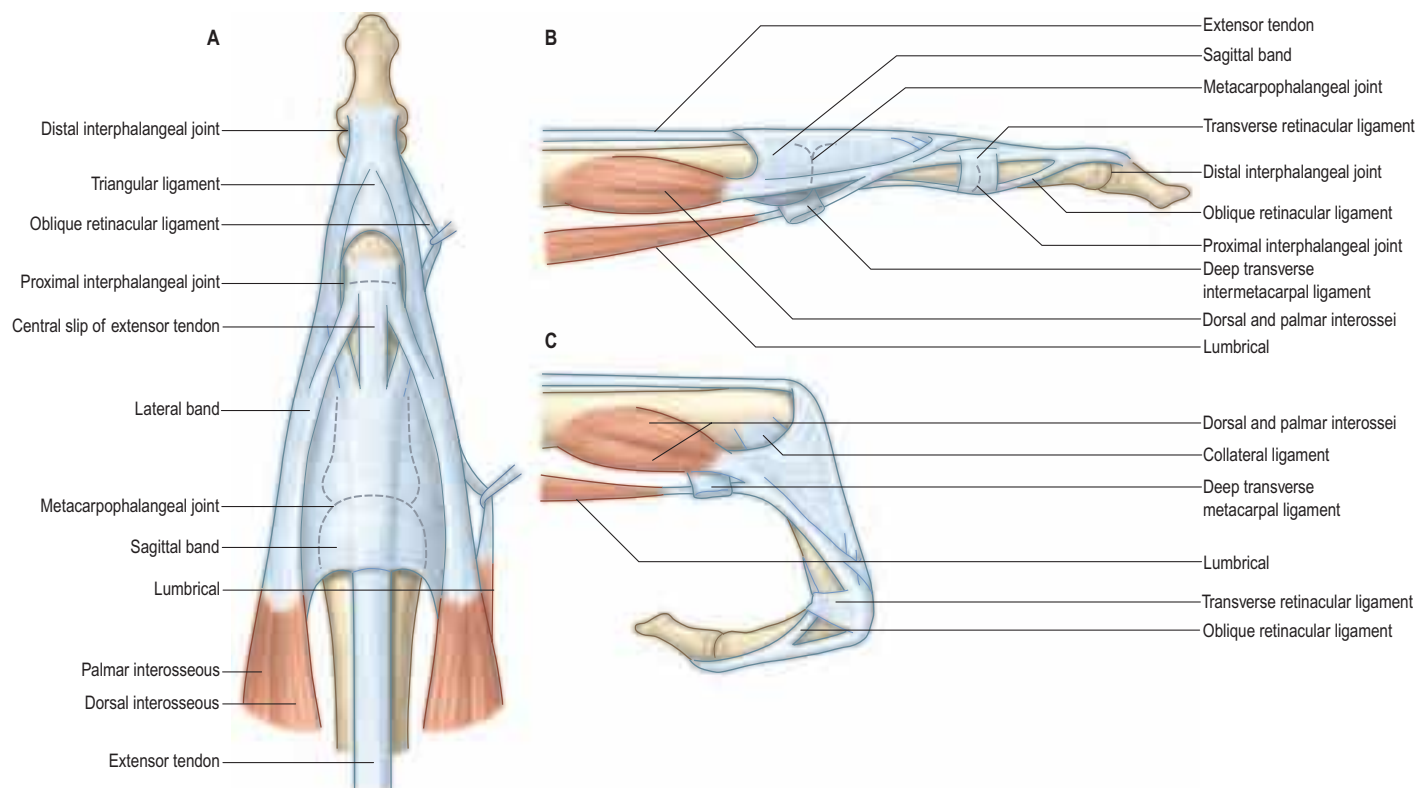


Fig. 50.29 The extensor mechanism of the finger. **A**, Dorsal view. **B**, Lateral view. **C**, Lateral view in flexion.

of rotation at the proximal interphalangeal joint. The extensor mechanism trifurcates into a central slip and two lateral bands just proximal to the proximal interphalangeal joint. The central slip receives a contribution from the tendons of the lumbricals and interossei via the lateral bands. Similarly, some fibres from the central region pass to each lateral band, producing a criss-cross arrangement of fibres. The central slip attaches to the base of the middle phalanx, while the lateral bands continue distally; they eventually fuse together and insert into the distal phalanx. The tension in the central slip and the lateral bands varies as the finger moves between flexion and extension and plays a crucial role in coordinating synchronous activity between the proximal and distal interphalangeal joints.

The transverse and oblique retinacular ligaments of Landsmeer connect the fibrous flexor sheath to the extensor apparatus. The transverse retinacular ligament passes from the A3 pulley of the fibrous flexor sheath at the level of the proximal interphalangeal joint to the lateral border of the lateral extensor band. The oblique retinacular ligament lies deep to the transverse retinacular ligament. It originates from the lateral aspect of the proximal phalanx and flexor sheath (A2 pulley) and passes volar to the axis of rotation of the proximal interphalangeal joint, but in a dorsal and distal direction, to insert into the terminal extensor tendon.

INTRINSIC MUSCLES OF THE HAND

The intrinsic muscles of the hand are organized into three groups plus a superficial muscle. The thenar muscles include flexor pollicis brevis, abductor pollicis brevis, opponens pollicis and adductor pollicis. The hypothenar muscles include abductor digiti minimi, flexor digiti minimi brevis and opponens digiti minimi. The interossei and lumbricals act on the fingers. Palmaris brevis is a superficial muscle that lies beneath the ulnar palmar skin.

Flexor pollicis brevis

Attachments Flexor pollicis brevis lies medial to abductor pollicis brevis (Fig. 50.30). It has superficial and deep parts. The superficial head arises from the distal border of the flexor retinaculum and the distal part of the tubercle of the trapezium, and passes along the radial side of the tendon of flexor pollicis longus. It is attached by a tendon that contains a sesamoid bone to the radial side of the base of the proximal phalanx of the thumb. The deep part arises from the trapezoid and capitate bones and from the palmar ligaments of the distal row of carpal bone, and passes deep to the tendon of flexor pollicis longus. It

unites with the superficial head on the sesamoid bone and base of the first phalanx.

The superficial head is frequently blended with opponens pollicis. The deep head varies considerably in size and may even be absent.

Relations Flexor pollicis brevis lies superficial in the thenar eminence and is distal to abductor pollicis brevis. It is crossed by the motor branch of the median nerve.

Actions Flexor pollicis brevis flexes the metacarpophalangeal joint.

Testing Flexor pollicis brevis is palpated whilst flexing the metacarpophalangeal joint, with the interphalangeal joint fully extended.

Abductor pollicis brevis

Attachments Abductor pollicis brevis is a thin, subcutaneous muscle in the proximolateral part of the thenar eminence (see Figs 50.25B, 50.30). It arises mainly from the flexor retinaculum, but a few fibres spring from the tubercles of the scaphoid bone and trapezium and from the tendon of abductor pollicis longus. Its medial fibres are attached by a thin, flat tendon to the radial side of the base of the proximal phalanx of the thumb, and its lateral fibres join the dorsal digital expansion of the thumb. The muscle may receive accessory slips from the long and short extensors of the thumb, opponens pollicis, or the styloid process of the radius.

Relations Abductor pollicis brevis lies proximomedial to flexor pollicis brevis in the superficial part of the thenar eminence.

Actions Abductor pollicis brevis draws the thumb ventrally in a plane at right angles to the palm of the hand (abduction).

Testing The patient abducts the thumb at right angles to the palm against resistance; the muscle can be seen and felt.

Opponens pollicis

Attachments Opponens pollicis lies deep to abductor pollicis brevis (see Fig. 50.30). It arises from the tubercle of the trapezium and the flexor retinaculum, and is attached to the whole length of the lateral border, and the adjoining lateral half of the palmar surface of the metacarpal bone of the thumb.

Relations Opponens pollicis lies deep between the other two muscles of the thenar eminence and is revealed only when they are retracted.

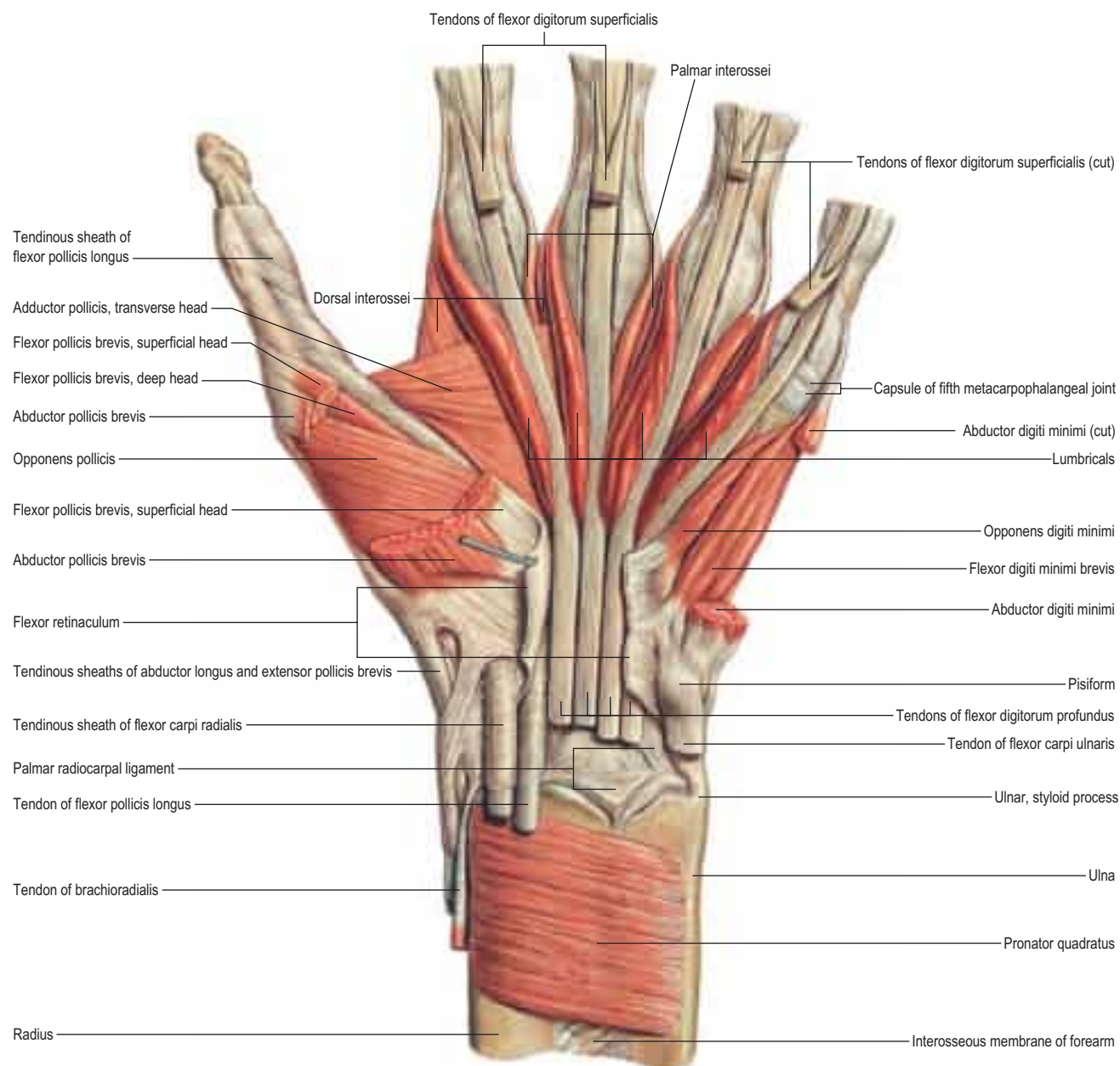


Fig. 50.30 The muscles of the left hand. The deep layer is shown after cutting the flexor retinaculum and partial removal of several superficial muscles: palmar aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Actions Opponens pollicis flexes the metacarpal bone of the thumb.

Testing The patient touches the base of the little finger against resistance.

Adductor pollicis

Attachments Adductor pollicis arises by oblique and transverse heads (see Figs 50.26, 50.28, 50.30). The oblique head is attached to the capitate bone, the bases of the second and third metacarpal bones, the palmar ligaments of the carpus, and the sheath of the tendon of flexor carpi radialis. Most of the fibres converge into a tendon (containing a sesamoid bone) that unites with the tendon of the transverse head and is attached to the ulnar side of the base of the proximal phalanx of the thumb. The deepest fibres may pass into the medial side of the dorsal digital expansion of the thumb. On the lateral side of the oblique head, a considerable fasciculus, described as the 'deep head' of flexor pollicis brevis, passes deep to the tendon of flexor pollicis longus to join flexor pollicis brevis. The transverse head is the deepest of the muscles of the thumb; it is triangular and arises from the distal two-thirds of the palmar surface of the third metacarpal. The fibres converge to be attached, with the oblique head and the first palmar interosseous, to the base of the proximal phalanx of the thumb. The two parts of the adductor vary in relative size and degree of connection.

Relations The deep palmar arch and the deep branch of the ulnar nerve pass between the two heads of the muscle. Anteriorly, adductor pollicis is crossed by the flexor tendons of the index finger and their sheath, and the first lumbrical, and is overlapped by flexor pollicis brevis. Posteriorly, it abuts against the first dorsal interosseous muscle; together, these muscles form the mass of the first web space of the hand.

Actions Adductor pollicis is the largest and most powerful of the intrinsic muscles and acts to approximate the thumb to the palm of the hand. It acts optimally when the abducted, rotated and flexed thumb is opposed to the fingers in gripping.

Testing The patient adducts the thumb at right angles to the palm against resistance.

Abductor digiti minimi

Attachments Abductor digiti minimi arises from the pisiform bone, the tendon of flexor carpi ulnaris and the pisohamate ligament (see Figs 50.25B, 50.30). It ends in a flat tendon that divides into two slips: one is attached to the ulnar side of the base of the proximal phalanx of the little finger, and the other to the ulnar border of the dorsal digital expansion of extensor digiti minimi. The muscle may have two or three slips and may be fused with flexor digiti minimi brevis. An additional slip may arise from the flexor retinaculum, antebrachial fascia, or tendons

of palmaris longus or flexor carpi ulnaris, and may be partly attached to the fifth metacarpal by a slip from the pisiform.

Relations Abductor digiti minimi lies along the ulnar border of flexor digiti minimi brevis and overlies opponens digiti minimi.

Actions Abductor digiti minimi abducts the little finger away from the fourth, e.g. in the habitual spreading of the digits when they are extended. Abduction is also possible when digits 2–4 are tightly adducted in flexion or extension.

Testing The little finger is abducted against resistance.

Flexor digiti minimi brevis

Attachments Flexor digiti minimi brevis arises from the convex surface of the hook of the hamate and the palmar surface of the flexor retinaculum, and inserts into the ulnar side of the base of the proximal phalanx of the little finger with abductor digiti minimi (see Fig. 50.30). It may be absent, or fused with the abductor, and it may attach to the distal end of the fifth metacarpal by a muscular slip.

Relations Flexor digiti minimi brevis lies lateral to the abductor. Its origin is separated from that of the abductor by the deep branches of the ulnar artery and nerve.

Actions Flexor digiti minimi brevis produces flexion of the little finger at its metacarpophalangeal joint, together with some lateral rotation.

Testing The subject flexes the metacarpophalangeal joint of the little finger against resistance with the interphalangeal joints in extension.

Opponens digiti minimi

Attachments Opponens digiti minimi is a triangular muscle lying under cover of the flexor and abductor (see Fig. 50.30). It arises from the convexity of the hook of the hamate and the contiguous portion of the flexor retinaculum, and inserts along the whole length of the ulnar margin of the fifth metacarpal bone and the adjacent palmar surface. The muscle is often divided into two lamellae by the deep branches of the ulnar artery and nerve. It blends to a variable degree with its neighbours.

Relations Opponens digiti minimi lies on the ulnar margin and adjacent palmar surface of the fifth metacarpal, overlapped by, and often partly fused with, the other two muscles of the hypothenar eminence.

Actions Opponens digiti minimi flexes the fifth metacarpal bone, drawing it forwards and rotating it laterally at the carpometacarpal joint; this deepens the hollow of the palm. These actions, together with flexion and some lateral rotation at the metacarpophalangeal and interphalangeal joints, bring the digit into opposition with the thumb.

Testing The patient opposes the tip of the little finger to the tip of the thumb with the interphalangeal joints of the little finger in extension.

Palmaris brevis

Attachments Palmaris brevis is a thin, quadrilateral muscle lying beneath the skin of the ulnar side of the palm (see Fig. 50.7A). It arises from the flexor retinaculum and the medial border of the central part of the palmar aponeurosis and is attached to the dermis on the ulnar border of the hand.

Relations Palmaris brevis is superficial to the ulnar artery and the superficial terminal branch of the ulnar nerve.

Actions Palmaris brevis wrinkles the skin on the ulnar side of the palm of the hand and deepens the hollow of the palm by accentuating the hypothenar eminence.

Testing The patient opposes the thumb to the little finger; the skin on the ulnar side of the palm is seen to pucker.

Interossei

The interossei occupy the intervals between the metacarpal bones and are divided into a palmar and a dorsal set.

Palmar interossei

Attachments Palmar interossei are smaller than dorsal interossei and lie on the palmar surfaces of the metacarpal bones rather than between



Fig. 50.31 The palmar interossei of the left hand: palmar aspect. Note that there is no first palmar interosseous in this specimen. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

them (see Fig. 50.30; Fig. 50.31). The middle finger has no palmar interossei. The remaining digits have palmar interossei on their aspects that face the middle finger. The first arises from the ulnar side of the palmar surface of the base of the first metacarpal bone and is inserted into a sesamoid bone on the ulnar side of the proximal phalanx; from here, it passes to the phalanx and, usually, also into the dorsal digital expansion. It is often rudimentary. The second arises from the ulnar side of the second metacarpal bone and is inserted into the same side of the digital expansion of the index finger. The third arises from the radial side of the fourth metacarpal bone and is inserted together with the third lumbrical. The fourth arises from the radial side of the fifth metacarpal bone and is attached with the fourth lumbrical and to the base of the proximal phalanx. The attachment of these muscles to the dorsal digital expansions (see Fig. 50.29) stabilizes the extensor tendons on the convex heads of the metacarpal bones during flexion and extension at the metacarpophalangeal joints. The interossei show little variation in their arrangement, although they may, occasionally, be duplicated.

Relations The first palmar interosseous lies anterior to the lateral head of the first dorsal interosseous and is overlapped anteriorly by the oblique head of adductor pollicis, which also crosses anterior to the second palmar interosseous. The third and fourth are overlapped by the long flexor tendons of the ring and little finger, respectively, within their flexor sheaths.

Actions Palmar interossei adduct the fingers towards the longitudinal axis of the middle finger. Interossei have a considerable cross-sectional area and, therefore, contribute strongly to metacarpophalangeal joint flexion and interphalangeal extension. When the interossei are paralysed, the grip strength of the hand is reduced and the arc of finger motion is abnormal, with a tendency for the fingers to claw. Each interosseous has a considerable ability to rotate the digit at the metacarpophalangeal joint. Generally, this is not obvious because interossei act in pairs, but it may occur where one interosseous is deficient as a result of injury or congenital deformity.

Testing The patient adducts the index ring and little fingers against resistance.

Dorsal interossei

Attachments Dorsal interossei consist of four bipennate muscles, each arising from the adjacent sides of two metacarpal bones, but more extensively from the metacarpal bone of the finger into which the muscle passes (see Figs 50.29–50.30; Fig. 50.32). They insert on the

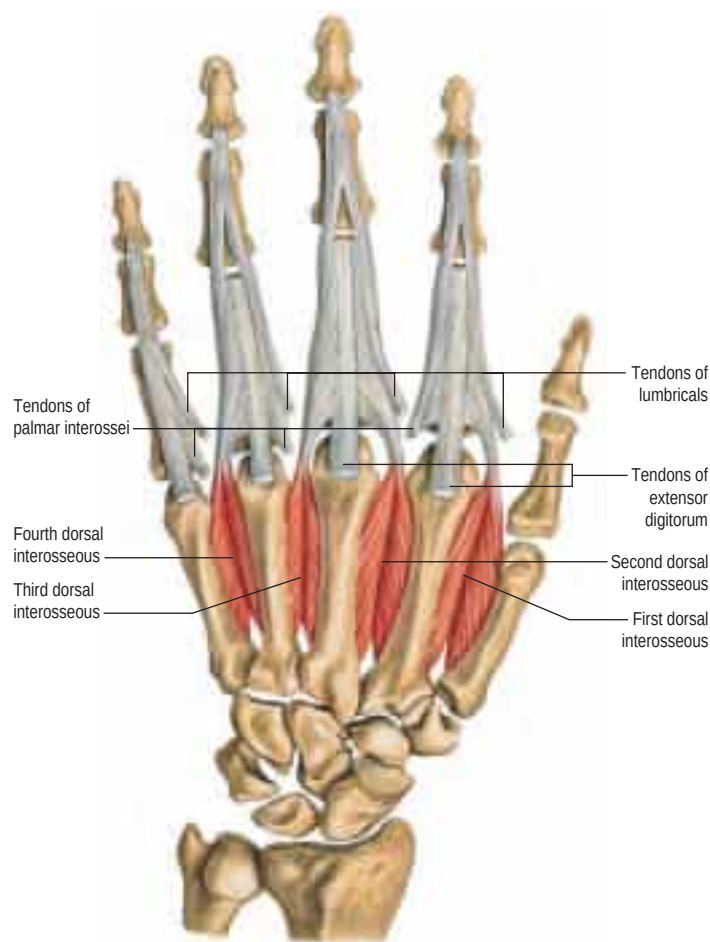


Fig. 50.32 The dorsal interossei of the left hand: dorsal aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

bases of the proximal phalanges and separately into the dorsal digital expansions. There is a narrow triangular interval between the double origin of each of these muscles; the radial artery passes through the first of these intervals, and a perforating branch from the deep palmar arch passes through each of the others. The first and largest muscle is sometimes called abductor indicis. It is attached to the radial side of the proximal phalanx of the index finger and to the capsule of the adjoining metacarpophalangeal joint. The second and third are attached to the radial and ulnar sides of the middle finger, respectively. Whereas the second generally reaches the digital expansion and the proximal phalanx, the third usually extends only to the digital expansion. The fourth may be wholly attached to the digital expansion, but it often sends an additional slip to the proximal phalanx.

Relations The first dorsal interosseous abuts anteriorly with adductor pollicis. The other dorsal interossei occupy the spaces between the second and the fifth metacarpal bones.

Actions Dorsal interossei abduct the fingers away from the longitudinal axis of the middle finger. See also the description of the actions of the palmar interossei above.

Testing The patient abducts the index, middle and ring fingers against resistance. The first dorsal interosseous is tested with the subject's fingers and palm flat on the table. The subject tries to abduct the index finger against the examiner's resistance. The muscle belly can be felt and seen.

Lumbricals

Attachments The four small lumbrical muscles arise from the tendons of flexor digitorum profundus (see Figs 50.29–50.30; Fig. 50.33). The first and second arise from the radial sides and palmar surfaces of the tendons of the index and middle fingers, respectively. The third arises from the adjacent sides of the tendons of the middle and ring fingers, and the fourth from the adjoining sides of the tendons of the ring and little fingers. Each passes to the radial side of the cor-

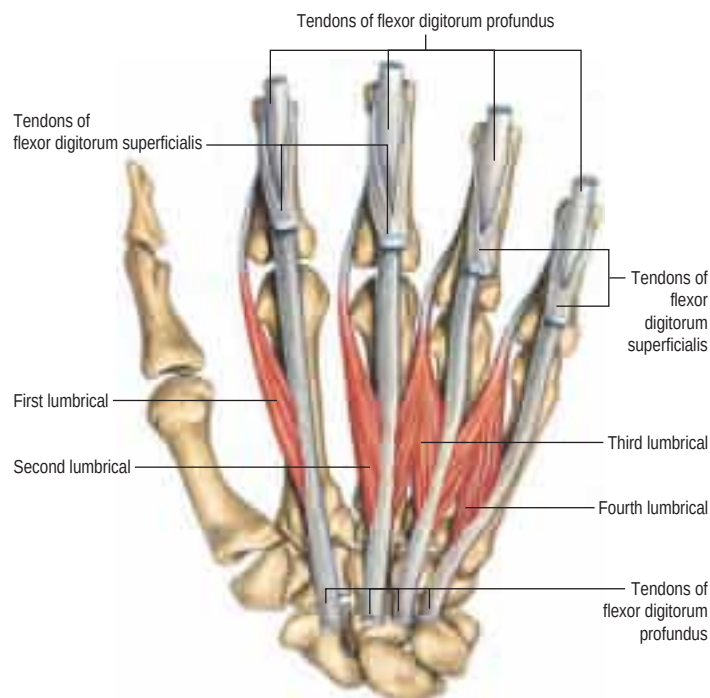


Fig. 50.33 Lumbricals of the left hand: palmar aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

responding finger and is attached to the lateral margin of the dorsal digital expansion of extensor digitorum covering the dorsal surface of the finger. Variations in the attachments of the lumbricals are common. Any of them may be unipennate or bipennate. When they are bipennate, the two heads arise from adjoining tendons of flexor digitorum profundus, and, in the case of the first lumbrical, from the tendon of flexor pollicis longus. Accessory lumbrical slips may be attached to an adjacent tendon of flexor digitorum superficialis.

Actions Lumbricals arise from flexor tendons and insert into the extensor apparatus. Since both attachments are mobile, they have the potential for producing movement at either. The action on the extensor apparatus extends both interphalangeal joints in a coordinated manner. The mode of action at the metacarpophalangeal joints is disputed, but if there is a flexor action, it is very weak. The effect on the flexor digitorum profundus attachment is to pull the tendon distally. The combined action on both origin and insertion is, therefore, to alter the posture of the finger to allow more interphalangeal extension. Pinching the index finger against the thumb without a lumbrical would result in a nail-to-nail contact; the addition of the lumbrical increases the interphalangeal joint extension and results in pulp-to-pulp pinch. Lumbricals contain many muscle spindles and have a long fibre length; it is, therefore, reasonable to assume that they play a role in proprioception.

Testing Lumbricals cannot be tested in isolation but only as the lumbrical–interosseous muscle complex. The examiner holds the metacarpophalangeal joint of the index finger in hyperextension and the subject is instructed to extend the proximal interphalangeal joint against resistance. This test is repeated on the other fingers.

Vascular supply to the intrinsic muscles of the hand

Flexor pollicis brevis is supplied by the superficial palmar branch of the radial artery, and branches from the princeps pollicis artery and radialis indicis artery. Abductor pollicis brevis is supplied by the superficial palmar branch of the radial artery and, often, by an independent branch directly from the radial artery lying on the radial aspect of the thumb. Opponens pollicis is supplied by the superficial palmar branch of the radial artery and by branches from the first palmar metacarpal artery (when present), arteria princeps pollicis, arteria radialis indicis and the deep palmar arch. Adductor pollicis is supplied by the arteria princeps pollicis and arteria radialis indicis (sometimes combined as the first palmar metacarpal artery) and by branches from the deep palmar arch. Abductor digiti minimi and flexor digiti minimi brevis are

supplied by the deep palmar branch of the ulnar artery and branches from the ulnar end of the superficial palmar arch; the medial border of the little finger is supplied by the palmar digital artery. Opponens digiti minimi is supplied by the deep palmar branch of the ulnar artery and branches from the medial end of the deep palmar arch. Palmaris brevis is supplied by branches from the ulnar end of the superficial palmar arch. The palmar interossei are supplied by branches from the deep palmar arch, arteria princeps pollicis, arteria radialis indicis, palmar metacarpal arteries, proximal and distal perforating arteries, and common and proper digital (palmar) arteries. The dorsal interossei are supplied by the dorsal metacarpal arteries (first to fourth), palmar metacarpal arteries (second to fourth), radial artery (first), arteria princeps pollicis, arteria radialis indicis, three perforating branches from the deep palmar arch (proximal perforating arteries) and three distal perforating branches. Their tendons are supplied by branches from the common and proper palmar digital arteries and the dorsal digital arteries.

The first and second lumbricals are supplied by the first and second dorsal metacarpal and dorsal digital arteries. The third and fourth lumbricals are supplied by the second and third common palmar digital arteries, and the third and fourth dorsal digital arteries and their anastomoses with the palmar digital arteries.

Innervation of the intrinsic muscles of the hand

The superficial head of flexor pollicis brevis is usually innervated by the recurrent motor branch of the median nerve, and the deep head is usually innervated by the deep branch of the ulnar nerve; variation is common. Abductor pollicis brevis is innervated by the recurrent motor branch of the median nerve and is the only thenar muscle that is constantly supplied by the median nerve. Opponens pollicis is innervated by the lateral terminal branch of the median nerve and, commonly, by a branch of the deep terminal branch of the ulnar nerve. Adductor pollicis, abductor digiti minimi, flexor digiti minimi brevis, opponens digiti minimi and all the interossei are innervated by the deep branch of the ulnar nerve; the first dorsal interosseous and adductor pollicis are supplied by its most distal portion. Palmaris brevis is innervated by the superficial branch of the ulnar nerve. The first and second lumbricals are innervated by the median nerve, and the third and fourth lumbricals by the deep terminal branch of the ulnar nerve. The third lumbrical frequently receives a supply from the median nerve. The first and second lumbricals are, occasionally, innervated by the deep terminal branch of the ulnar nerve.

MOVEMENTS OF THE HAND

The apparently simple human functions of closing the hand to grasp an object, or opening the palm to release it, are, in reality, tasks of considerable mechanical complexity, requiring the simultaneous contraction of many individual muscles. The isolated action of a single muscle may be inferred from the positions of its origin and insertion, and the estimated line of action (usually, the centre line of the muscle) in relation to the axes of all the joints traversed by the muscle and its tendon. The limb can be regarded as a chain of joints crossed by muscles. If it is known which muscles are active, then the reason why one joint moves and others do not is a matter of simple mechanical relationships. For example, flexor pollicis longus is considered to have a major role as a flexor of the interphalangeal joint of the thumb. However, the position of its tendon relative to more proximal joints in the limb gives it the potential for producing flexion at the metacarpophalangeal joint and also at the trapeziometacarpal and wrist joints. In the living subject, the actual motion that takes place depends on which other muscle groups are acting, and so the potential for movement must be considered for each joint in the chain in turn. Motion at the wrist is generally balanced by wrist extensors. Motion at the trapeziometacarpal joint is balanced by abductor pollicis longus. Flexor pollicis longus will then have an action as a flexor of the metacarpophalangeal and interphalangeal joints only.

The factor that determines whether one or both of two joints will move is the turning moment at each. The greater the perpendicular distance from the line of muscle or tendon pull to the axis of the joint, the stronger is the turning effect of the muscle at the joint, but the smaller the range of joint motion that can be produced. In the case of flexor pollicis longus, the tendon is situated further from the axis of the metacarpophalangeal joint than from the axis of the interphalangeal joint; it will, therefore, tend to produce flexion preferentially at the metacarpophalangeal joint, unless that joint is restrained by extensor

pollicis brevis. In this way, different postures of the thumb can be produced by the interplay of flexor and extensor forces. These simple guiding principles should provide an understanding of muscle action in the hand that is sufficient for most purposes.

In considering the role of a particular muscle, there is a tendency to concentrate on motion. Indeed, many muscles are named on the basis of the movements that they generate, although others, often those whose actions are the most difficult to interpret, are described according to their morphology or situation. A more important function may be the nature of the force generated. For example, although flexor pollicis longus flexes the thumb (see above), a large range of flexion is actually required in only a few activities, such as certain ripping tasks. In most pinch and manipulative tasks, the role of the thumb is to apply isometric force, which it does with such precision that it is possible to pick up an egg and neither crush nor drop it. Thus, for much of the time, flexor pollicis longus behaves as an extremely sophisticated mechanism for the application of force, in which contraction and proprioception are equally important.

The anatomical position of the hand (palm flat and pointing anteriorly, forearm supinated) is a convenient standard for studying structural relationships. The hand in the relaxed (anaesthetized) position adopts a posture of partial flexion and mid-supination/pronation (the reader can verify this by relaxing completely and observing forearm and hand position).

SPECIAL FUNCTIONS OF THE HAND

Closing the hand



Available with the *Gray's Anatomy* e-book

Role of the long digital flexors



Available with the *Gray's Anatomy* e-book

Role of the wrist



Available with the *Gray's Anatomy* e-book

Making a tight fist



Available with the *Gray's Anatomy* e-book

Opening the hand



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Movements of the thumb

The thumb does not easily assume the classical anatomical position. Therefore, the normal descriptive anatomical terms, i.e. anterior, posterior, medial and lateral, do not readily apply. The terms 'palmar', 'dorsal', 'ulnar' and 'radial' have been adopted in clinical practice.

An opposable thumb requires a different system of control from the other digits. Since the metacarpal is much more mobile than in the digits, muscles are needed to control the extra freedom of movement. The basic active movements are flexion–extension, abduction–adduction, rotation and circumduction. In the resting position of the first metacarpal, flexion and extension are parallel with the palmar plane, and abduction and adduction occur at right angles to this.

Flexion and extension should be confined to motion at the interphalangeal or metacarpophalangeal joints (see Video 50.1). Palmar abduction (see Video 50.1), in which the first metacarpal moves away from the second at right angles to the plane of the palm, and radial abduction (see Video 50.1), in which the first metacarpal moves away from the second with the thumb in the plane of the palm, occur at the carpometacarpal joint. The opposite of radial abduction is ulnar adduction, or transpalmar adduction, in which the thumb crosses the palm towards its ulnar border. In clinical practice, the term adduction is generally used without qualification. Circumduction describes the angular motion of the first metacarpal, solely at the carpometacarpal joint, from a position of maximal radial abduction in the plane of the palm towards the ulnar border of the hand, maintaining the widest possible angle between the first and second metacarpals (see Video



It is clear that the fingers and palm of the hand flex in gripping, grasping or making a fist, but there are subtle differences in hand posture in these various activities. The basic mechanisms of hand closure will be described before special grips are considered.

As the digits flex, the wrist usually extends (dorsiflexes) at the same time. The involvement of the long digital flexors in this movement will be considered first, followed by an analysis of the role of the wrist.

Flexor digitorum superficialis acts principally to flex the proximal interphalangeal joints, through its insertions into the middle phalanges. However, in each digit, it also has an action on the metacarpophalangeal joint because the tendon passes anterior to that joint. The muscle has the potential to produce flexion at the wrist for the same reason. The fact that each tendon arises from an individual muscle slip allows the clinician to test one finger at a time. The reader can verify this by attempting to flex each digit individually while using the other hand to keep the distal interphalangeal joints of the remaining fingers in extension. This test is frequently used in clinical practice and is useful for the middle and ring fingers, where flexion of one finger alone must be attributed to flexor digitorum superficialis. The index finger, however, has its own profundus musculotendinous unit and may, therefore, move independently under the action of this tendon. Many individuals cannot flex the proximal interphalangeal joint of the little finger alone, usually because of linkage between the tendons of the ring and little finger flexor digitorum superficialis, but occasionally because superficialis is deficient. Most individuals can flex the metacarpophalangeal joint of the little finger using flexor digiti minimi.

Flexor digitorum profundus is the only flexor of the distal interphalangeal joint. It also contributes, together with superficialis, to flexion at the proximal interphalangeal and metacarpophalangeal joints. These two long flexors (sometimes called extrinsic flexors because the muscle bellies are outside the hand) can be considered to act together to flex the finger. However, their action alone would wind up the interphalangeal before the metacarpophalangeal joints and the finger would not move in a normal arc of flexion. This is precisely what happens in an ulnar nerve paralysis, in which the interossei and lumbricals are not functioning. These small (intrinsic) muscles have been described earlier in terms of their individual actions. For their role in coordinated activity, it is sufficient to appreciate that their contribution changes the arc produced by the long flexors, increasing flexion at the metacarpophalangeal joint and reducing flexion at the proximal interphalangeal joint. All three joints are then angulated to the same degree and the fingers form a normal arc of flexion. As the finger flexes, the long extensor tendons (extensor digitorum, extensor indicis and extensor digiti minimi) aid the process by relaxing and allowing the extensor apparatus to glide distally on the dorsum of the phalanges.

As the fingers wind up to make a fist, the wrist tends to extend, particularly when force is applied. This extension has a marked effect on the excursion of the long flexor tendons. On its own, digital flexion would require the long tendons to move proximally in their sheaths and the flexor muscles in the forearm would shorten. Extension of the wrist tends to produce a lengthening of the same muscles, which, in normal use, is almost enough to balance the shortening due to finger flexion; the net effect is a very slight shortening (approximately 1 cm) of the long flexors in the forearm. The wrist can, therefore, be seen as a mechanism for maximizing force because it allows the fingers to flex while maintaining the resting length of the extrinsic muscles near to the peak of the force-length curve. It is, of course, possible to wind up the fingers with the wrist held in a neutral position but the grip is somewhat weaker. With the wrist in full flexion, it is not possible to flex the fingers fully.

Flexion of the fingers on gripping tends to result in a distal excursion of the long extensors. However, this tendency is counteracted by dorsiflexion of the wrist. The net effect is a very small proximal excursion of the long extensor tendons on gripping, mirroring the effect on the flexor surface. If the movement of the wrist is exaggerated so that the wrist is a little flexed on opening the hand, and fully dorsiflexed on closing it, the net excursion of long flexors and extensors is zero, i.e. this whole movement sequence can be completed with the forearm flexor and extensor muscles contracting isometrically.

The reader can observe the relationship between digits and wrist by performing the following manoeuvre. The wrist is held in a relaxed, mid-supinated position, with the elbow flexed at 90°. If the forearm is now rotated into pronation, the wrist will fall into flexion and the fingers will automatically extend. If the forearm is rotated into supination, the wrist will extend and the fingers flex. The finger movements compensate for the wrist movements and are entirely automatic; they are made without the need for any excursion of forearm flexor or extensor tendons. This test, the wrist tenodesis test, is a useful way of examining the limb for tendon injury. The pointing finger (which does not move with wrist motion) 'points to' a tendon injury.

Wrist motion is controlled principally by two wrist flexors (flexor carpi radialis and flexor carpi ulnaris) and three extensors (extensors carpi radialis longus and brevis, and extensor carpi ulnaris). Although the radiocarpal joint has some functional similarity to a ball and socket joint, it is possible to think of the wrist as a variable hinge joint, the axis of which may be set in a number of inclinations. For example, in using a hammer, it is useful to rotate the wrist backwards and forwards about an axis that permits not only wrist flexion but also ulnar deviation. It would be very restricting to have a pure hinge joint with collateral ligaments of fixed length. In this context, the wrist flexors and extensors may be regarded as variable collateral ligaments that allow the joint to be set about a number of different axes.

It is possible to observe and to palpate the muscle groups that are active in making a tight fist. Flexor digitorum profundus and flexor digitorum superficialis are active, and flexor carpi ulnaris contracts strongly, as do the wrist extensors. Palpation of the long digital extensors on the dorsum of the wrist, the first dorsal interosseous in the thumb web, and of the other interossei and the thenar and hypothenar muscles, will confirm that all these muscles are contracting. As a firm fist is swung forwards in anger, brachioradialis stands out; at the moment of impact, virtually every muscle in the limb is active.

The hand is opened from its relaxed balanced posture, e.g. when stretching out to reach an object. This motion is made up of extension of the distal interphalangeal, proximal interphalangeal and metacarpophalangeal joints. The hand is provided with an ingenious mechanism that allows this to happen. The laws of mechanics would suggest that one motor would be required for every joint in a chain, together with some sort of controlling mechanism to ensure that the chain of joints moved together in a coordinated fashion. In the hand, this is achieved through an extensor apparatus that minimizes the number of motors required for movement by allowing the muscles to act on more than one joint, and by linking different levels in the mechanism so that the arc of motion is controlled.

The tendons of extensor digitorum run distally over the metacarpal heads, forming the major component of the extensor apparatus. Extensor digitorum has no insertion into the proximal phalanx and, therefore, exerts its extensor action on the metacarpophalangeal joint indirectly through more distal insertions. The first point of insertion is at the base of the middle phalanx ('central slip', in clinical practice). Acting at this insertion alone, extensor digitorum can extend both metacarpophalangeal and proximal interphalangeal joints together. The interossei are also active in hand opening, since they will tend to increase extension of the proximal interphalangeal joint. There is, therefore, a range of possibilities. At one extreme, with no contribution from the interossei, the long extensor will exert all of its action at the metacarpophalangeal joint; this leads to full extension, and even hyperextension, while the proximal interphalangeal joint remains flexed (the typical 'intrinsic minus', or claw hand, of ulnar nerve paralysis). At the other extreme, when the intrinsics act strongly together with extensor digitorum, the proximal interphalangeal joint will extend completely while the metacarpophalangeal joint remains flexed ('intrinsic plus' hand). Thus, the hand possesses, in the proximal part of the extensor apparatus, a variable mechanism that allows different amounts of relative metacarpophalangeal or proximal interphalangeal joint motion.

In contrast, the more distal part of the extensor apparatus acts as an automatic or fixed mechanism that determines that the two interphalangeal joints, proximal and distal, will move together. The lateral slips of the extensor apparatus arise from extensor digitorum and pass distally on either side of the central slip and thus over the proximal interphalangeal joint; being further lateral, they are nearer the joint axis because the dorsal surface curves away on each side. A helpful analogy that has been suggested for this arrangement is to consider it as two pulleys of different size on one axle. The central slip can be regarded as a cord that passes over the larger wheel, and each lateral slip as a cord that passes over the smaller wheel. Since these latter pulleys are smaller, there is less longitudinal excursion for a given rotation of the wheel, and this allows some of the excursion to be used for another function, namely: extension at the distal joint. There is an additional mechanism by which the lateral slips move laterally during flexion of the proximal interphalangeal joint. The effect of this lateral movement is to reduce further the distance between the lateral slips and the joint axis, thereby reducing the amount of excursion at the proximal interphalangeal joint still more and allowing more excursion at the distal joint. When the hand flexes, this mechanical linkage system allows both interphalangeal joints to flex together in a coordinated way.

The extensor expansion also receives contributions from the interossei and lumbricals, which approach the digits from the webs and join the corresponding expansion in the proximal segment of the digit. These muscles can, therefore, act on the extensor apparatus at two levels: they can extend the proximal interphalangeal joint through fibres that radiate towards the central slip, and they can act on the distal interphalangeal joint through fibres that join the lateral slip.

Apart from the components of the extensor expansion that are concerned with joint function, the whole structure requires additional anchorage. This must be arranged in such a way that it is not displaced from the underlying skeleton, yet it must not restrict longitudinal movement. These difficult requirements are met by transverse retinacular ligaments at the level of the joints, the transverse ligaments running to relatively fixed attachment points in the region of the joint axis. As the expansion glides backwards and forwards, the transverse fibres move like bucket handles. Smooth gliding layers are required under the expansion and retinacular ligaments to allow motion to occur without friction.

50.1). Lateral inclinations of the first phalanx maximize the extent of excursion of the circumduction arc. Opposition is a composite position of the thumb achieved by circumduction of the first metacarpal, internal rotation of the thumb ray, and maximal extension of the metacarpophalangeal and interphalangeal joints (see Video 50.1). Retroposition is the opposite to opposition (see Video 50.1). Flexion and adduction is the position of maximal transpalmar adduction of the first metacarpal; the metacarpophalangeal and interphalangeal joints are flexed and the thumb is in contact with the palm (see Video 50.1).

Rotary movements occur during circumduction. The simple angular movements described above combine with rotation about the long axis of the metacarpal shaft. In opposition, the shaft must rotate medially into pronation. In retroposition, the thumb must rotate laterally into supination. Axial rotation of the thumb metacarpal is produced by muscle activity (which moves the thumb through its arc of circumduction); the geometry of the articular surfaces of the trapeziometacarpal joint; and tensile forces in the ligaments (which combine with forces exerted by the muscles of opposition and retroposition to produce axial rotation). The stability of the first metacarpal is greatest after complete pronation in the position of full opposition, when the tension in the ligaments, muscular contraction and joint congruence combine to maximal effect.

Position of rest

The hand has a well-recognized position of rest, with the wrist in extension and the digits in some degree of flexion. The precise position of the thumb in the position of rest appears to be rather variable. Typically, it is considered as the midpoint between maximal palmar abduction and maximal retroposition. In this position, the carpometacarpal joint lies within 20° of radial abduction and 30° of palmar abduction; from clinical observations, it seems that the metacarpophalangeal joint lies within approximately 40° of flexion and the interphalangeal joint between extension and 10° of flexion.

Grips

From the position of rest, the tip of the thumb can approach the radial aspect of the fingers without incurring axial rotation because the palmar and dorsal trapeziometacarpal ligaments remain relaxed (see below). From different positions of the arc of circumduction, numerous different types of pinch grip are possible (see Video 50.1). In clinical practice, these have been classified into two main types: tip pinch and lateral (or key) pinch. Many forces contribute to these configurations.

The thumb is activated by monoarticular muscles (abductor pollicis longus and opponens pollicis), biarticular muscles (extensor pollicis brevis, adductor pollicis, abductor pollicis brevis and flexor pollicis brevis) and polyarticular muscles (extensor pollicis longus and flexor pollicis longus). It appears, however, that even a monoarticular muscle can change posture in all three joints by altering the overall balance of forces and it is, therefore, very difficult to attribute function to the individual intrinsic muscles. However, the thumb muscles do seem to provide two broad functions. They position the metacarpal, an activity automatically accompanied by rotation, and also control the axial stability of the skeleton of the thumb.

The thumb muscles may be classified into those used for retroposition, opposition and pinch grip.

Retroposition muscles

The muscles that bring about retroposition are extensor pollicis longus, extensor pollicis brevis and abductor pollicis longus. As the thumb moves into retroposition, automatic axial rotation produces supination of the first metacarpal. This is produced by the off-axis action of two parallel, but oppositely directed, forces: one exerted by extensors pollicis longus and brevis, and the other by abductor pollicis longus and the anterior oblique carpometacarpal ligament.

Opposition muscles

A succession of activity occurs in the thenar muscles during the movement of opposition. Three subgroups of radial (abductor pollicis longus and extensor pollicis brevis), central (abductor pollicis brevis and opponens pollicis) and ulnar (flexor pollicis brevis) muscles are involved. These forces act simultaneously but with different intensities, depending on the situation of the thumb. As the thumb moves into opposition, there is automatic axial rotation of the first metacarpal shaft to produce pronation. This is produced by the paired action of oppositely directed forces: the opposition muscles provide one force, and the posterior oblique carpometacarpal ligament provides the other.

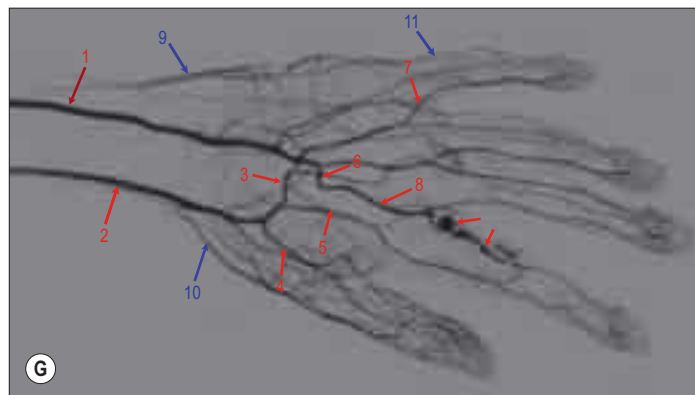


Fig. 50.34 A digital subtraction macroangiogram between capillary and venular phases. A small arterial-venous malformation shown on the index finger is suitable for radiologically guided embolization (arrows). Key: 1, ulnar artery; 2, radial artery; 3, deep palmar arch; 4, princeps pollicis artery; 5, arteria radialis indicis; 6, superficial palmar arch (incomplete); 7, proper palmar digital artery (little finger); 8, common palmar digital artery; 9, basilic vein; 10, cephalic vein; 11, digital vein.

(A-F, H-K, continued online)

Pinch grip muscles

The muscles of pinch grip may be divided into lateral, medial and intermediate subgroups. The lateral subgroup (opposition muscles) moves the first metacarpal into palmar abduction. The metacarpal shaft rotates medially into pronation. Radial angulation at the metacarpophalangeal joint increases the span of the hand. The metacarpophalangeal joint is stabilized principally by extensor pollicis brevis and flexor pollicis brevis. Flexion of the proximal and distal phalanges is controlled. Muscles of the medial subgroup (abductor pollicis brevis and first dorsal interosseous) produce an approach of the first metacarpal towards the palm. Since they act with the lateral group, they have a strong controlling effect on the position and rotation of the first metacarpal. The intermediate subgroup consists of flexor pollicis longus, which flexes the interphalangeal or metacarpophalangeal joint. Palpating the thenar eminence during tip and lateral pinch provides some appreciation of the action of the pinch grip muscles.

The paralysed hand



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VASCULAR SUPPLY

ARTERIES

Anastomoses occur between the radial and ulnar arteries at the wrist (via the palmar and dorsal carpal arches) and in the hand (via the superficial and deep palmar arches), and between their digital and metacarpal branches (Fig. 50.34; Wong and Higgins 2015).

Radial artery

At the wrist, the radial artery passes on to the dorsal aspect of the carpus between the lateral carpal ligament and the tendons of abductor pollicis longus and extensor pollicis brevis (Fig. 50.35). It crosses the scaphoid bone and trapezium (in the anatomical snuff-box), where again, its pulsation is obvious; as it passes between the heads of the first dorsal interosseous, it is crossed by the tendon of extensor pollicis longus. Between the thumb extensors, it is crossed by the start of the cephalic vein and the branches of the radial nerve that pass to the thumb and index finger (see Fig. 50.42). Occasionally, it gives off a distal superficial dorsal branch that crosses the radial extensor tendons at the wrist, together with the superficial radial nerve. Branches of the lateral cutaneous nerve of the forearm run along its distal part as it curves round the carpus.

In the hand, the radial artery passes through the first interosseous space between the heads of the first dorsal interosseous and crosses the palm. At first, it lies deep to the oblique head of adductor pollicis and then passes between its oblique and transverse heads or through the transverse head. At the fifth metacarpal base, it anastomoses with the deep branch of the ulnar artery, completing the deep palmar arch (Fig. 50.36).

The wrist and hand may be paralysed by many conditions, the most common of which are spinal cord, brachial plexus and peripheral nerve injury or compression, poliomyelitis, cerebral palsy and leprosy. Each of these will cause different patterns of paralysis and contracture. The main deformities in the cerebral palsied hand are pronation of the forearm, flexion of the wrist and fingers, and flexion-adduction of the thumb. Flexion-adduction deformity of the thumb reflects an imbalance between its intrinsic and extrinsic musculature – principally powered by adductor pollicis, the first dorsal interosseous, flexor pollicis brevis and abductor pollicis brevis – which produces flexion, adduction and lateral rotation of the carpometacarpal joint, flexion and lateral rotation of the metacarpophalangeal joint and flexion of the interphalangeal joint. In an attempt to counteract this, the patient tends to hold the wrist in extreme flexion and ulnar deviation ([Matev 1963](#)). In leprosy, the intrinsic muscles of the hand are paralysed. The metacarpophalangeal joints have no primary flexor but retain the powerful extensor digitorum. The interphalangeal joints have no primary extensor and rely on the long flexor to initiate movement. This results in four main disabilities: the fingers cannot be extended fully at the interphalangeal joints; abduction and adduction of the digits are lost; when grasping an object, flexion begins at the terminal joints, closing the fingers on themselves rather than on the palm; and the effective part of the finger becomes the tip and nail rather than the pulp ([Brand 1958](#)).

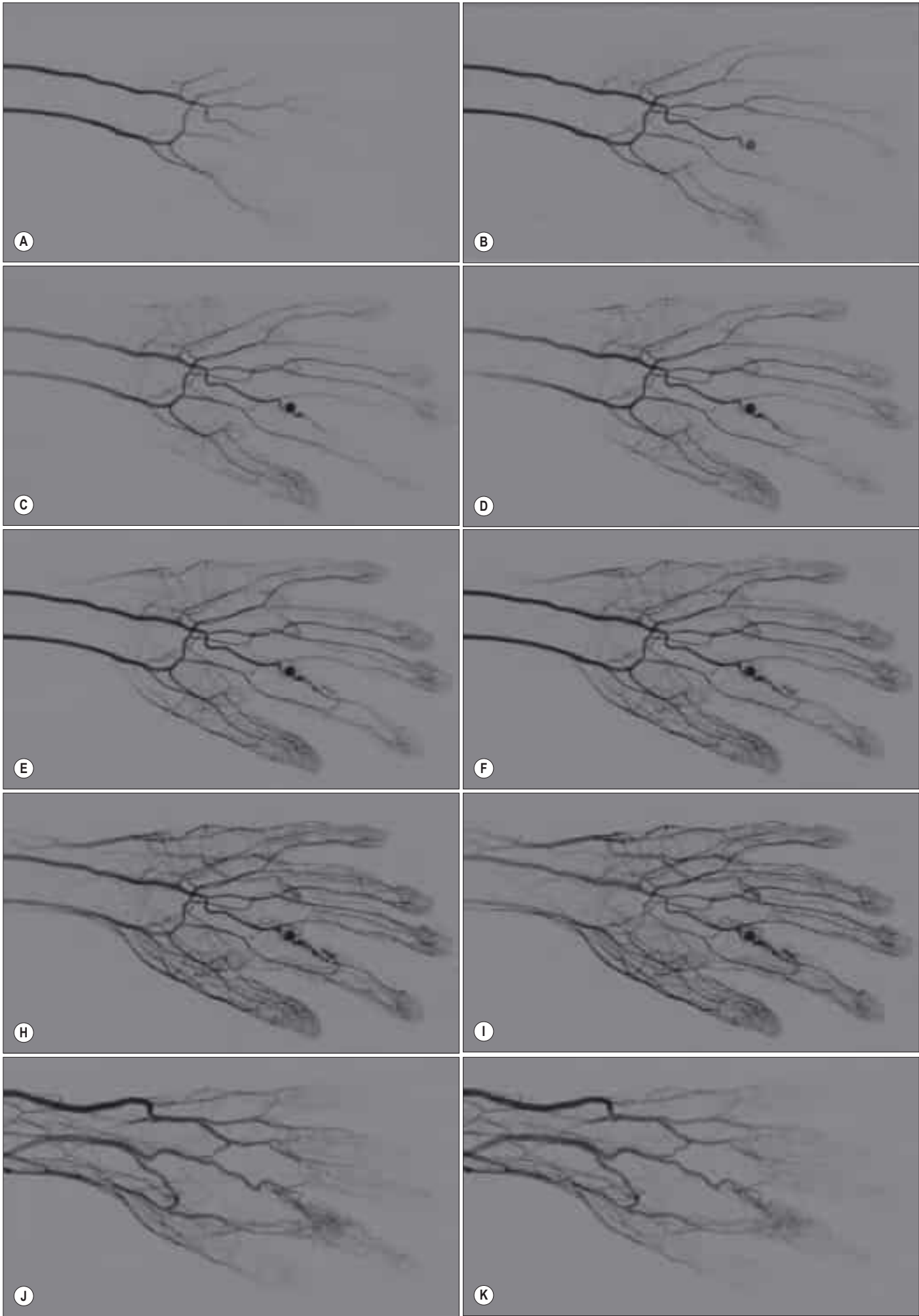


Fig. 50.34 Hand digital subtraction macroangiogram. Predominant phases. A, B arterial phase; C, D, arteriolar phase; E, F capillary phase; H, I venular phase; J, K venous phase. A small arterial-venous malformation shown on the index finger is suitable for radiologically guided embolization.

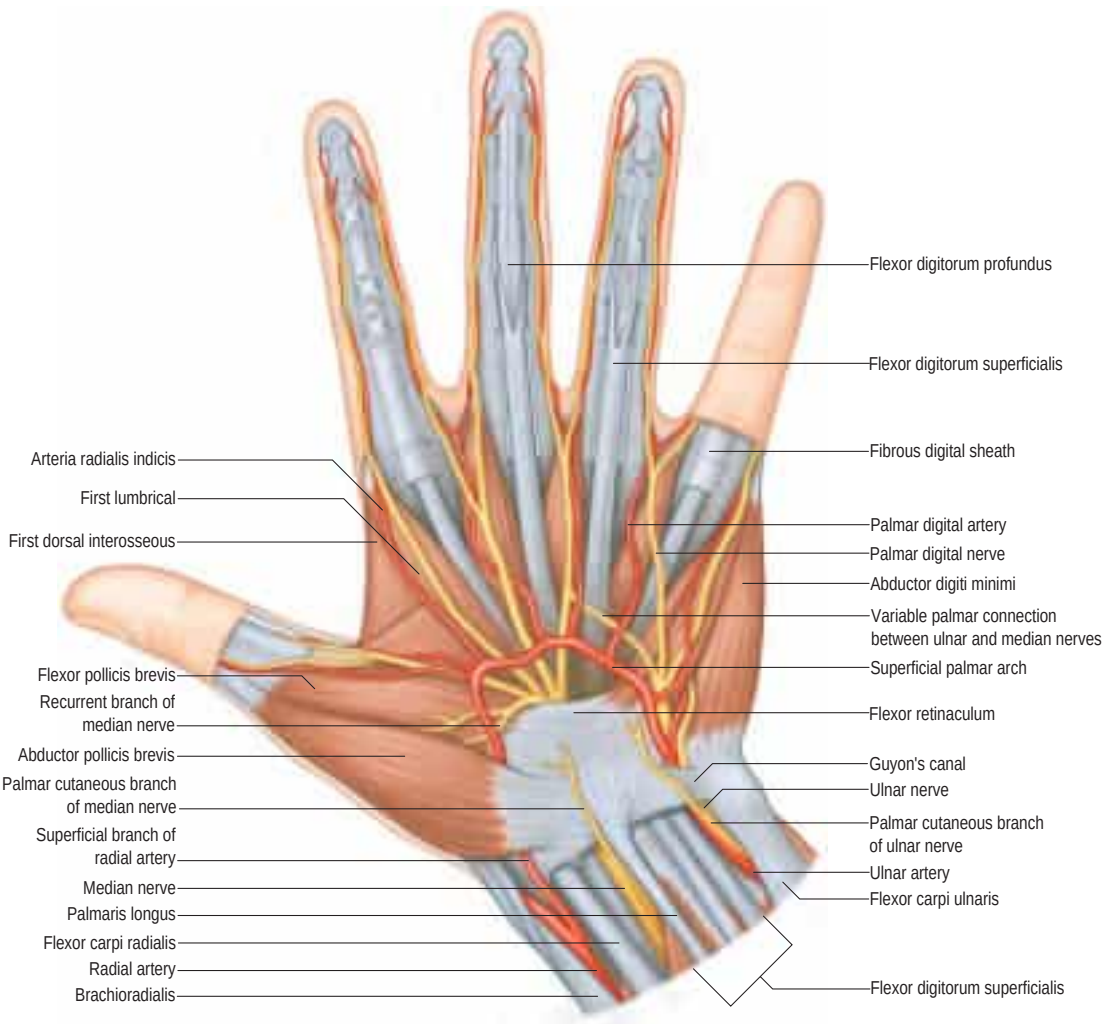


Fig. 50.35 Superficial structures of the left palm and wrist. See also [Figure 50.25B](#).

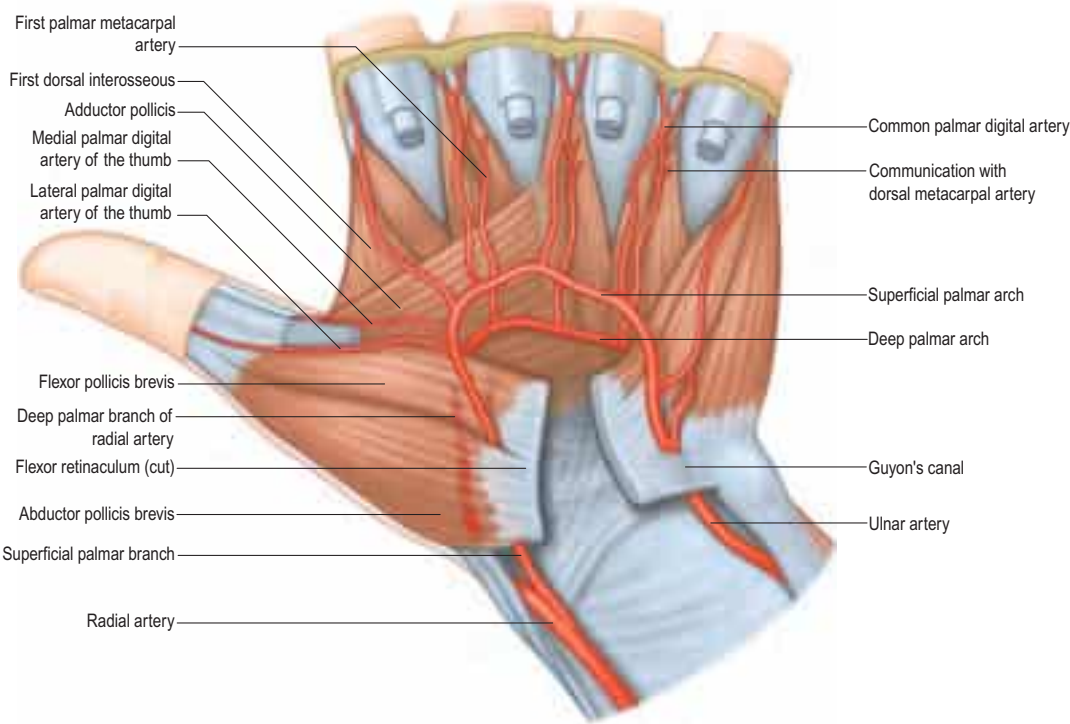


Fig. 50.36 The deep palmar arch and its branches. Note that the palmar digital arteries to the thumb have a separate origin from the superficial palmar arch in this specimen (as sometimes occurs). The arteria princeps pollicis is absent in this specimen.

Palmar carpal branch The palmar carpal branch arises near the distal border of pronator quadratus (see Fig. 49.9B). It crosses the anterior surface of the distal end of the radius, near the palmar carpal surface, and passes medially to anastomose behind the long flexor tendons with the palmar carpal branch of the ulnar artery. This transverse anastomosis is joined by longitudinal branches from the anterior interosseous artery and recurrent branches from the deep palmar arch, forming a cruciate palmar radiocarpal arch that supplies the carpal articulations and bones by descending branches.

Superficial palmar branch The superficial palmar branch arises from the radial artery just before it curves round the carpus. It passes through, and occasionally over, the thenar muscles, which it supplies, and usually anastomoses with the end of the ulnar artery to complete a superficial palmar arch (see Figs 50.35–50.36).

Dorsal carpal branch The dorsal carpal branch arises deep to the tendons of the thumb extensors. It runs medially across the dorsal carpal surface under the tendons and anastomoses with the ulnar dorsal carpal branch, and also with the anterior and posterior interosseous arteries, to form a dorsal radiocarpal arch (see Figs 49.9A, 49.9C). The carpal arches are both close to bone and supply the distal epiphyseal parts of the radius and ulna. Three dorsal metacarpal arteries descend from the dorsal intercarpal arch on the second to fourth dorsal interosseus, and bifurcate into dorsal digital branches to supply the adjacent sides of all four fingers.

First dorsal metacarpal artery The first dorsal metacarpal artery arises from the radial artery just before it passes between the heads of the first dorsal interosseous. It divides, almost at once, into two branches that supply the adjacent sides of the thumb and index finger as far distally as the proximal interphalangeal joint. The radial side of the thumb receives a branch direct from the radial artery proper. In most cases, the first dorsal metacarpal artery follows a fascial course overlying the first dorsal interosseous and parallel to the second metacarpal bone. Occasionally, it may follow an intramuscular course. Distally, it anastomoses with the dorsal branches of the radiopalmar digital artery of the index finger over the proximal phalanx.

Second, third and fourth dorsal metacarpal arteries The second to fourth dorsal metacarpal arteries arise from the dorsal carpal arch. Near their origins, they anastomose with the deep palmar arch by proximal perforating arteries and, near their bifurcation, with dorsal perforating branches from the palmar metacarpal arteries that pass between the metacarpal necks. They also anastomose distally at the level of the web spaces with dorsal perforating branches from the palmar digital arteries from the superficial palmar arch. The third and fourth dorsal metacarpal arteries are much smaller than the first and second. Cutaneous branches from the dorsal metacarpal arteries supply the dorsal skin as far distally as the proximal interphalangeal joint. At the level of the neck of the second, third and fourth metacarpals, a direct cutaneous branch is given off that passes proximally and supplies an area of skin between the two adjacent metacarpals (Fig. 50.37).

Arteria princeps pollicis The arteria princeps pollicis arises from the radial artery as it turns into the palm to form the deep palmar arch. It descends on the palmar aspect of the first metacarpal under the

oblique head of adductor pollicis lateral to the first palmar interosseous. At the base of the proximal phalanx, deep to the tendon of flexor pollicis longus, the artery divides into two branches that appear between the medial and lateral attachments of the oblique head of adductor pollicis and run along both sides of the thumb. On the palmar surface of the distal phalanx, the branches form a pollicial arch that supplies the skin and subcutaneous tissue. The arteria princeps pollicis is the usual nutrient of supply to the first metacarpal bone.

Arteria radialis indicis The arteria radialis indicis is often a proximal branch of the arteria princeps pollicis, or may arise from the superficial arch or from the first dorsal metacarpal artery, or may even combine with the arteria princeps pollicis to form the first palmar metacarpal artery. It descends between the first dorsal interosseous and transverse head of adductor pollicis, and runs along the lateral side of the index finger to its tip (see Fig. 50.35). It anastomoses with the index medial digital artery and with the arteria princeps pollicis and the superficial palmar arch at the distal border of the transverse head of adductor pollicis.

Deep palmar arch The deep palmar arch is formed by anastomosis of the end of the radial artery with the deep palmar branch of the ulnar artery (see Figs 50.36, 50.34C, 49.9B). It crosses the bases of the metacarpal bones and interossei, covered by the oblique head of adductor pollicis, the tendons of the digital flexors and the lumbricals. In its concavity, running laterally, is the deep branch of the ulnar nerve. Rarely, the arch may be incomplete.

Palmar metacarpal artery The three palmar metacarpal arteries run distally from the convexity of the deep palmar arch on the interossei of the second to fourth spaces, and join the common digital branches of the superficial arch at the digital clefts (see Fig. 50.36). They supply nutrient branches to the medial four metacarpals.

Perforating branches Three perforating branches from the deep palmar arch cross the second to fourth interosseous spaces between the heads of the corresponding dorsal interossei and anastomose with the dorsal metacarpal arteries.

Recurrent branches Recurrent branches ascend proximally from the deep palmar arch anterior to the carpus to supply the carpal bones and intercarpal articulations. They end in the palmar carpal arch.

Ulnar artery

At the wrist, the ulnar artery (see Figs 50.35–50.36) is covered by skin, fasciae and palmaris brevis. It lies between the superficial and main parts of the flexor retinaculum, lateral to the ulnar nerve and pisiform.

Dorsal cutaneous branch A constant dorso-ulnar perforator vessel is given off distally. It arises 2–5 cm proximal to the pisiform and accompanies the dorsal cutaneous branch of the ulnar nerve. It emerges between flexor carpi ulnaris and extensor carpi ulnaris.

Palmar carpal branch The palmar carpal branch crosses the distal ulna deep to the tendons of flexor digitorum profundus and anastomoses with the palmar carpal branch of the radial artery to make a palmar radiocarpal arch.

Dorsal carpal branch The dorsal carpal branch arises just proximal to the pisiform. It curves deep to the tendon of flexor carpi ulnaris to reach the carpal dorsum, which it crosses laterally beneath the extensor tendons. It anastomoses with the dorsal carpal branch of the radial artery to complete the dorsal carpal arch. Near its origin, it sends a small digital branch along the ulnar side of the fifth metacarpal to supply the medial side of the dorsal surface of the fifth finger.

Deep palmar branch The deep palmar branch is often double. It passes between abductor and flexor digiti minimi, through or deep to opponens digiti minimi, and anastomoses with the radial artery, completing the deep palmar arch. The deep palmar branch accompanies the deep branch of the ulnar nerve.

Superficial palmar arch

The superficial palmar arch is an anastomosis fed mainly by the ulnar artery (see Figs 50.35–50.36). The latter enters the palm with the ulnar

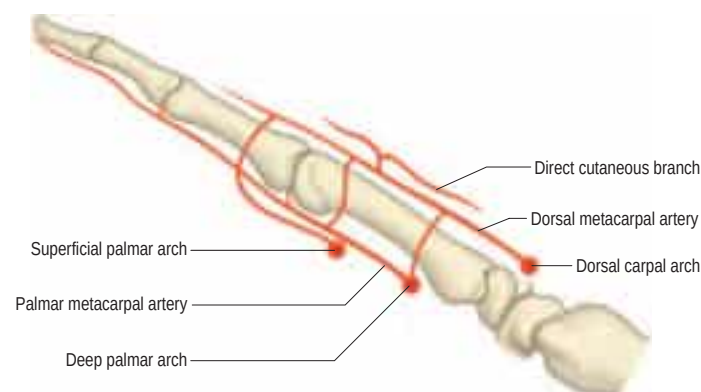


Fig. 50.37 Communication between the palmar and dorsal metacarpal arteries and the direct cutaneous branch given off the dorsal metacarpal artery, 0.5–1.0 cm proximal to the metacarpophalangeal joint: sagittal view.

nerve, anterior to the flexor retinaculum and lateral to the pisiform. It passes medial to the hook of the hamate, then curves laterally to form an arch that is convex distally and level with a transverse line through the distal border of the fully extended base of the thumb. About one-third of superficial palmar arches are formed by the ulnar artery alone, a further third are completed by the superficial palmar branch of the radial artery, and a third by the arteria radialis indicis, or a branch of either arteria princeps pollicis or the median artery. The superficial palmar arch is covered by palmaris brevis and the palmar aponeurosis, and it is superficial to flexor digiti minimi, branches of the median nerve and the tendons of the long flexors and the lumbricals.

Common and proper palmar digital arteries Three common palmar digital arteries arise from the convexity of the superficial palmar arch (see Fig. 50.36). They pass distally on the second to fourth lumbricals, each joined by a corresponding palmar metacarpal artery from the deep palmar arch, and divide into two proper palmar digital arteries. These run along the contiguous sides of all four fingers, dorsal to the digital nerves, between Grayson's and Cleland's ligaments, anastomosing in the subcutaneous tissue of the fingertips and near the interphalangeal joints. Each digital artery has two dorsal branches that anastomose with the dorsal digital arteries and supply the soft parts dorsal to the middle and distal phalanges, including the matrices of the nails. The palmar digital artery for the medial side of the little finger leaves the arch under palmaris brevis. Palmar digital arteries supply the metacarpophalangeal and interphalangeal joints and nutrient rami to the phalanges. They are the main digital supply because the dorsal digital arteries are minute. The origins of the palmar digital arteries of the thumb are quite variable. Both may arise from a single arteria princeps pollicis or they may arise separately from the superficial palmar arch. The ulnar digital artery may arise from the first dorsal metacarpal artery.

The terminal branches of the digital arteries contribute to a number of vascular arcades that provide a rich vascular supply for the distal elements of each digit (see Fig. 50.37; Fig. 50.38). Three distal phalan-

geal dorsal arterial arcades anastomose with each other and with those from the other side of the digit. The superficial arcade occurs at the level of the proximal nail fold and is supplied primarily by a dorsal branch from the palmar digital artery, which is given off at the level of the middle phalanx. The proximal subungual arcade is at the level of the lunula and is supplied by a terminal branch of the digital artery, which passes dorsally. The distal subungual arcade occurs more distally in the nail bed. It is supplied by a dorsal vessel that emerges from the point of confluence of the 'H'-shaped anastomosis between the terminal portions of both digital arteries, and passes from volar to dorsal aspect of the digit under the interosseous ligament connecting the proximal and distal parts of the distal phalanx.

Variations The ulnar and radial arteries to the hand may occasionally be supplemented by a median artery that contributes to, or supplies, the superficial palmar plane and by the anterior interosseous artery, which may make a contribution to the deep palmar plane.

Rodriguez-Niedenführ et al (1999) have confirmed that the median artery may persist in two different forms, palmar and antebrachial. The palmar type arises from the junction of the ulnar artery and its common interosseous trunk; it is long and large, and reaches the palm, usually ending as the first or second common digital artery, or both. The antebrachial type is slender and short, usually arising from the anterior interosseous artery and finishing in the forearm. These variations are seen in about 20% of individuals and tend to be unilateral.

Kleinert et al (1989) demonstrated by plethysmography that 5% of hands have ulnar artery dominance in all digits, compared with 28% with complete radial digital dominance.

Glomus tumours Glomus tumours are very painful tumours of the glomus bodies (small arteriovenous anastomoses involved in the regulation of peripheral skin temperature control (see Fig. 6.16)). They are often tiny and sometimes difficult to identify. These tumours typically occur in the proximal nail fold/subungual regions of the fingertips in association with the dorsal digital arterial arcades, although they can occur anywhere. They are diagnosed by exquisite point tenderness over the swelling, with reduction of tenderness when the finger is exsanguinated.

VEINS

Superficial veins of the hand

Dorsal and palmar digital veins Dorsal digital veins pass along the sides of the fingers, joined by oblique branches. They unite from the adjacent sides of the digits into three dorsal metacarpal veins that form a dorsal venous network over the metacarpus (Fig. 50.39). This is joined laterally by a dorsal digital vein from the radial side of the index finger and both dorsal digital veins of the thumb, and is prolonged proximally as the cephalic vein. Medially, a dorsal digital vein from the ulnar side of the little finger joins the network, which ultimately drains proximally into the basilic vein. A vein often connects the central parts of the network to the cephalic vein near the mid-forearm. Palmar digital veins connect to their dorsal counterparts by oblique veins that pass between metacarpal heads. They also drain to a plexus superficial to the palmar aponeurosis, extending over both thenar and hypothenar regions.

Cephalic vein The cephalic vein forms over the anatomical snuff-box from the radial extremity of the dorsal venous plexus and runs proximally over the distal lateral aspect of the radius, where it is easily visible. Its further course is described on pages 858 and 830.

Deep veins of the hand

Superior and deep palmar venous arches Superficial and deep palmar venous arches accompany their arterial counterparts and receive the corresponding branches; common palmar digital veins join the superficial arch and palmar metacarpal veins join the deep arch.

Palmar and dorsal metacarpal veins Deep veins accompanying the dorsal metacarpal arteries receive perforating branches from the palmar metacarpal veins. They end in the radial veins and the dorsal venous network over the metacarpus. This network is joined laterally by a dorsal digital vein from the radial side of the index finger and by both digital veins of the thumb, and is prolonged proximally as the cephalic vein.

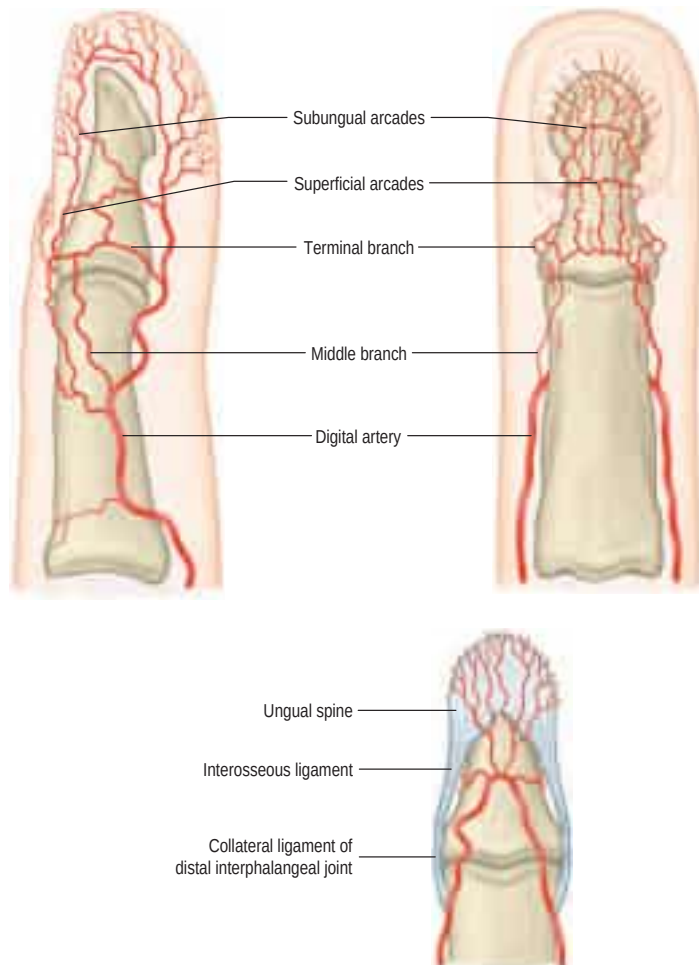


Fig. 50.38 Terminal vascular arcades over the distal phalanx. (With permission from Flint MH 1956 Some observations on the vascular supply of the nail bed and terminal segments of the fingers. BJPS 8:186–95.)

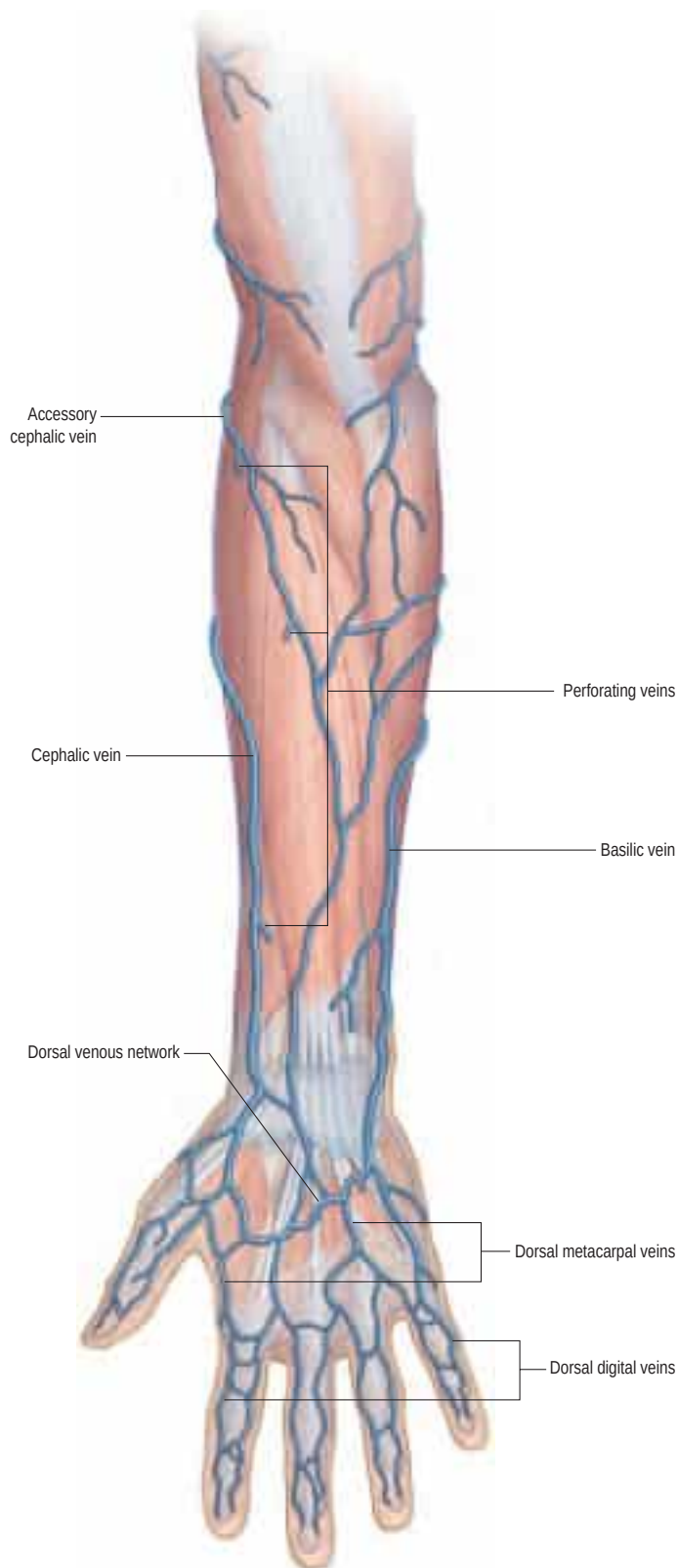


Fig. 50.39 The veins of the dorsum of the hand. (Redrawn with permission from Drake RL, Vogl AW, Mitchell A et al (eds), *Gray's Atlas of Anatomy*, Elsevier, Churchill Livingstone. Copyright 2008.)

INNERVATION

Median nerve

The median nerve proximal to the flexor retinaculum is lateral to the tendons of flexor digitorum superficialis and lies between the tendons of flexor carpi radialis and palmaris longus (Figs 50.40A–50.41). It passes under the retinaculum in the 'carpal tunnel', where it may be compressed in the carpal tunnel syndrome (see Fig. 46.11). Distal to the retinaculum, the nerve enlarges and flattens, and usually divides into five or six branches; the mode and level of division are variable.

Muscular branch (motor or recurrent branch)

The muscular branch is short and thick, and arises from the lateral side of the nerve; it may be the first palmar branch or a terminal branch that arises level with the digital branches. It runs laterally, just distal to the flexor retinaculum and with a slight recurrent curve beneath the part of the palmar aponeurosis covering the thenar muscles (see Fig. 50.40A). It turns round the distal border of the retinaculum to lie superficial to flexor pollicis brevis, which it usually supplies, and either continues superficial to the muscle or traverses it. It gives a branch to abductor pollicis brevis that enters the medial edge of the muscle, and then passes deep to it to supply opponens pollicis, entering its medial edge. Its terminal part occasionally gives a branch to the first dorsal interosseous, and may be its sole or partial supply. The muscular branch may arise in the carpal tunnel and pierce the flexor retinaculum. Anomalies of the median nerve occur in approximately 10% of patients undergoing a carpal tunnel release. These branches usually arise from the ulnar side of the nerve and may be motor or sensory.

Palmar digital branches

The median nerve usually divides into four or five digital branches (see Figs 50.40A, 50.35). It often divides first into a lateral ramus, providing digital branches to the thumb and the radial side of the index finger, and a medial ramus, supplying digital branches to adjacent sides of the index, middle and ring fingers. Other modes of termination can occur.

Digital branches are commonly arranged as follows. They pass distally, deep to the superficial palmar arch and its digital vessels, at first anterior to the tendons of the long flexors. Two proper palmar digital nerves, sometimes from a common stem, pass to the sides of the thumb; the nerve supplying its radial side crosses in front of the tendon of flexor pollicis longus. The proper palmar digital nerve to the lateral side of the index finger also supplies the first lumbrical. Two common palmar digital nerves pass distally between the tendons of the long flexors. The lateral divides in the distal palm into two proper palmar digital nerves that traverse adjacent sides of the index and middle finger. The medial divides into two proper palmar digital nerves that supply adjacent sides of the middle and ring fingers. The lateral common digital nerve supplies the second lumbrical, and the medial receives a communicating twig from the common palmar digital branch of the ulnar nerve and may supply the third lumbrical. In the distal part of the palm, the digital arteries pass deeply between the divisions of the digital nerves; the nerves lie anterior to the arteries on the sides of the digits. The median nerve usually supplies palmar cutaneous digital branches to the radial three and a half digits (thumb, index, middle and the lateral side of the ring); sometimes, the radial side of the ring finger is supplied by the ulnar nerve. Occasionally, there is a communicating branch between the common digital nerve to the middle and ring fingers (derived from the median nerve) and the common digital nerve to the ring and little fingers (derived from the ulnar nerve); this may explain variations in sensory patterns that do not conform to the classic pattern.

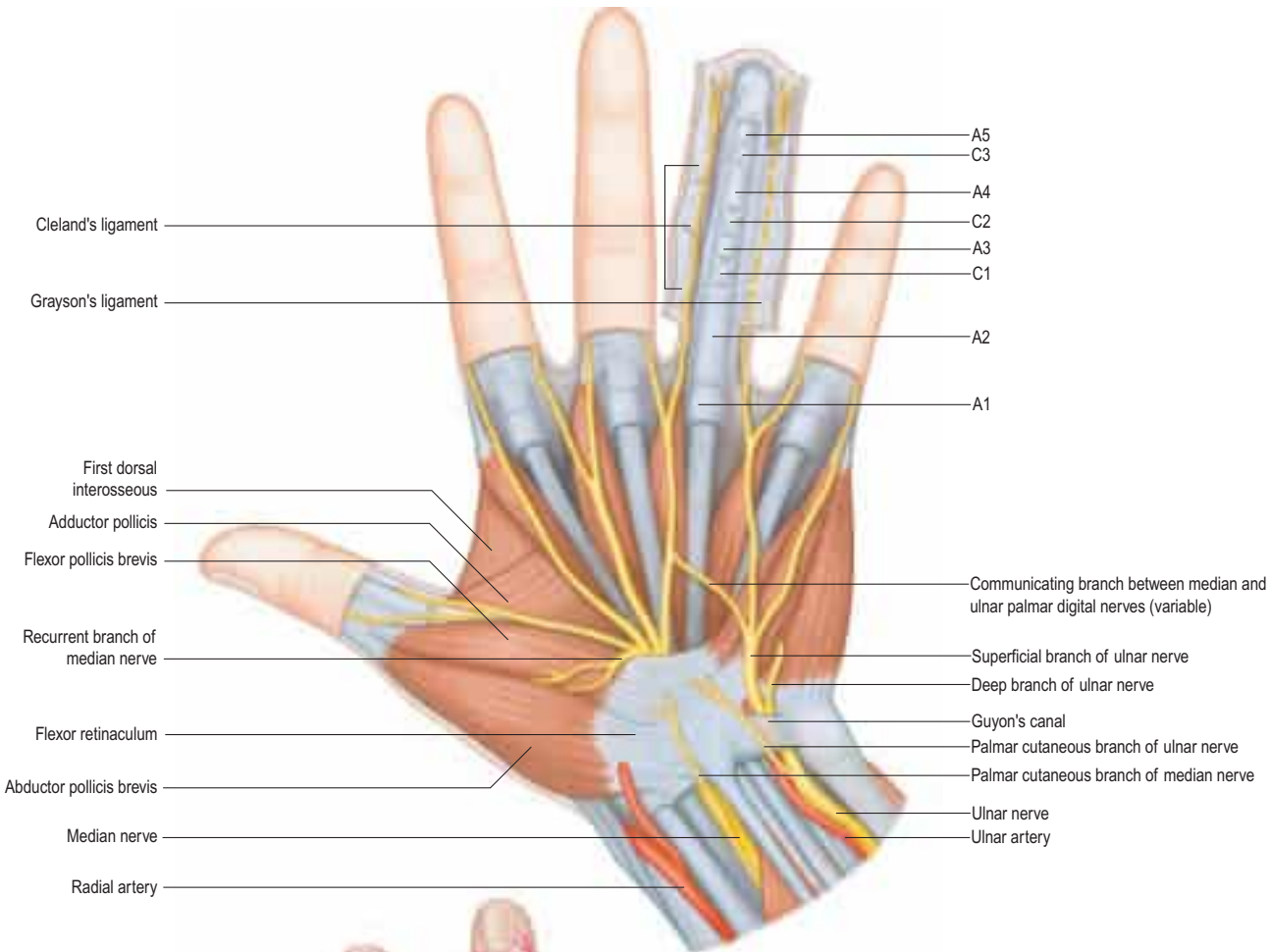
The proper palmar digital nerves pass along the medial side of the index finger and both sides of the middle and the lateral side of the ring finger. They enter these digits in fat between slips of the palmar aponeurosis. Together with the lumbricals and palmar digital arteries, they pass dorsal to the superficial transverse metacarpal ligament and ventral to the deep transverse metacarpal ligament. In the digits, the nerves run distally beside the long flexor tendons (outside their fibrous sheaths), level with the anterior phalangeal surfaces and anterior to the digital arteries, between Grayson's and Cleland's ligaments. Each nerve gives off several branches to the skin on the front and sides of the digit and sends branches to the metacarpophalangeal and interphalangeal joints.

The digital nerves supply the fibrous sheaths of the long flexor tendons, blood vessels (vasomotor) and sweat glands (sudomotor). Distal to the base of the distal phalanx, each digital nerve gives off a branch that passes dorsally to the nail bed. The main nerve frequently trifurcates to supply the pulp and skin of the terminal part of the digit. Distal to the base of the proximal phalanx, each proper digital nerve also gives off a dorsal branch to supply the skin over the back of the middle and distal phalanges. The proper palmar digital nerves to the thumb and the lateral side of the index finger emerge with the tendons of the long flexors from under the lateral edge of the palmar aponeurosis. They are arranged in the digits as described above, except that, in the thumb, small distal branches supply the skin on the back of the distal phalanx only.

Other branches

In addition to the branches of the median nerve described above, variable vasomotor branches supply the radial and ulnar arteries and their branches. Some of the intercarpal, carpometacarpal and

A



B

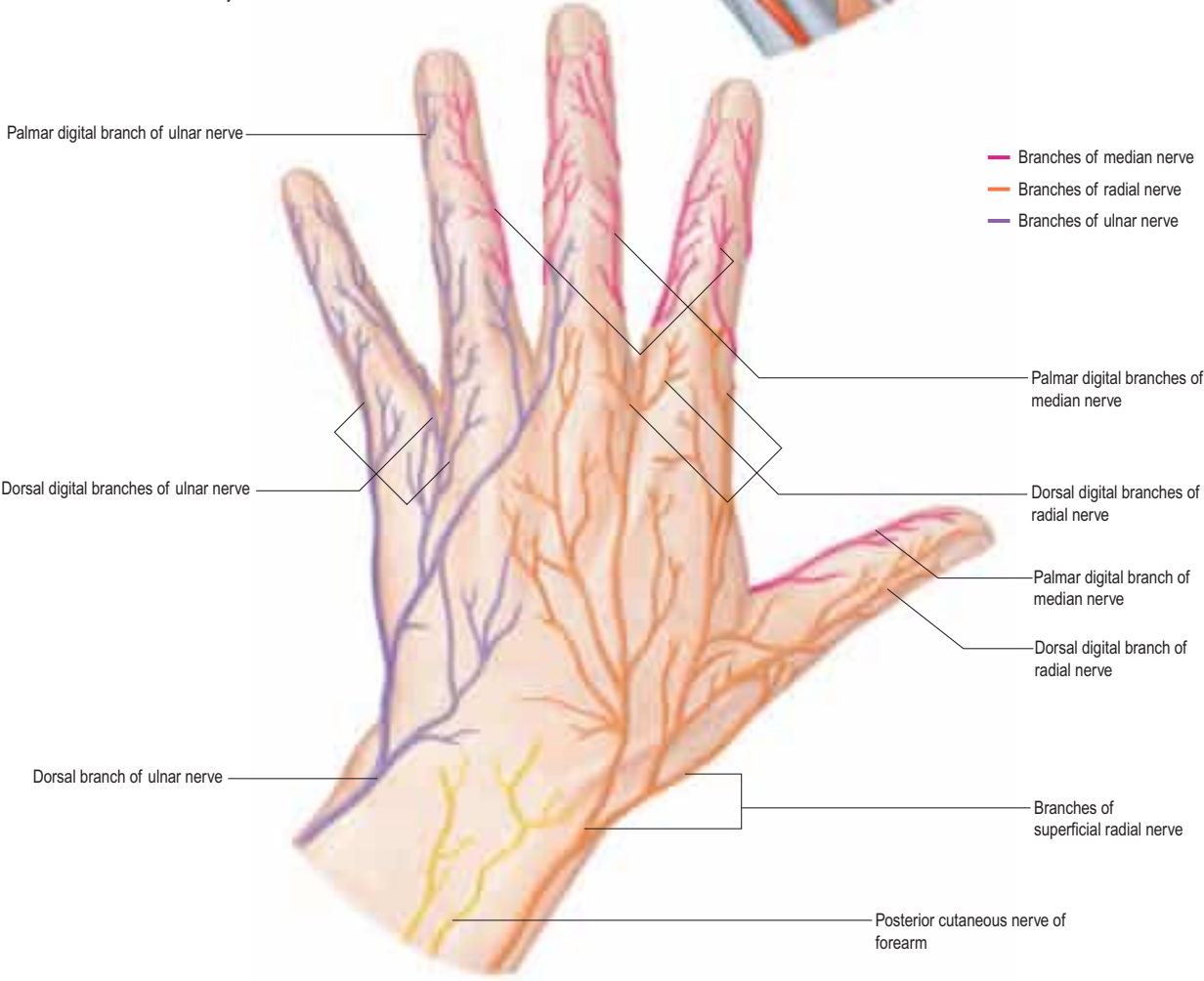


Fig. 50.40 The cutaneous nerves of the hand. **A**, Palmar aspect. The anular and cruciate pulleys are shown schematically in the ring finger. **B**, Dorsal aspect.

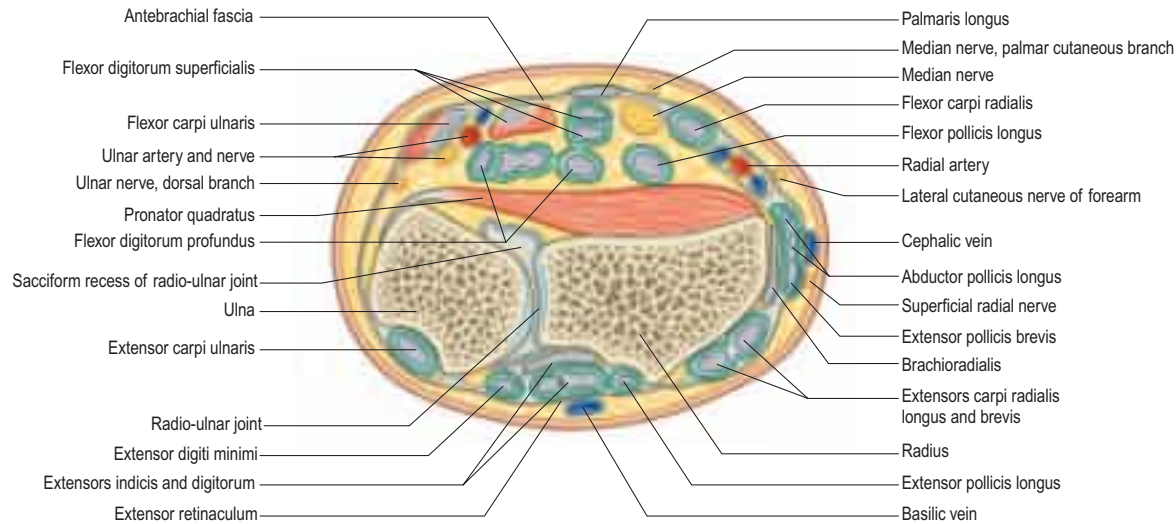


Fig. 50.41 A transverse section through the supinated left wrist at the level of the distal radio-ulnar joint, showing the muscle tendons (in grey) and their synovial sheaths (in green). (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

intermetacarpal joints are said to be supplied by the median nerve or its anterior interosseous branch; the precise details are uncertain.

Carpal tunnel syndrome

Carpal tunnel syndrome is the most common entrapment mononeuropathy. It is caused by compression of the median nerve as it passes through the fibro-osseous tunnel beneath the flexor retinaculum (see Fig. 50.28).

Median nerve division at the wrist

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Ulnar nerve

At the wrist, the ulnar nerve passes under the superficial part of the flexor retinaculum (in Guyon's canal) with the ulnar artery, and divides into superficial and deep terminal branches (see Fig. 50.5).

Dorsal branch

The dorsal branch arises approximately 5 cm proximal to the wrist (see Fig. 49.23). It passes distally and dorsally, deep to flexor carpi ulnaris, perforates the deep fascia, descends along the medial side of the back of the wrist and hand, and then divides into two, or often three, dorsal digital nerves. The first supplies the medial side of the little finger; the second, the adjacent sides of the little and ring fingers; while the third, when present, supplies adjoining sides of the ring and middle fingers. The latter may be replaced, wholly or partially, by a branch of the radial nerve, which communicates with it on the dorsum of the hand (see Fig. 50.40). In the little finger, the dorsal digital nerves extend only to the base of the distal phalanx; in the ring finger, they extend only to the base of the middle phalanx. The most distal parts of the little finger and of the ulnar side of the ring finger are supplied by dorsal branches of the proper palmar digital branches of the ulnar nerve. The most distal part of the lateral side of the ring finger is supplied by dorsal branches of the proper palmar digital branch of the median nerve.

Superficial and deep terminal branches **Superficial terminal branch**

The superficial terminal branch supplies palmaris brevis and the medial palmar skin. It divides into two palmar digital nerves, which can be palpated against the hook of the hamate bone. One supplies the medial side of the little finger; the other (a common palmar digital nerve) sends a branch to the median nerve and divides into two proper digital nerves to supply the adjoining sides of little and ring fingers. The proper digital branches are distributed like those derived from the median nerve.

Deep terminal branch

The deep terminal branch accompanies the deep branch of the ulnar artery as it passes between abductor digiti minimi and flexor digiti minimi, and then perforates opponens digiti minimi to follow the deep palmar arch dorsal to the flexor tendons (see Fig. 50.40A). At its origin, it supplies the three short muscles of the little finger. As it crosses the hand, it supplies the interossei and the third and fourth lumbricals. It

ends by supplying adductor pollicis, the first dorsal interosseous and, usually, flexor pollicis brevis.

The medial part of flexor digitorum profundus is supplied by the ulnar nerve, as are the third and fourth lumbricals, which are connected with the tendons of this part of the muscle. Similarly, the lateral part of flexor digitorum profundus and the first and second lumbricals are supplied by the median nerve. The third lumbrical is often supplied by both nerves. The deep terminal branch gives branches to some intercarpal, carpometacarpal and intermetacarpal joints; precise details are uncertain. Vasomotor branches, arising in the forearm and hand, supply the ulnar and palmar arteries.

Ulnar tunnel syndrome

Ulnar tunnel syndrome is an entrapment neuropathy of the ulnar nerve as it passes through Guyon's canal at the wrist.

Ulnar nerve division at the wrist

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Radial nerve

Branches of the superficial branch of the radial nerve reach the hand by curving around the wrist over the tendons of abductor pollicis longus and extensor pollicis brevis, and dividing into dorsal digital nerves (see Fig. 50.40B, Fig. 50.42). They usually communicate with the posterior and lateral cutaneous nerves of the forearm on the dorsum of the hand.

Dorsal digital nerves

There are usually four or five small dorsal digital nerves. The first supplies the skin of the radial side of the thumb and the adjoining thenar eminence, and communicates with branches of the lateral cutaneous nerve of the forearm. The second supplies the medial side of the thumb; the third, the lateral side of the index finger; the fourth, the adjoining sides of the index and middle fingers. The fifth communicates with a ramus of the dorsal branch of the ulnar nerve and supplies the adjoining sides of the middle and ring fingers, where it is frequently replaced by the dorsal branch of the ulnar nerve. The digital nerves of the thumb reach only to the root of the nail; those in the index finger, midway along the middle phalanx; and those to the middle and the lateral part of the ring finger may reach no further than the proximal interphalangeal joints. The remaining distal dorsal areas of the skin in these digits are typically supplied by palmar digital branches of the median and ulnar nerves. However, the superficial terminal branch of the radial nerve may supply the whole dorsum of the hand.

SURFACE ANATOMY OF THE WRIST AND HAND

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See Video 50.2.

The carpal tunnel may be narrowed by arthritic changes in the wrist joint, particularly from rheumatoid arthritis, and by soft tissue thickening, as may occur in myxoedema and acromegaly; it may also be associated with oedema, obesity or pregnancy. Usually, the condition is idiopathic. Normally, the median nerve slides smoothly in and out of the carpal tunnel during flexion and extension of the wrist, but when the nerve is compressed, additional damage may be produced during these movements. The dominant hand is usually affected first, probably because this hand is used more frequently and more vigorously. Typically, the syndrome produces pain, paraesthesia and numbness in the thumb, index, middle and medial side of the ring fingers, which is worse at night and on gripping objects. The palmar branch of the median nerve is spared because it does not pass through the carpal tunnel. With time, the compression leads to wasting and weakness of abductor pollicis brevis. Treatment is usually surgical decompression of the nerve by dividing the flexor retinaculum, either as an open procedure or with endoscopic assistance. The most common hazards of decompression are: inadequate decompression because of failure to divide the most distal part of the flexor retinaculum; damage to the motor, or palmar cutaneous, or one or more terminal sensory branches; prolapse of the nerve through the gap in the retinaculum; and damage to the superficial palmar arch causing haematoma, compression, pain and, ultimately, fibrosis or even a late false aneurysm (Birch 2011). The distal ulnar nerve and its branches may also be damaged.

Anatomical anomalies of the median nerve occur at the level of the carpal tunnel in up to 12% of hands. The most common are variations in the course of the recurrent motor branch in relation to the flexor retinaculum: transligamentous (23%), subligamentous (31%) or extraligamentous (46%) (Kretschmer et al 2009). Damage may also occur to the palmar cutaneous branch of the median nerve and to the superficial connections between the median and ulnar nerves (the so-called Berrettini branch), division of which will leave the patient with numbness in the middle and ring fingers. Stancić et al (1999) noted that this connection was present in 81% of dissected hands and proposed that it should be regarded as a normal anatomical finding.

Median nerve division at the wrist leads to paralysis of the lumbricals to the index and middle fingers and of the thenar muscles (apart from flexor pollicis brevis and adductor pollicis), as well as loss of sensation to the thumb, index, middle and radial half of the ring fingers. The radial half of the hand becomes flattened as a result of wasting of the thenar muscles and the adducted posture of the thumb. Division of the median nerve distal to the origin of its palmar cutaneous branch, which arises somewhere between 3 and 7 cm proximal to the proximal edge of the flexor retinaculum and the distal transverse wrist crease, will leave the branch intact to supply the skin over the thenar eminence and radial side of the proximal part of the hand. Careful clinical examination of the sensibility of the skin in this region should help to clarify the level of the lesion. High division of the nerve (Ch. 49) will, as with compressive lesions at this level, result in weakness of pronation of the forearm, flexion of the wrist and the lateral two fingers.

Causes of compression at this site include a ganglion, trauma, and proximity of aberrant or accessory muscles. The symptoms include pain in the hand or forearm and sensory changes in the palmar aspect of the little and ulnar half of the ring fingers; sensation on the ulnar aspect of the dorsum of the hand is normal. In addition, there may be weakness and wasting of the intrinsic muscles of the hand supplied by the ulnar nerve, with clawing posture in extreme cases.

Surgical treatment involves decompression of the nerve by division of the roof of Guyon's canal and removal of the causative lesion, such as a ganglion. Ulnar tunnel syndrome can be difficult to distinguish from more proximal causes of ulnar nerve compression, of which the most common is entrapment of the nerve in cubital canal at the elbow (Ch. 49). Sensory changes may be present more proximally and, in severe cases, weakness may be noted in flexor carpi ulnaris and the ulnar part of flexor digitorum profundus. In the absence of obvious clinical indicators, local tenderness over the site of compression and nerve conduction studies may aid diagnosis.

Ulnar nerve division at the wrist paralyses all the intrinsic muscles of the hand (apart from the radial two lumbricals, abductor pollicis brevis, and part of flexor pollicis brevis and opponens pollicis). The intrinsic muscle action of flexing the metacarpophalangeal joint and extending the interphalangeal joints is lost. The unopposed action of the long extensors and flexors of the fingers cause clawing, with extension of the metacarpophalangeal joints and flexion of the interphalangeal joints. The clawing is less intense in the index and middle fingers because their lumbricals are supplied by the median nerve. (For a detailed account of the hand posture adopted in ulnar nerve lesions, see Smith (2002).) There is sensory loss over the little finger and the ulnar half of the ring finger. The skin over the ulnar aspect of the dorsum of the hand is spared because the dorsal branch of the ulnar nerve is given off approximately 5 cm proximal to the wrist joint. A combined median and ulnar nerve palsy at the wrist results in a full claw hand with thenar and hypothenar flattening and thumb adduction and flexion (a simian hand).

The superficial cutaneous branches of the radial nerve are at risk of injury from any procedure or injury in their immediate vicinity. Incisions or injections in or around the radial border of the wrist are particularly implicated. The resultant painful neuromata can prove to be resistant to treatment.

FLEXOR/VOLAR SURFACE

The wrist joint is identified between the distal ends of the radius and ulna, and the proximal carpus, on flexion and extension of the wrist. The line of the wrist joint corresponds to a line, convex superiorly, joining the radial and ulnar styloid processes. Its position is also marked by the proximal of the two transverse anterior wrist creases (see Fig. 46.30A). The radial styloid process normally sits 1 cm distal to the ulnar styloid process. Alteration of this arrangement can indicate fracture.

When the forearm is pronated, the head of the ulna forms a rounded surface elevation on the medial side of the posterior aspect of the wrist (see Figs 46.29C, 46.30B), and the ulnar styloid process projects distally from its posteromedial aspect. The radial styloid process projects inferiorly from the distal radius along the lateral aspect of the wrist. The posterior aspect of the lower end of the radius is partly obscured by the extensor tendons but is palpable. The prominent dorsal radial (Lister's) tubercle is both visible and palpable, and aligns with the web space between digits 2 and 3 (see Fig. 46.30B). The tendon of extensor pollicis longus in extensor compartment 3 sits medial to the tubercle, and the tendons of extensors carpi radialis longus and brevis sit lateral to the tubercle in compartment 2.

Four of the carpal bones are palpable. The pisiform is both visible and palpable on the palmar aspect of the medial wrist at the base of the hypothenar eminence. It is often overlain by the distal wrist crease. It can be moved over the articular surface of the triquetrum when the wrist is passively flexed. The hook of the hamate lies approximately 2.5 cm distal to the pisiform, in line with the ulnar border of the ring finger. It can be felt on deep palpation, and the superficial division of the ulnar nerve can be rolled from side to side over the tip of the hook. The tubercle of the scaphoid is situated at the base of the thenar eminence, in line with the tendon of flexor carpi radialis and, in many individuals, forms a small, visible elevation. Immediately distal to it, but covered by the muscles of the thenar eminence, the tubercle of the trapezium can be identified on deep palpation.

Proximal to the wrist creases, the prominent tendon of flexor carpi radialis can be seen and palpated when the wrist is flexed against resistance (see Fig. 46.30A). The radial artery lies on its lateral side and crosses the anterior margin of the expanded distal end of the radius, where it is palpable. It then passes posteriorly, deep to the tendons of abductor pollicis longus and extensor pollicis brevis, and enters the anatomical snuff-box, where its pulsations are palpable (see Fig. 46.30C).

Palpation lateral to flexor carpi radialis, 3–4 cm proximal to the wrist crease, reveals the muscle belly of flexor pollicis longus (flexing and extending the thumb will confirm that the examining finger is correctly placed). The area on the ulnar side of flexor carpi radialis tendon is packed with functionally important structures. The median nerve is either covered by, or situated just lateral/radial to, the tendon of palmaris longus. The latter can be identified by gently pinching the tips of the digits together and partially flexing the wrist (see Fig. 46.30A). Near to the wrist, the median nerve lies very close to the skin and can, therefore, be injured by relatively superficial lacerations. When palmaris longus is absent, only a thin covering of subcutaneous fat and deep fascia separate skin and nerve. The four tendons of flexor digitorum superficialis lie deep to the median nerve; the tendons to the middle and ring fingers lie anterior to those for the index and little fingers as they pass deep to the flexor retinaculum, and can be felt and usually seen to move during finger flexion/extension. Deeper still are the tendons of flexor digitorum profundus. The broad tendon of flexor carpi ulnaris is easily palpated passing to the pisiform on the ulnar side of the anterior wrist; the ulnar nerve, artery and venae comitantes lie along its lateral (radial) edge. Any sharp injury that cuts through the tendon usually has enough energy to sever the ulnar neurovascular bundle.

ANATOMICAL SNUFF-BOX

When the thumb is fully extended, the depression of the anatomical snuff-box is seen on the lateral aspect of the wrist immediately distal to the radial styloid process. The snuff-box is bounded on its anterolateral (radial) side by the tendons of abductor pollicis longus (laterally) and extensor pollicis brevis (immediately medial), and on its posteromedial/ulnar side by the tendon of extensor pollicis longus. Running a finger along extensor pollicis brevis enables palpation of the superficial radial nerve, which can be rolled from side to side on the tendon. The cephalic vein passes through the roof of the snuff-box, where it is visible and palpable, and the pulsation of the radial artery can be palpated deeply on its floor.

Three bones are palpable distal to the radial styloid process along the floor of the anatomical snuff-box: the convex, ovoid proximal articular surface of the scaphoid (best palpated during alternating wrist abduction/adduction), the radial aspect of the trapezium and the expanded base of the first metacarpal. The first carpometacarpal joint forms a palpable depression distal to the trapezium. When wrist stability is clinically assessed throughout its range of movements, the scaphoid may be compressed bidigitally, between index finger and thumb, along its oblique long axis between the tubercle and articular surface. Similarly, the trapezium may be compressed between its tubercle and radial aspect.

EXTENSOR/DORSAL SURFACE

The tendons located within the six extensor compartments of the wrist are subcutaneous and palpable (see Fig. 46.30B). Compartment 1 (containing abductor pollicis longus and extensor pollicis brevis) sits over the lateral radius. Regional tenderness to palpation occurs in de Quervain's tenosynovitis. Compartments 2 (containing extensors carpi radialis longus and brevis) and 3 (containing extensor pollicis longus) flank the lateral and medial sides of the dorsal radial (Lister's) tubercle, respectively. Compartments 5 (containing extensor digiti minimi) and 6 (containing extensor carpi ulnaris) flank the posterior and medial sides of the ulnar head, respectively. Compartment 4 (containing extensors digitorum and indicis) sits between the ulnar head and compartment 3. The tendons of extensors carpi radialis longus and brevis can be identified on the dorsal carpus when the fist is clenched and relaxed. The tendons of extensor digitorum are visible and palpable on the dorsal hand with the fingers fully extended. The tendon of extensor carpi ulnaris is palpable distal to the ulnar styloid as it crosses the wrist, with the wrist in extension and adduction. The first dorsal interosseous forms a visible fleshy mass on the lateral part of the dorsal hand between the index finger and thumb, especially whilst abducting the index finger against resistance.

HAND

The palmar skin is marked by a number of creases but they are of little value as points of reference. Transverse skin creases cross the palmar aspect of the fingers in three places: the most proximal crease sits at the junction of the digit with the palm and lies approximately 2 cm distal to the metacarpophalangeal joint; the intermediate crease sits opposite the proximal interphalangeal joint; and the distal crease sits just proximal to the distal interphalangeal joint.

The first metacarpal is both visible and palpable on the dorsolateral hand; its dorsal aspect forms the lateral border of the hand. The fleshy elevation of the thenar eminence, produced by abductor and flexor pollicis brevis overlying opponens pollicis, is located on the lateral palm (see Fig. 46.48A). The less prominent hypothenar eminence is located on the medial palm and is formed by the comparable muscles of the little finger. The medial aspect of the hypothenar eminence forms the medial border of the hand.

The dorsal aspects of the trapezium and of each of the metacarpal bones are partially palpable around the margins of the corresponding extensor tendons. The heads of the metacarpal bones form the prominences of the knuckles and are most obvious during digit flexion. On the dorsal hand, the metacarpophalangeal joint line forms a palpable depression lateral to the digital extensor tendon, and alternating digit flexion/extension aids identification. The joint can be aspirated/injected from this perspective. Palpation distal to the metacarpal head reveals the flared base of the corresponding proximal phalanx. The interphalangeal joints are palpable on the dorsal aspect of a flexed digit just distal to the prominences formed by the heads of the proximal and middle phalanges.

The flexor retinaculum, and therefore the surface position of the carpal tunnel, can be mapped according to its palpable bony attachments (see Fig. 46.30A). The distal border is concave inferiorly and is marked by a curved line that joins the tubercle of the trapezium to the hook of the hamate. Its proximal border is marked by a curved line, concave superiorly, which joins the tubercle of the scaphoid to the pisiform. Flexion of the wrist produces a number of transverse skin creases (usually two, sometimes three) at the wrist. The dominant, most distal, wrist crease normally sits distal to the lunate and overlies pisiform and the proximal edge of the flexor retinaculum. The carpal tunnel passes distal to this crease for 2.5–3 cm.

The position of the superficial palmar arterial arch is indicated by a horizontal line approximately 4 cm long, drawn at the level of the fully extended and partially abducted thumb (see Fig. 46.30A). A horizontal line, approximately 4 cm long and running from a point just distal to the hook of the hamate, indicates the position of the deep palmar arterial arch. The deep arch is located approximately 1 cm proximal to the superficial arch. The digital neurovascular bundles pass along the medial and lateral sides of the dorsal and volar surfaces of the digits, and can be anaesthetized in these positions via the web space.



Bonus e-book images and videos

Fig. 50.5 Guyon's canal at the wrist.

Fig. 50.21 The effect of axial loading on the rotatory movements of the bones of the wrist.

Fig. 50.22 Scapholunate dissociation.

Fig. 50.34 Cont'd Hand digital subtraction macroangiogram. Predominant phases.

Video 50.1 Movements of the hand.

Video 50.2 Wrist block: surface anatomy.

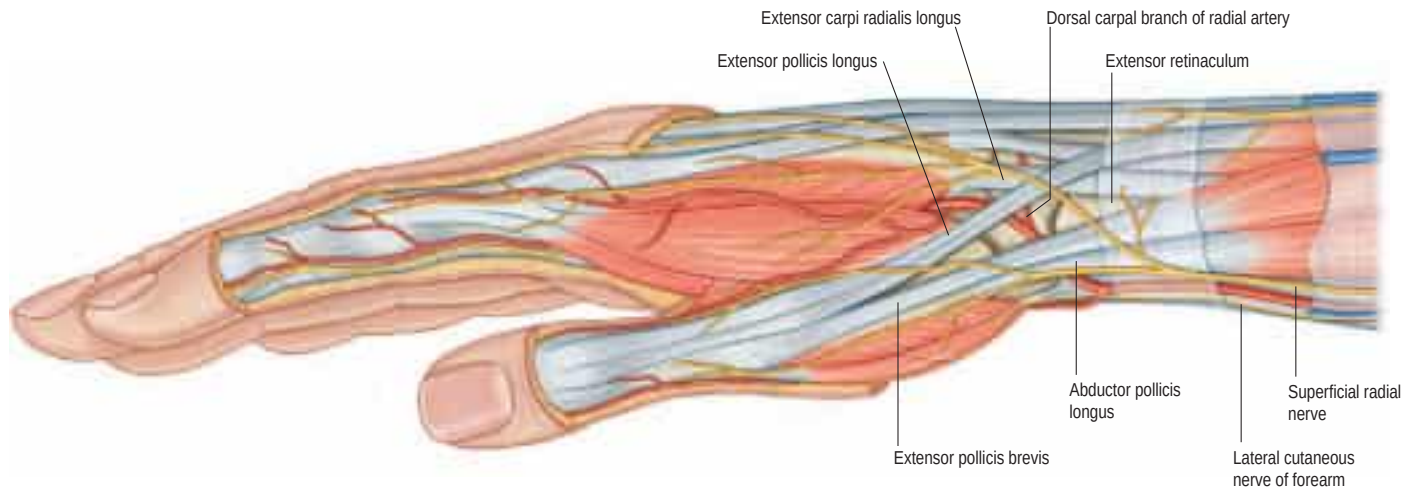


Fig. 50.42 The radial side of the hand. Note the course of the terminal branches of the superficial radial nerve and the lateral cutaneous nerve of the forearm. (Courtesy of Rolfe Birch, all rights reserved, published in Birch R, *Surgical Disorders of the Peripheral Nerve*, 2nd ed, 2011. Springer-Verlag, London.)

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A chapter describing the various ways in which surgeons and others can injure the peripheral nerves of the body. Essential reading for all orthopaedic surgeons.

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